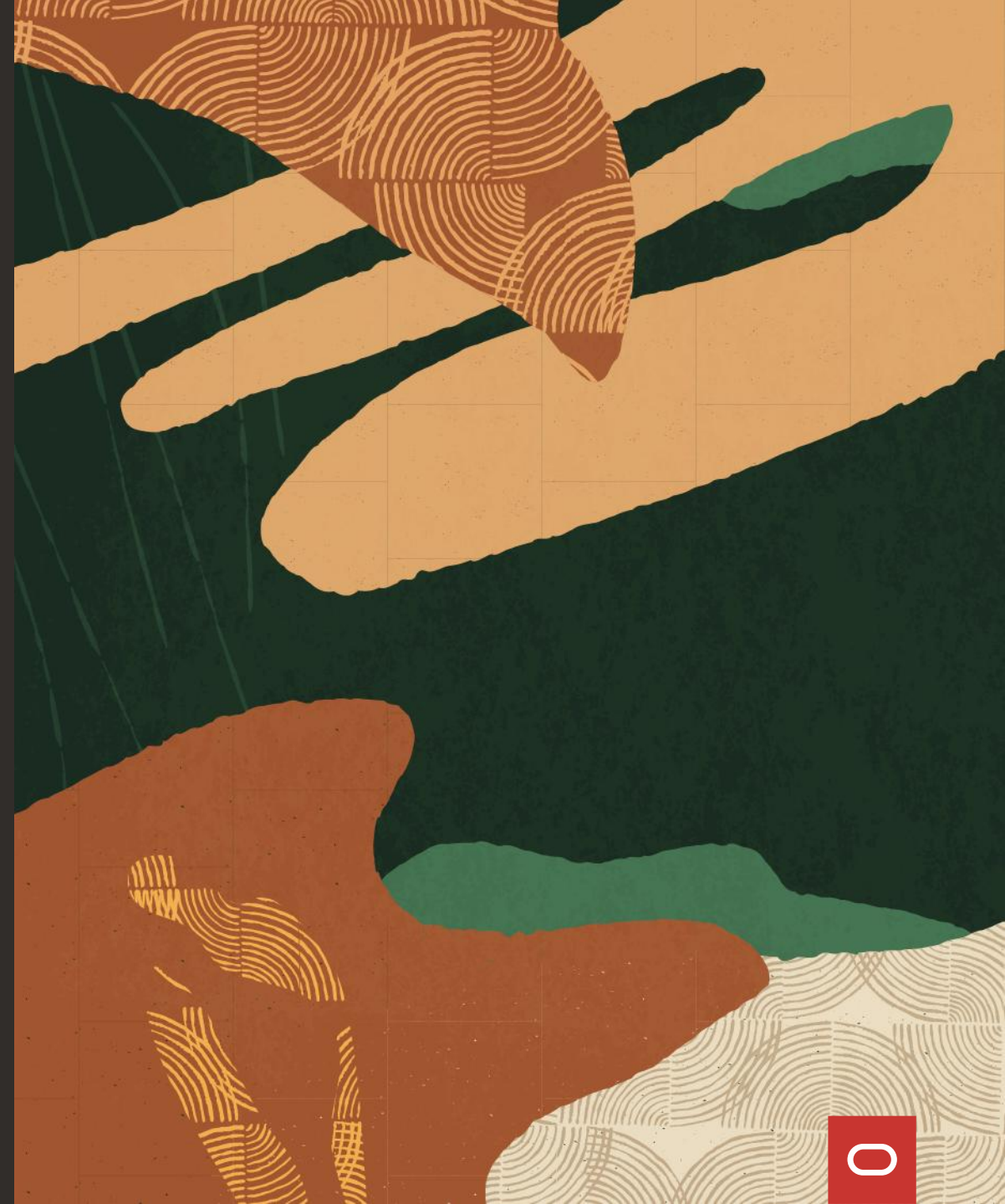


Oracle Database@Azure

Technical Deck

OCI Multicloud



Safe Harbor Statement

The following is intended to outline our general product direction. It is intended for information purposes only and may not be incorporated into any contract. It is not a commitment to deliver any material, code, or functionality, and should not be relied upon in making purchasing decisions. The development, release, timing, and pricing of any features or functionality described for Oracle's products may change and remains at the sole discretion of Oracle Corporation.

Statements in this presentation relating to Oracle's future plans, expectations, beliefs, intentions and prospects are "forward-looking statements" and are subject to material risks and uncertainties. A detailed discussion of these factors and other risks that affect our business is contained in Oracle's Securities and Exchange Commission (SEC) filings, including our most recent reports on Form 10-K and Form 10-Q under the heading "Risk Factors." These filings are available on the SEC's website or on Oracle's website at <http://www.oracle.com/investor>.



Agenda

01

Service Overview

02

Exadata Database Service (Exa-D)

03

Autonomous Database (ADB-S)

04

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05

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06

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08

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Roadmap

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Automation

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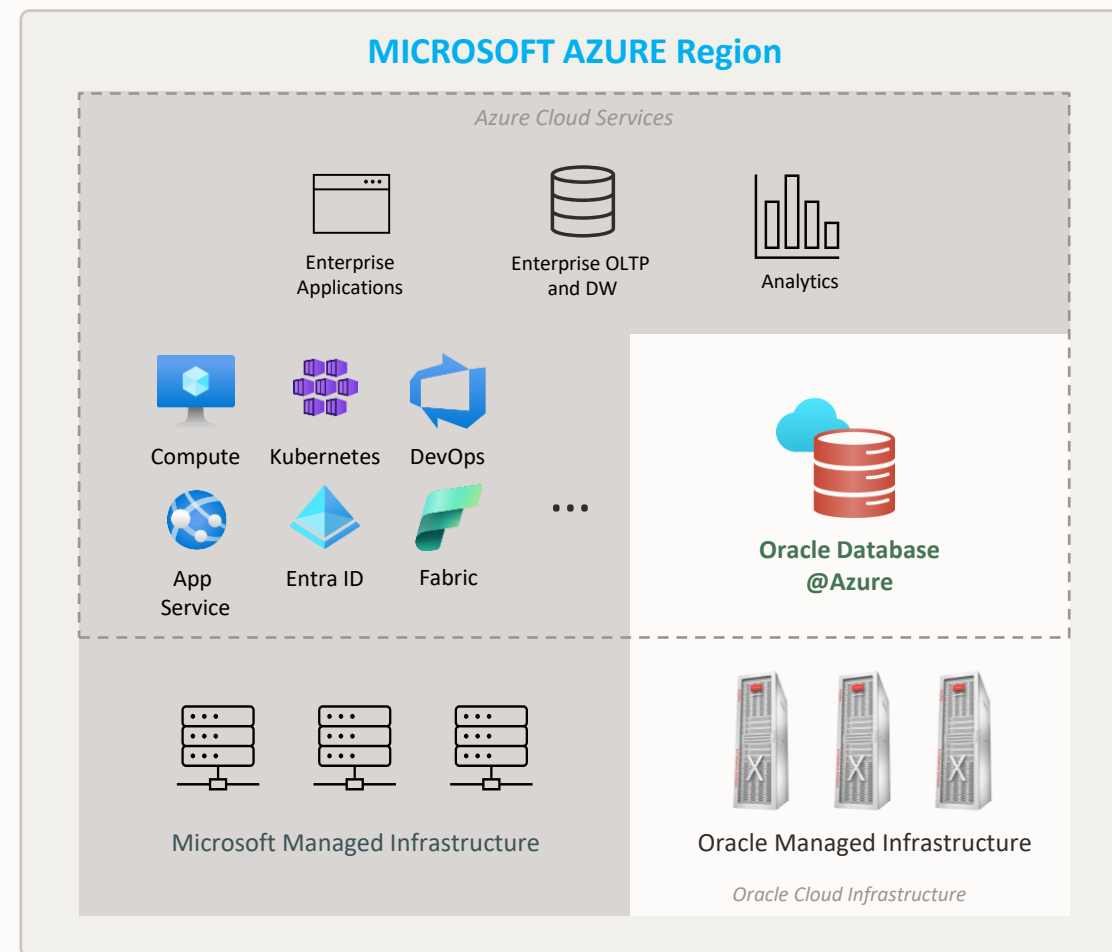
Support

Service Overview

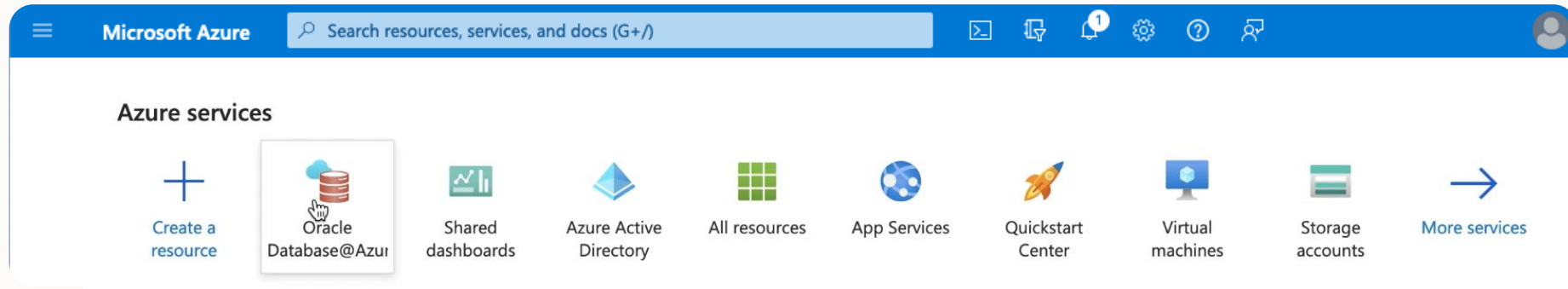
Oracle Database@Azure – Multicloud done right

Oracle database services on OCI in Microsoft Azure datacenters enable you to:

- Migrate and modernize Oracle databases in the cloud
- Innovate with OCI and Azure services
- Simplify purchasing and operations



Oracle Database@Azure offerings



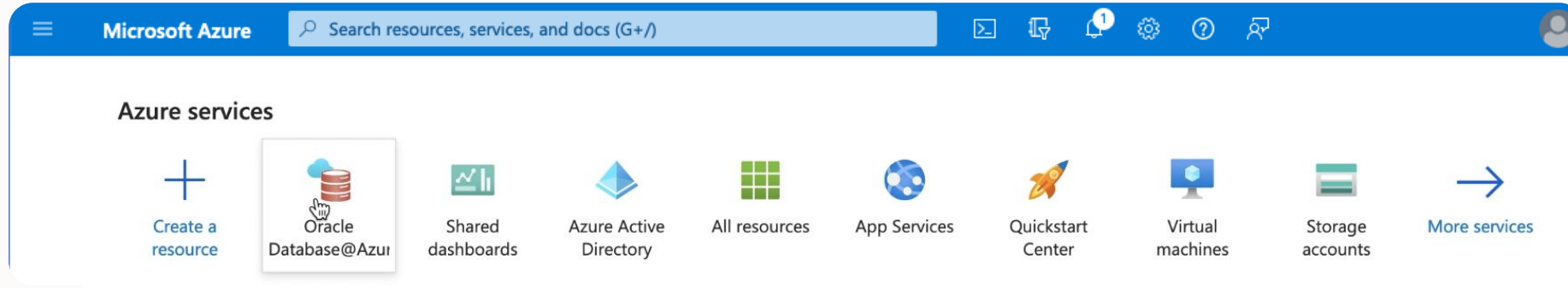
Exadata Database Service

- Dedicated, Oracle-managed infrastructure for most challenging workloads and performance/isolation
- Scale-out design, database-optimized compute, networking, storage
- Full control of database management and control plane
- Algorithm-driven OLTP, analytics, and consolidation

Autonomous Database

- Built on Oracle Exadata infrastructure
- Fully managed and automated provisioning, patching, upgrades, hardware refreshes with enterprise-class business continuity
- Elastic resources and pay-for-use billing
- Built-in low-code APEX, ELT, AI/ML, graph and spatial analytics

Oracle Database@Azure offerings



Zero Dataloss Autonomous Recovery Service

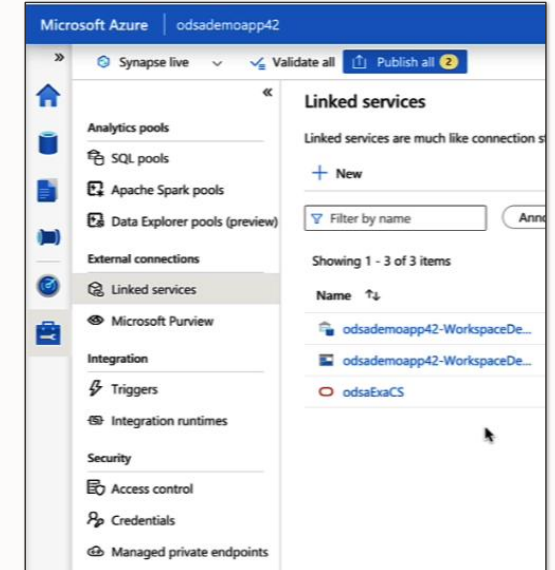
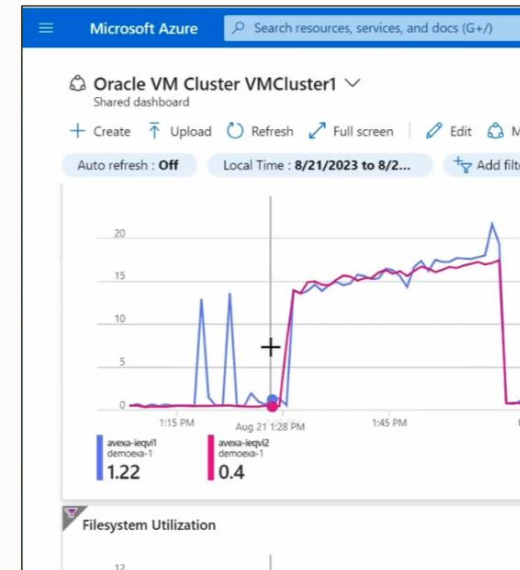
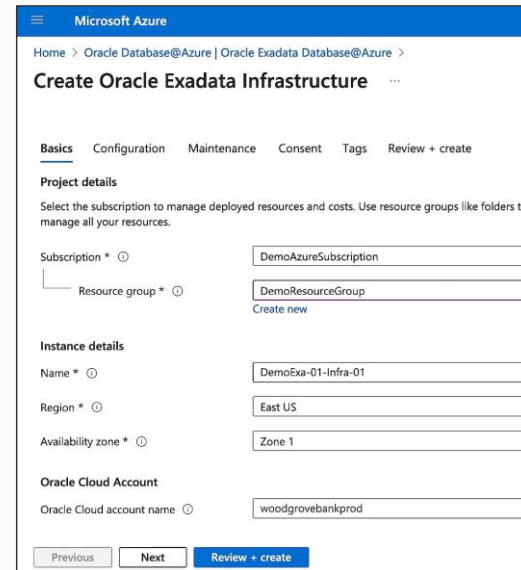
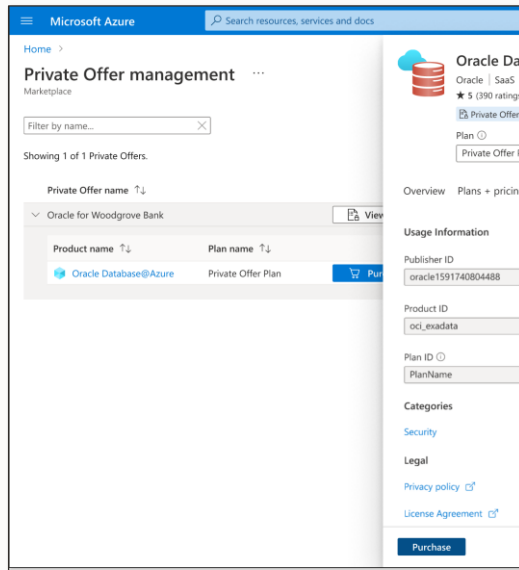
- Oracle Database Zero Data Loss Autonomous Recovery Service is a fully managed data protection service for Oracle databases running on Oracle Database@Azure Exadata Database Service
- Backup automation and enhanced data protection capabilities to offload all backup processing and storage requirements to Oracle Database Autonomous Recovery Service,
- Eliminating backup infrastructure costs and manual administration overhead. control plane

The best of Oracle and Microsoft, all in one place

Purchase in Azure Marketplace and parity with OCI

Deploy, manage, and monitor through Azure Portal and APIs

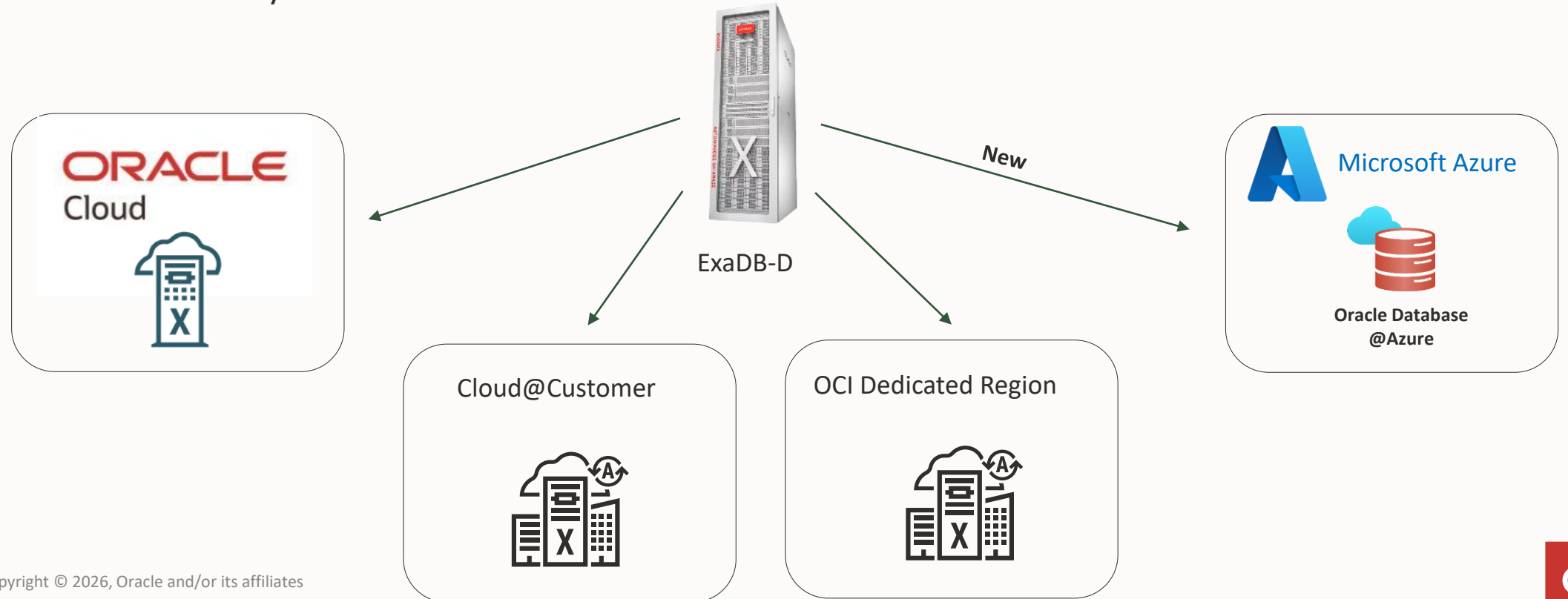
Combine with your choice of Azure services



Exadata Database Service on Dedicated Infrastructure (ExaDB-D)

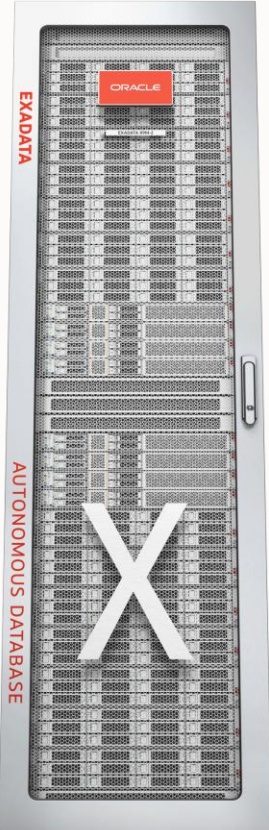
Deployment models

- ExaDB-D that is available in Oracle Database@Azure is the same that is offered in other deployment models (OCI, Cloud@Customer and Dedicated Region)
- This new solution is optimized for those customer who need the performance, availability and resiliency of an ExaDB-D but want to continue with application tier in Azure and need a very short network latency to database



Exadata Vision

Extreme Performance and Availability, Lowest Cost, Available Everywhere



Ideal Database Hardware

Scale-out, database optimized compute, networking, and storage

Database Aware System Software

Unique algorithms vastly improve OLTP, Analytics, and Consolidation

Automated Management

Fully automated and optimized end-to-end

Available

Oracle Cloud

Oracle
Cloud@Customer

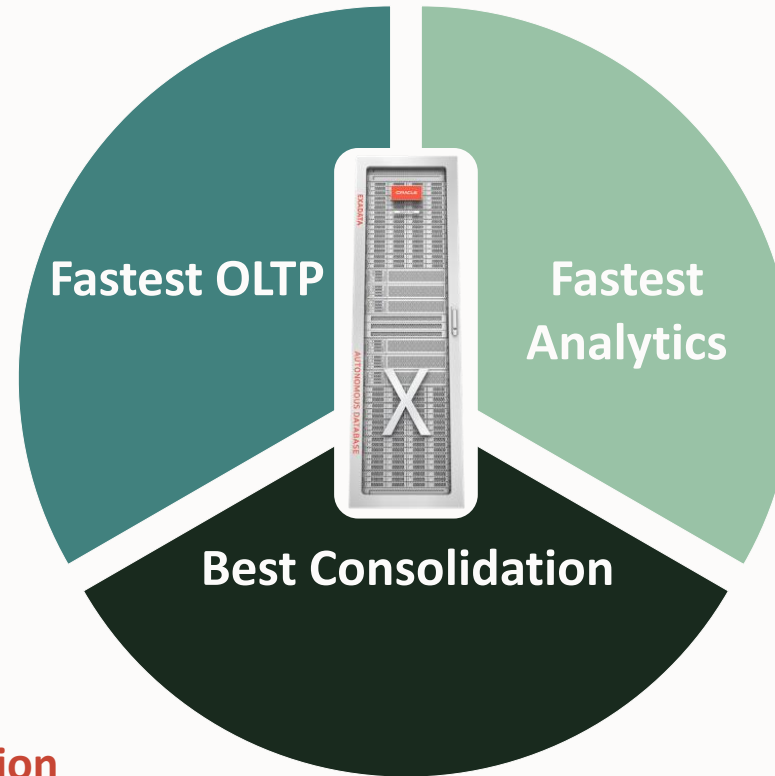
Oracle
Database@Azure

On Premises

Exadata Smart System Software: OLTP | Analytics | Consolidation

Fastest OLTP

- **Fastest OLTP I/O** with scale-out storage, RDMA, shared memory in storage servers, and NVMe flash
- **Fastest scale-out** with unique RDMA algorithms for inter-node cluster coordination
- **Fastest recovery** from unplanned downtime and component failure



Fastest Analytics

- **Unique Smart Scan** automatically offloads data intensive SQL operations to storage
- **Unique Smart Flash Cache** and **Storage Index** automatically accelerate database I/O
- **Unique Columnarization** automatically converts data to fast In-Memory Columnar format in flash

Best Consolidation

- **Unique Prioritization** of latency sensitive or important workloads
- **Unique Workload Isolation** of multiple tenants or workloads

Exadata Database Service on Dedicated Infrastructure

All the Power and Functionality of Oracle Database and Exadata in the Public Cloud

Enterprise-Grade Oracle Database as a Service

- Same architecture as on-premises Exadata
 - Runs Oracle Real Applications Clusters
- Dedicated Exadata platform for performance/isolation
- Includes all Oracle Database options or Bring Your Own License
- 100% compatible with existing applications that use Oracle Database

Simplicity, Elasticity, Flexibility

- Infrastructure managed by Oracle experts
- Root access to customer VM
- Customer controlled cloud automation
- Online, elastic resource scaling
- Pay-per-use database cores



Exceptional Outcomes

- Performance
- Availability
- Security
- Consolidation



Elastic Scaling

- Add CPU and Storage Online
- No DB/Workload too Large



Highest Value

- Best Price/Performance
- Pay-per-use Pricing
- One DB Service for all workloads



Exadata X9M Infrastructure in Public Cloud

Fully-elastic compute and storage provisioning and expansion

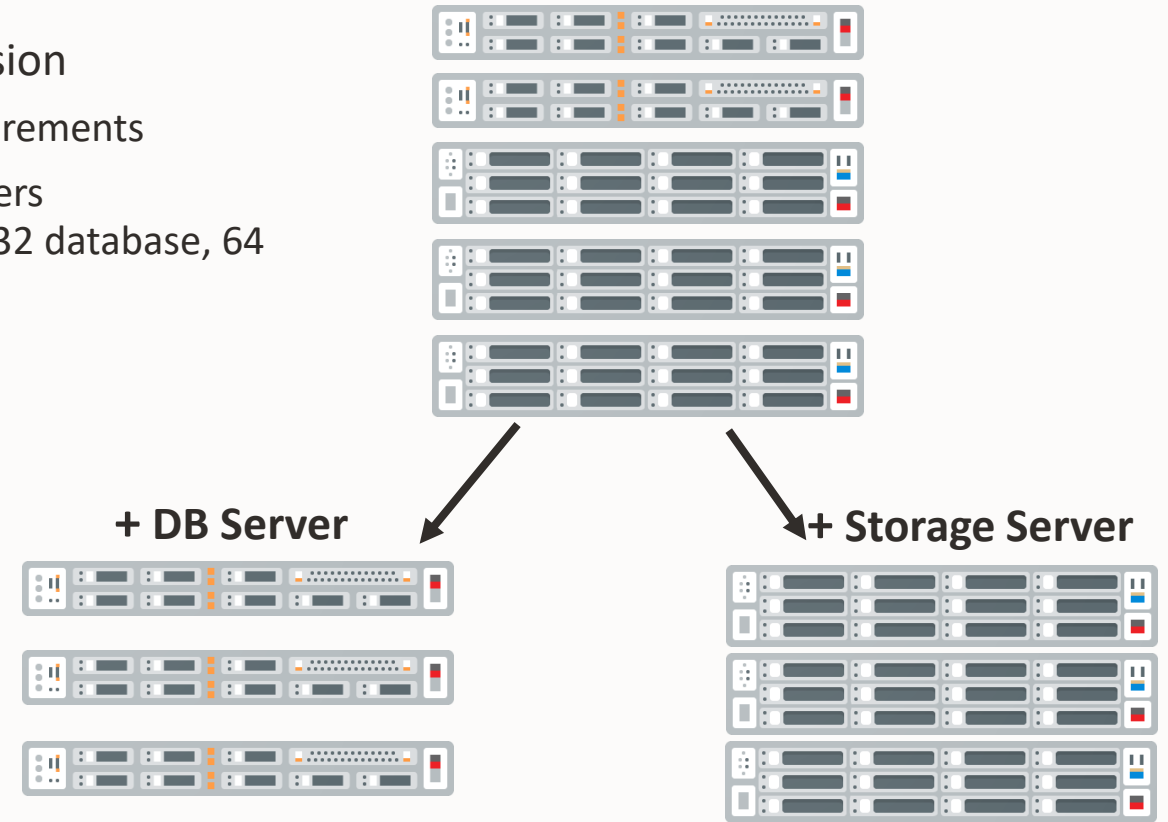
- Properly size hardware configurations to meet workload requirements
- Minimum configuration of 2 Database Servers, 3 Storage Servers
 - Add as little as one database or one storage server up to 32 database, 64 storage servers
- Dedicated to you – no noisy or malicious neighbors

Scale-Out X9M Database Servers

- 126 Maximum OCPUs
- 1,390 GB Memory
- 50 Gbps Network (Shared for Client/Backup)

Scale-Out Intelligent X9M Storage Servers

- 48 processor cores
- 1.5 TB PMem
- 25.6 TB NVMe Flash
- 63.6 TB HDD Usable Storage



Add Database or Storage Servers as needed



Exadata Database Service on Dedicated Infrastructure Architecture

Exadata Components in the Cloud

- Scale-out database and storage servers connected by secure fabric
- Database clustering with Real Applications Clusters
- Fully redundant hardware and software for high availability

Cloud Controls and Networking

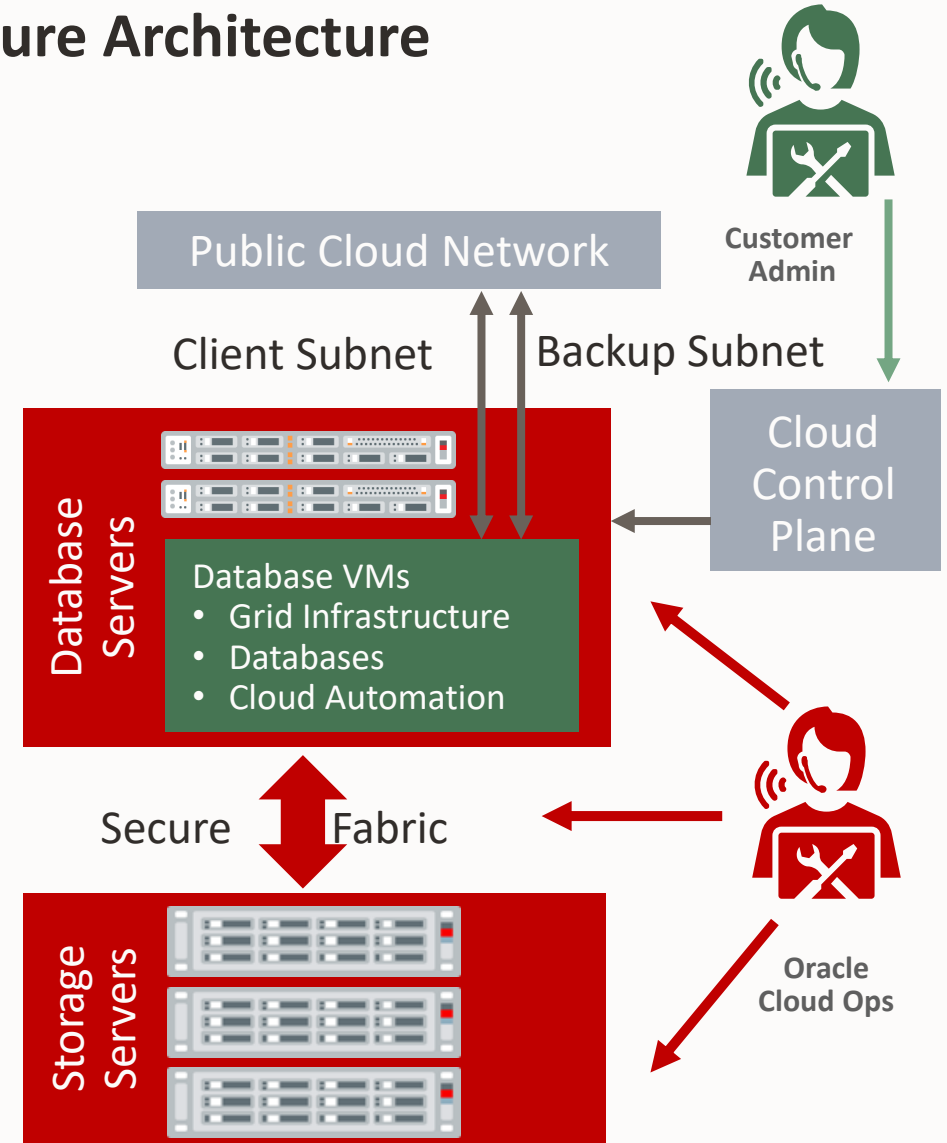
- Identity management and policies to control access
- Software defined networks with isolation

Secure Isolation

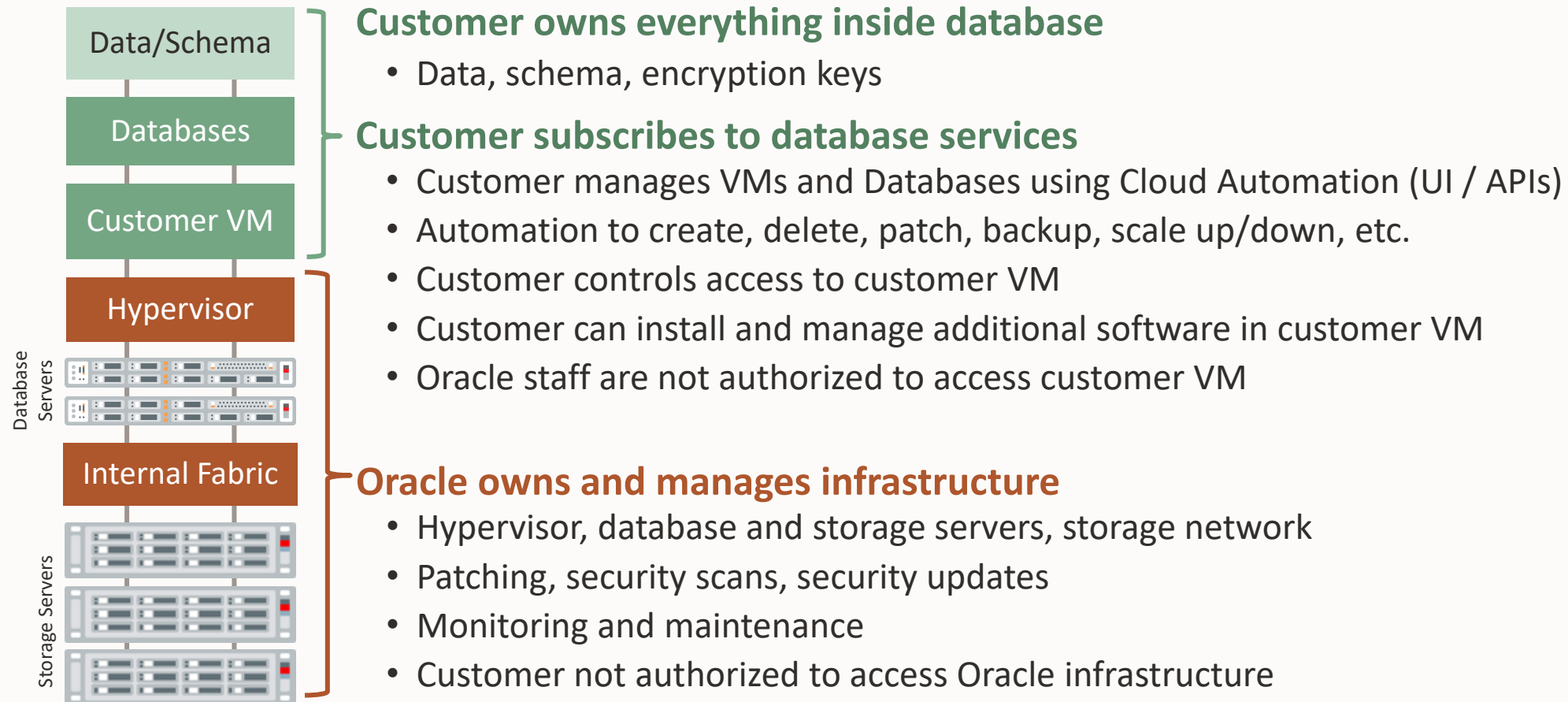
- Databases run in secure virtual machines isolated from underlying infrastructure

Control Plane

- Controls and manages Exadata infrastructure



Simple Cloud Management Model in Public Cloud



Benefits of Oracle Database@Azure with ExaDB-D

1

Run your applications with the best platform for Oracle databases in Azure

2

Achieve fastest performance and elastic scalability with Exadata

3

Attain maximum availability uniquely featuring Oracle RAC in Azure

4

Consolidate on Exadata to Reduce Costs

5

Migrate to cloud with no disruption and zero downtime



Oracle Database@Azure Autonomous Database - Serverless

Oracle Autonomous Database is complete and simple

Fully-managed, cloud native, Database as a Service

Oracle Data Management Vision:

Make it **simple** to **develop** and **run** modern operational and analytic apps for **all** use cases at **any** scale



**Autonomous
Database**

Autonomous Database delivers the complete Converged Database

- Best price, best performance
- Most cost-effective for all use cases at any scale
- Simplest for operations, developers, and analysts

Oracle Autonomous Database delivers the best data experience

With the lowest total cost of ownership



**Mission-Critical
Database**



**High Performance
Engineered System**



**Automation,
Best Practices
and Data Tools**



**Autonomous
Database**

Oracle Autonomous Database is complete and simple

Fully-managed, cloud native, Database as a Service



High cost and risk to run and maintain

Cost to upgrade and secure

Cost to run and optimize

Cost to manage, integrate, and connect data



**Autonomous
Database**

Spend less, improve performance, mitigate risk



Auto-patching, built-in security and compliance

Auto-scale and enterprise business continuity

Built-in data tools and open data sharing

Autonomous Database automatically secures, tunes, and scales



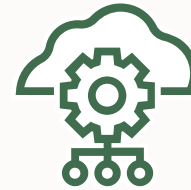
**Automated
patching and
security**



**Automated
scaling**



**Automated
backups**



**Automated
provisioning**



**Automated
configuration**



**Automated
tuning**

✓ Eliminates manual
tasks

✓ Eliminates human
error

✓ Eliminates
downtime

✓ Eliminates scaling
complexity

✓ Eliminates
performance
tuning complexity



Simplification of day-to-day operational tasks

Automated security



- Automatically and frequently applies the latest and greatest security patches
- Automatically encrypts all data at rest, in motion, and when backed up to protect from bad actors or malware
- Automatically protects against unauthorized user access to sensitive data as well as unauthorized database changes
- Offers built-in Database Vault for meeting industry, regulatory, and corporate security standards

Automated scaling



- Automatically scale for changing workloads
- Zero downtime with no impact to performance
- Lower costs—pay-for-use model
- Shut off compute for idle systems, restart instantly
- Schedule automatic shutdown-startup

Automated backups



- No manual backups required
- Fully-managed, automated backups
- Full backup every 60 days along with weekly cumulative and daily incremental
- Restore or recover a database to any point-in-time within the last 60 days

Simplification of day-to-day operational tasks

Auto-provisioning



- Deploys mission-critical databases which are fault-tolerant and highly available
- Automatically scales compute resources when needed and enables true pay-per-use
- Option to select patch level when provisioning ADB
- View maintenance and patch history

Auto-configuration



- Automatically configures the database to optimize for data warehouse workloads
- Automatic partitioning and automatic indexing simplify and automate design decisions - removes guess-work from data management
- Predefined maintenance windows for automatically applying patches

Automated tuning

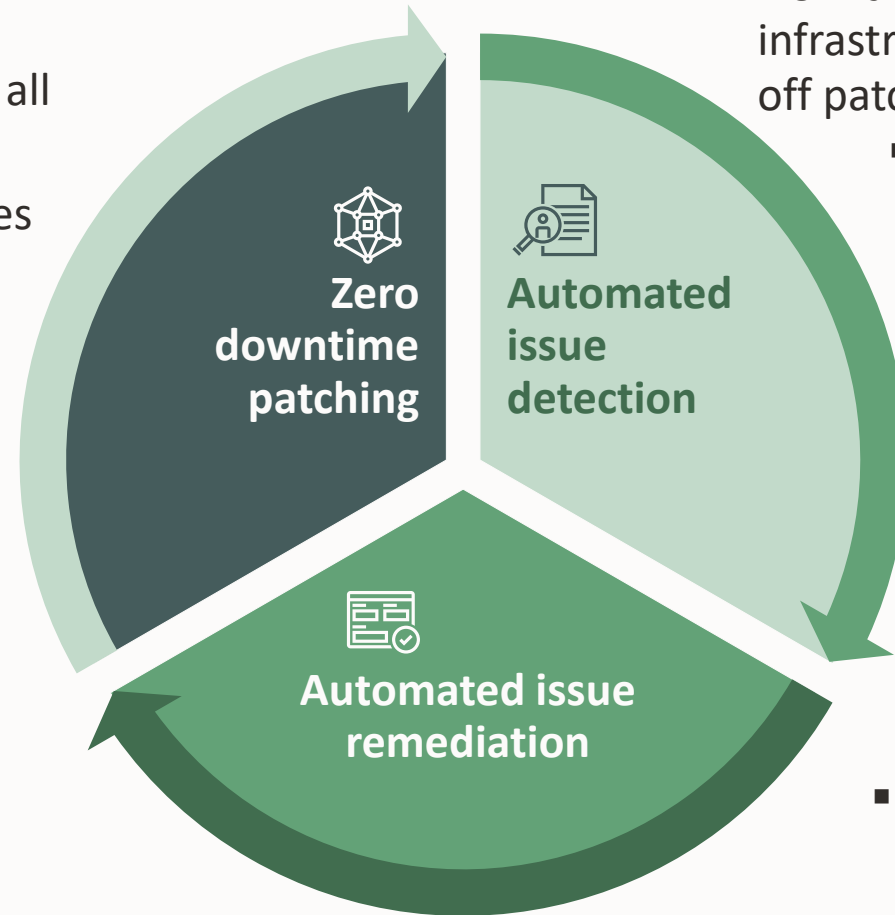


- ADB captures, identifies, verifies and monitors SQL statement execution
- Automatically optimizes SQL statements for better performance
- Use Performance Hub to view time range for performance data (ie Active Session History (ASH) Analytics details across different dimensions

Automated security and bug fixes - Auto-patching

Reduce costs and risks of manual patching

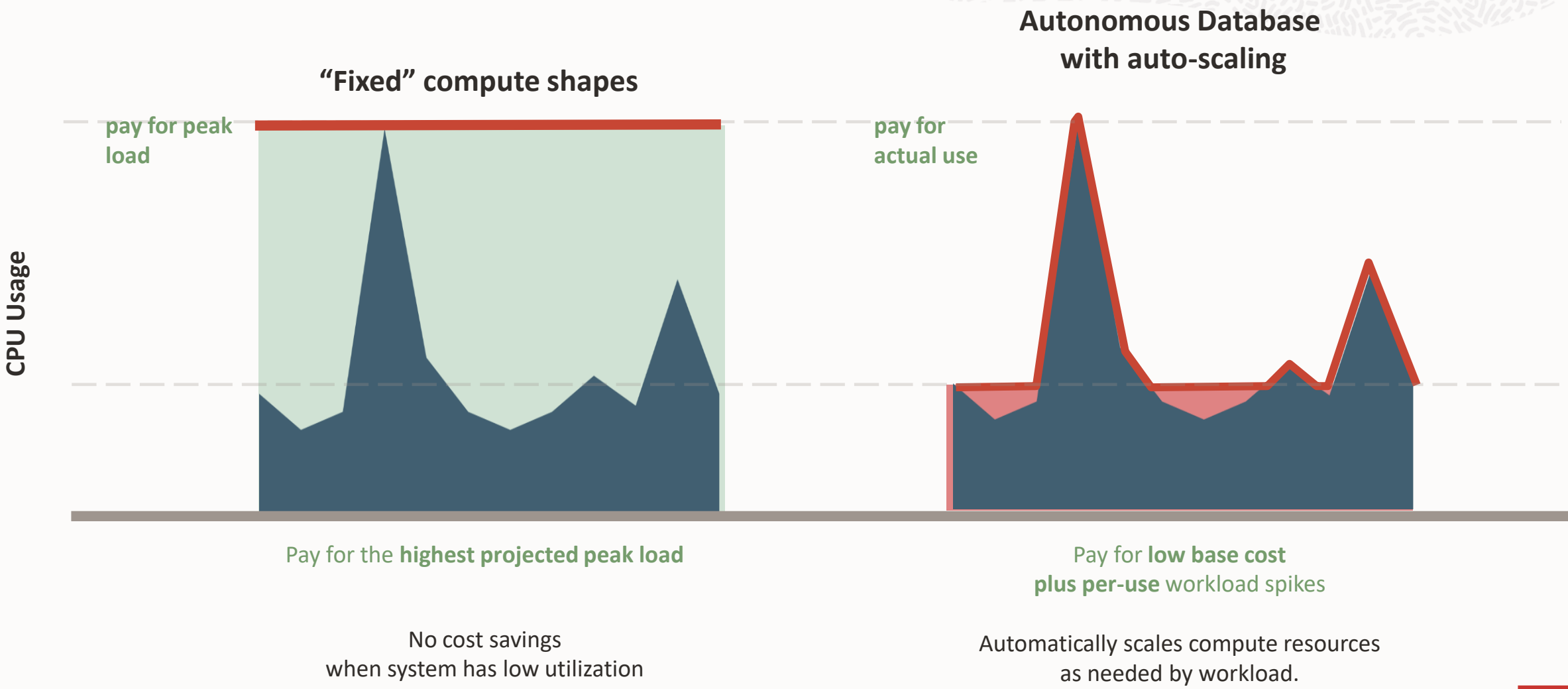
- **Zero downtime and no interruption of service** – all systems continue to run
- Continuous testing of fixes through CI/CD pipeline



- **No manual patch management** for infrastructure, release update or one-off patches
 - Continuously monitors the entire infrastructure for security issues and bugs
 - Automatically generates Service Requests for each issue
- **88% of issues are automatically resolved** with zero customer actions
- Immediate investigation and resolution by Cloud Ops
- Root-cause analysis for every issue

Lower costs with elastic resources and pay-for-use billing

Comparing fixed compute shapes to auto-scaling



Lower costs with flexible licensing

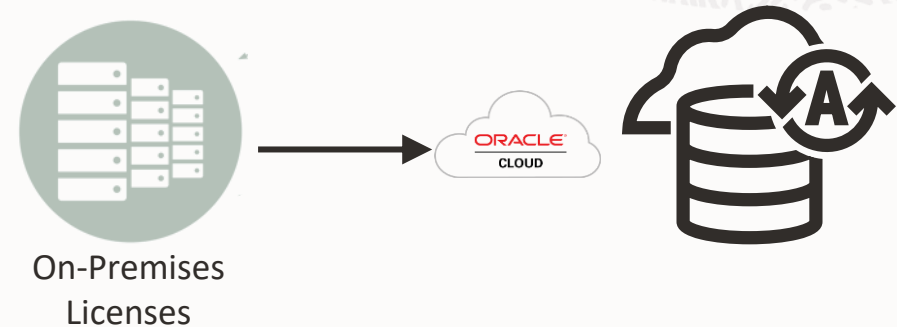
Up to **75% cost savings**



Everything is built in

1. License Included

- Ideal for organizations seeking to build new applications or expand functionality
- Includes Oracle Database, analytics, operations and management packs at one low price
- **Pay-per-use pricing, all paid using Universal Credits**



2. Bring Your Own License

- **Ideal for organizations moving existing workloads**
- Utilize **existing on-premises licenses** and benefit from 75% cost savings
- **Oracle Support Rewards** further reduces OCI costs
 - Cloud consumption provides credits toward technology support
- Includes Transparent Data Encryption, Data Safe, Partitioning, Advanced Compression, and select management packs at **no additional cost**



Higher reliability and availability through automation

Highly-available, fully automated infrastructure

- Automatic recovery for compute node failures
- Triple-mirrored disks for disk failures

Online patching for complete protection

- Frequent software and security patching cycle
- Patches applied with zero database downtime

Automatic backups

- Fully automated daily backups to Cloud Object Storage
- Point in time recovery to anytime in last 60 days



*"I'd rather build new features and functionality than worry about our business operating. Oracle Autonomous Database provides autoscaling, self-tuning, self-repairing, and self-encryption so **we have zero downtime to securely process mission-critical financial transactions** while we focus on delivering new capabilities to our customers at the same time."*

Ray Grove, Vice President, Product Management,
Thomson Reuters

Automation removes time spent on operational tasks

DBA's now focus on innovation and increasing business value

Tasks and responsibilities to **generate business value**

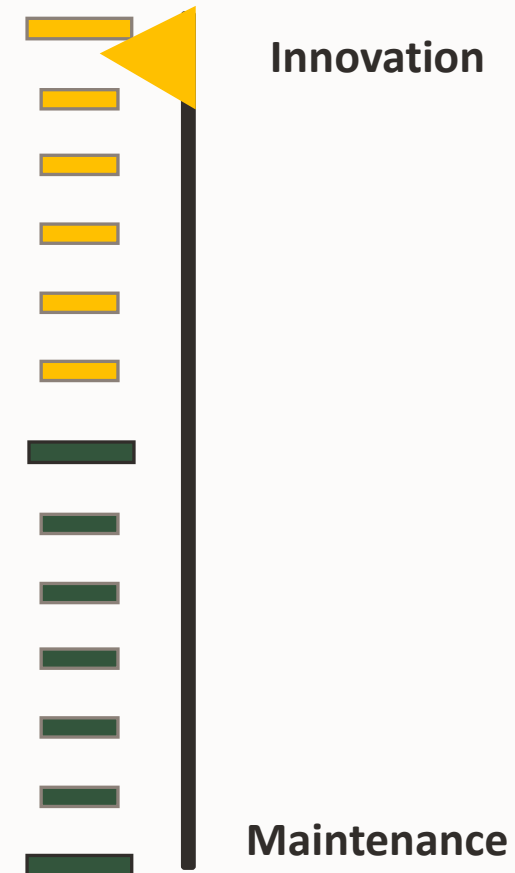
- Architecture, planning, data modeling
- Data security and lifecycle management
- Application related tuning
- End-to-end service level management

Tasks and responsibilities for ~~maintenance and administration~~

- ~~• Configuration and tuning of systems, network, storage~~
- ~~• Database provisioning, patching~~
- ~~• Database backups, H/A, disaster recovery~~
- ~~• Internal helpdesk and call center~~

Welcome to the future: Autonomous Database

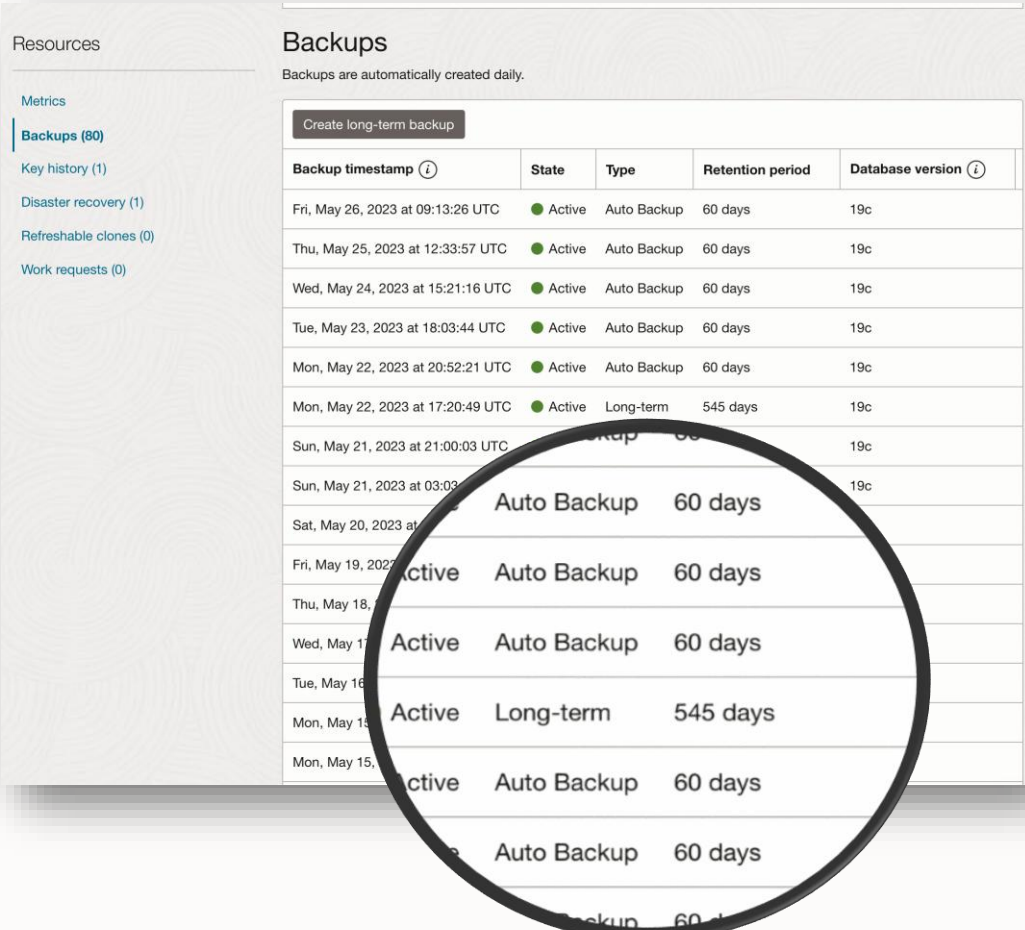
Value Scale



Minimize risk with **automated backups**

Top reasons why automated backups beat customer-managed backups

- 1 No customer action required**
Fully automated daily backup
Configurable retention from 0 to 60 days
Point in time recovery
- 2 Automated backup testing**
Validation ensures future data restoration
- 3 Optional long-term backup retention**
Up to 10 years for regulatory compliance
- 4 Create a new database from backup**
Deploy to any cloud region
Restore to any point in time



Resources

Metrics

Backups (80)

Key history (1)

Disaster recovery (1)

Refreshable clones (0)

Work requests (0)

Backups

Backups are automatically created daily.

Create long-term backup

Backup timestamp ⓘ	State	Type	Retention period	Database version ⓘ
Fri, May 26, 2023 at 09:13:26 UTC	Active	Auto Backup	60 days	19c
Thu, May 25, 2023 at 12:33:57 UTC	Active	Auto Backup	60 days	19c
Wed, May 24, 2023 at 15:21:16 UTC	Active	Auto Backup	60 days	19c
Tue, May 23, 2023 at 18:03:44 UTC	Active	Auto Backup	60 days	19c
Mon, May 22, 2023 at 20:52:21 UTC	Active	Auto Backup	60 days	19c
Mon, May 22, 2023 at 17:20:49 UTC	Active	Long-term	545 days	19c
Sun, May 21, 2023 at 21:00:03 UTC	Active	Auto Backup	60 days	19c
Sun, May 21, 2023 at 03:03:03 UTC	Active	Auto Backup	60 days	19c
Sat, May 20, 2023 at 03:03:03 UTC	Active	Auto Backup	60 days	19c
Fri, May 19, 2023 at 03:03:03 UTC	Active	Auto Backup	60 days	19c
Thu, May 18, 2023 at 03:03:03 UTC	Active	Auto Backup	60 days	19c
Wed, May 17, 2023 at 03:03:03 UTC	Active	Auto Backup	60 days	19c
Tue, May 16, 2023 at 03:03:03 UTC	Active	Auto Backup	60 days	19c
Mon, May 15, 2023 at 03:03:03 UTC	Active	Long-term	545 days	19c
Mon, May 15, 2023 at 03:03:03 UTC	Active	Auto Backup	60 days	19c
Sun, May 14, 2023 at 03:03:03 UTC	Active	Auto Backup	60 days	19c

Oracle Database@Azure BaseDB

Oracle Base Database Service

Standard and Enterprise Database Services

Full-featured Oracle Database cloud instances

- Deploy the features required from Standard Edition 2, Enterprise Edition, and Oracle Database options
- Run single-instance or enhance availability with 2-node RAC

Efficient database lifecycle management

- Eliminate infrastructure management
- Increase administrator productivity with user-controlled cloud automation
 - Web and API-driven lifecycle operations
 - Provisioning, updates, backups, etc.

Cloud economics

- User initiated rolling OCPU scaling to align costs with allocated compute resources
- Choice of 4-tiers of Oracle Database License-included options or Bring Your Own License (BYOL)

Base Database Service



Traditional

Oracle Database service
in VMs on shared compute

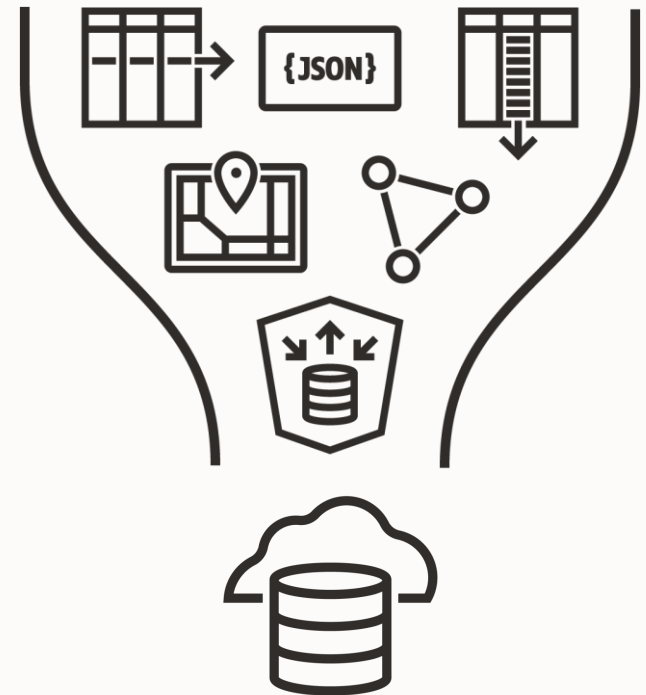
Co-managed

Full administrative control

Supports all DB features incl. RAC

Oracle Database on Base Database Service

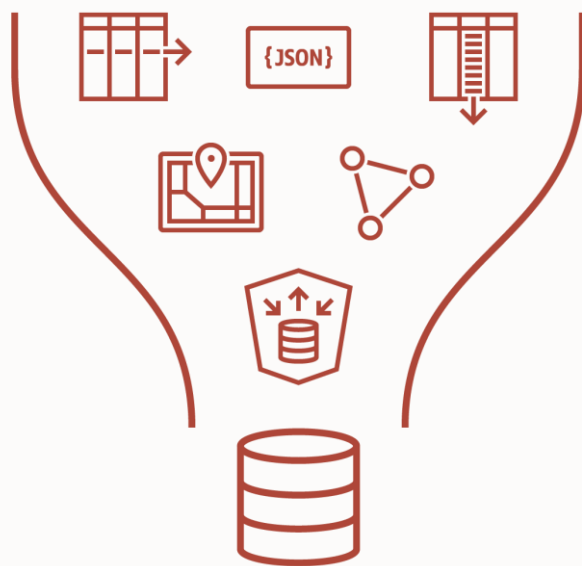
- Any data and workload type
- Application compatibility with on-premises and other cloud deployments
- More productive development
 - Standardized APIs for rapid development
 - Reusability of code, libraries and frameworks
- Maximum availability with RAC and Data Guard
- Defense in-depth security from database to cloud
- Simple database migration with zero downtime
- Consistent pricing model with predictable spend
- Lower TCO



Oracle Database | World's Best Converged Database Engine

Enterprise-class performance, scalability, reliability & security for all data-driven workloads

Any Data



Converged, open, SQL database for multiple data types and data models

Any Workload



Transparent scale-up, scale-out, sharding, in-memory, parallel SQL

Most Productive



Developers & Architects, Data Analysts/Scientists and DBAs



License-included Oracle Database Consumption Options

Functionality tiers to meet application-specific requirements

Standard Database Service

Standard Edition 2 Database features plus

- Tablespace Encryption
- Machine Learning
- Spatial and Graph
- Up to 3 PDBs

Enterprise Database Service

Enterprise Edition Database features plus

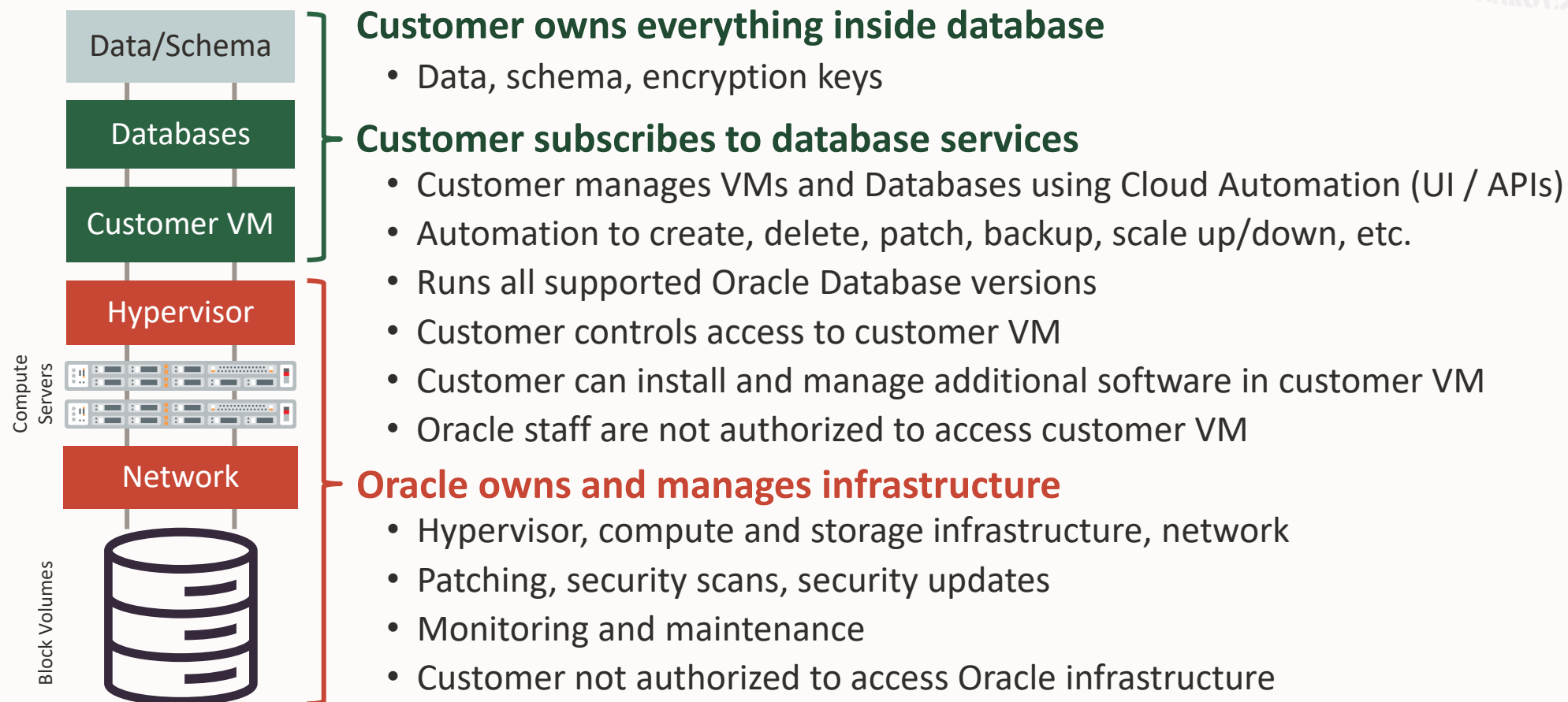
- Data Guard
- Real Application Testing
- Data Masking and Subsetting Pack
- Tuning Pack and Diagnostic Packs
- True Cache

Enterprise Database Service high performance

Enterprise Database Service features plus

- Partitioning
- Advanced Compression
- Advanced and Label Security, DB Vault
- Multitenant
- OLAP
- Lifecycle and Cloud Management Packs
- SQL Firewall

Simple Cloud Management Model in Public Cloud



Deploy, Scale and Manage Databases in Oracle Cloud



oracle.com/database/base-database-service

1

Oracle Database as a Service

- Same database software in Oracle Cloud as on-premises
- Full-stack redundancy and integrated disaster recovery
- Full database control with defense in-depth security
- Simple database migration with zero downtime

2

Simple and Rapid Development

- Single converged database for any data and workload type
- Rapid application development with Oracle APEX and other included tools

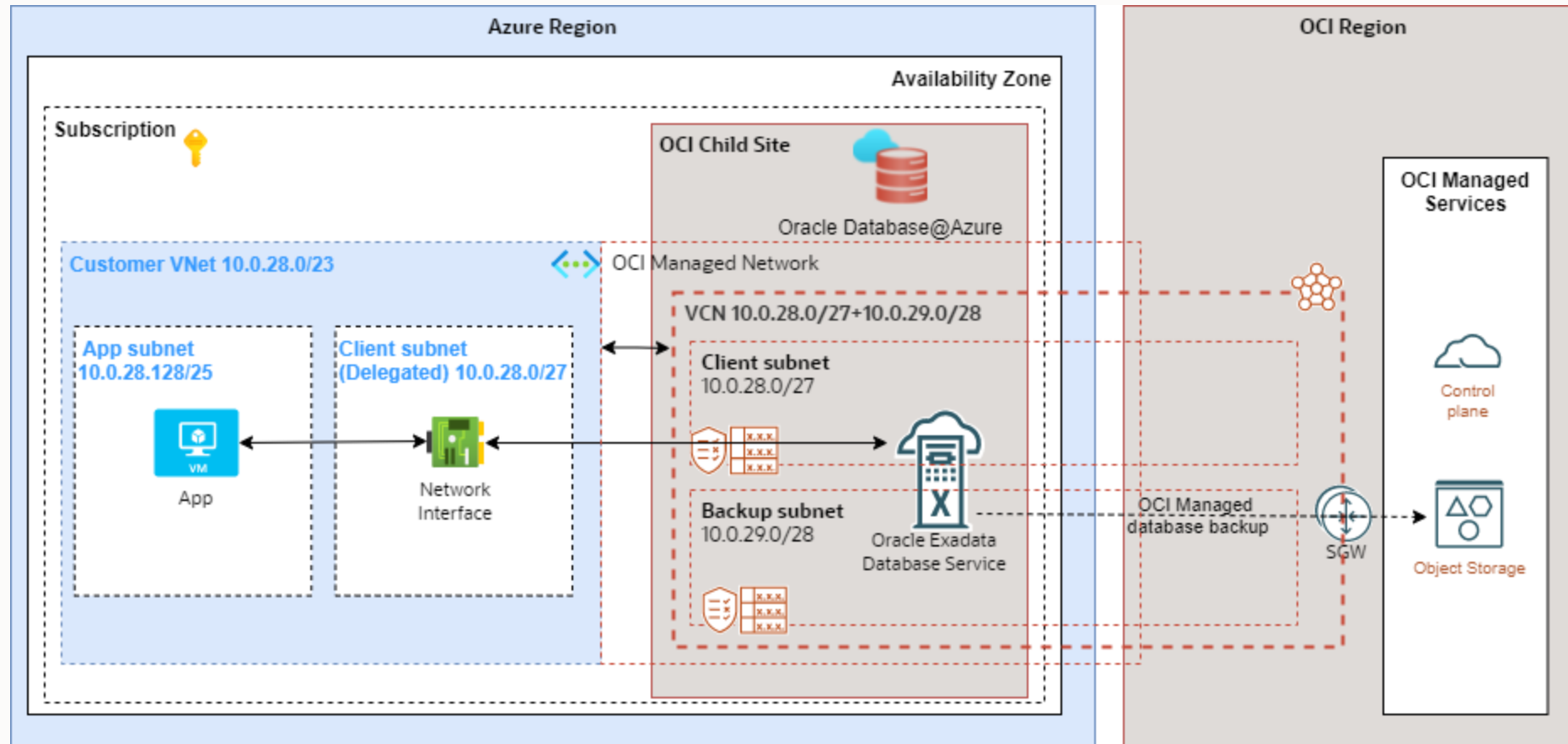
3

Low Cost

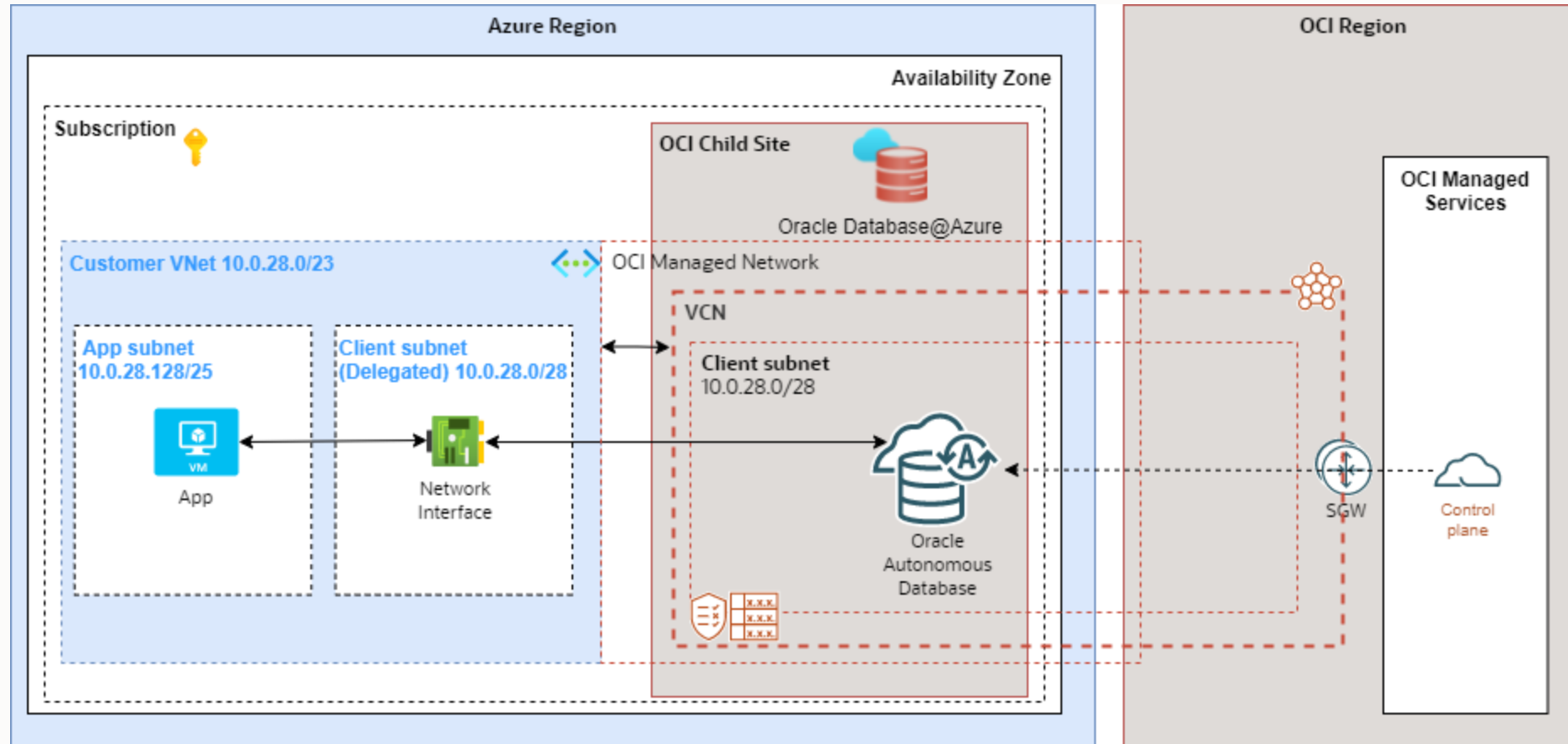
- Flexible shapes with low entry price and pay only for what you use
- License-included or Bring Your Own License (BYOL)
- Cloud automation for database lifecycle management

Architecture

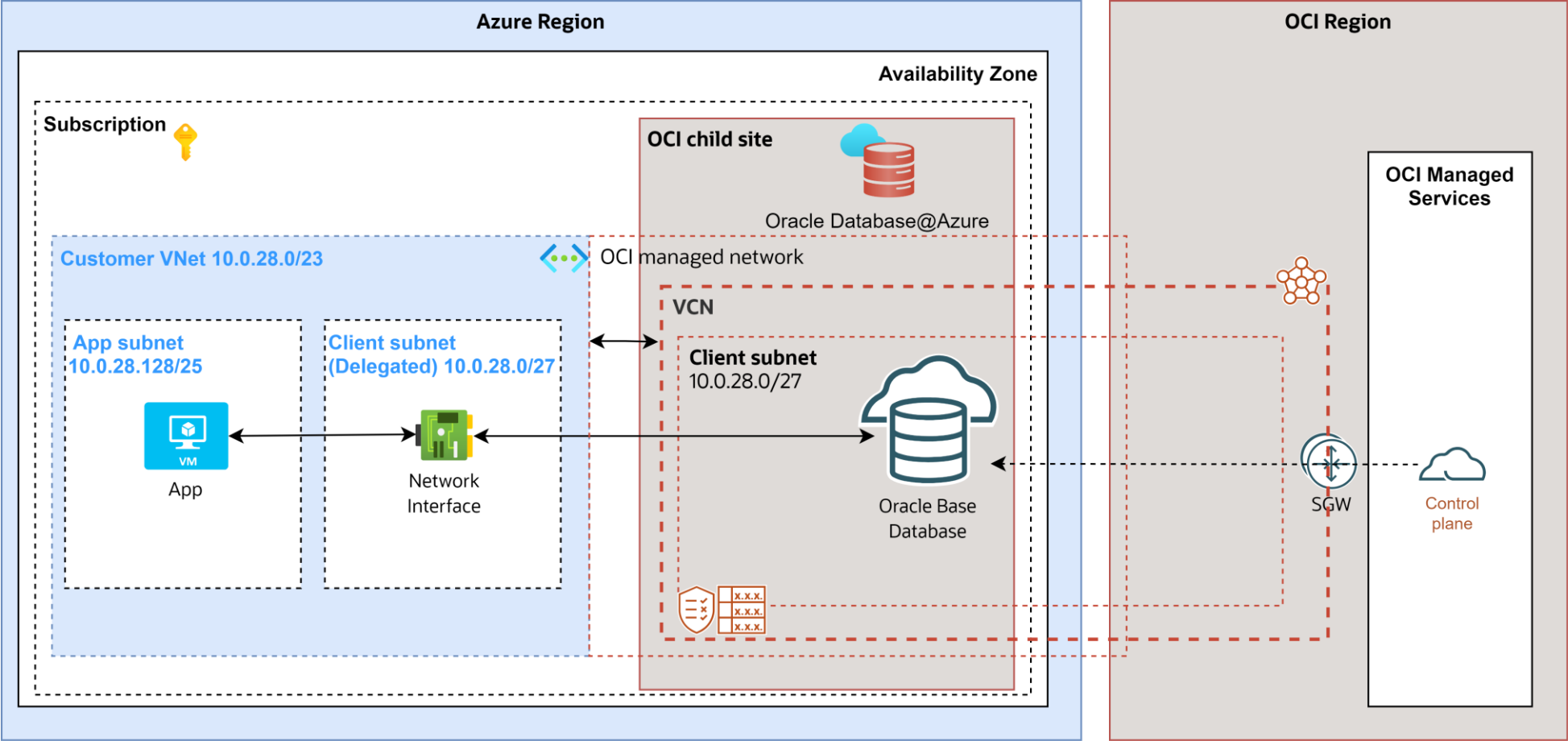
Exadata Database Service – Single AZ Architecture



Autonomous Database – Serverless – Single AZ Architecture

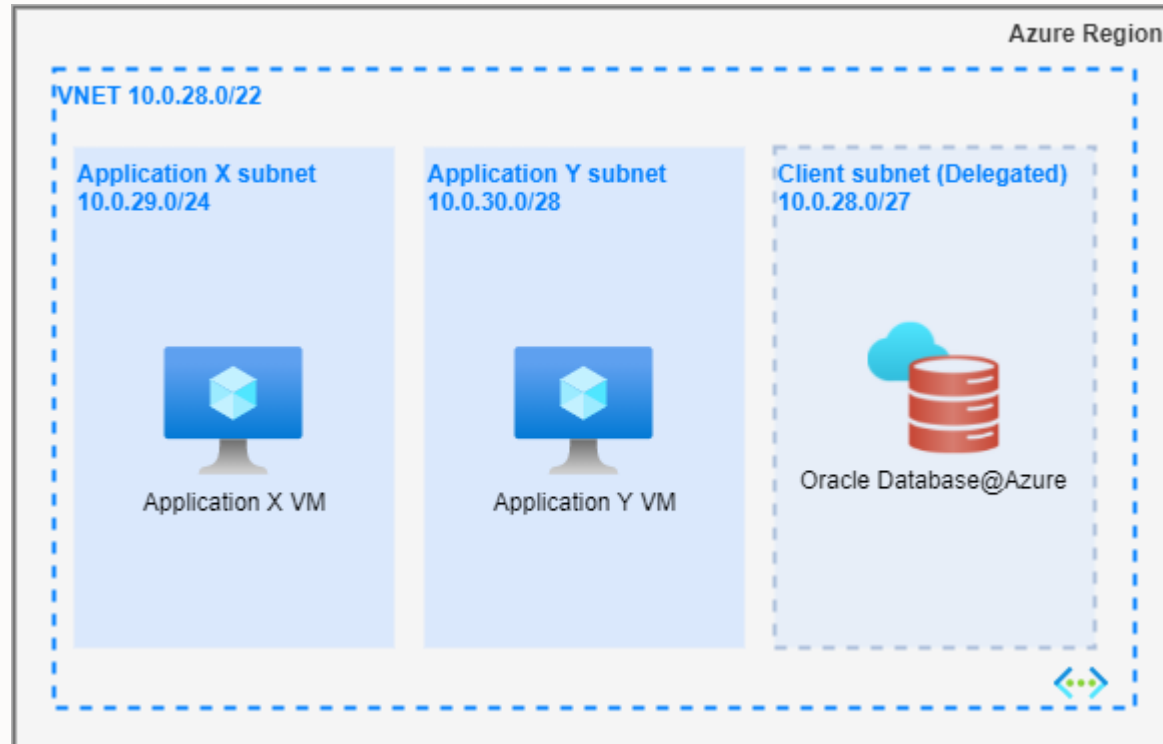


Base Database – Single AZ Architecture



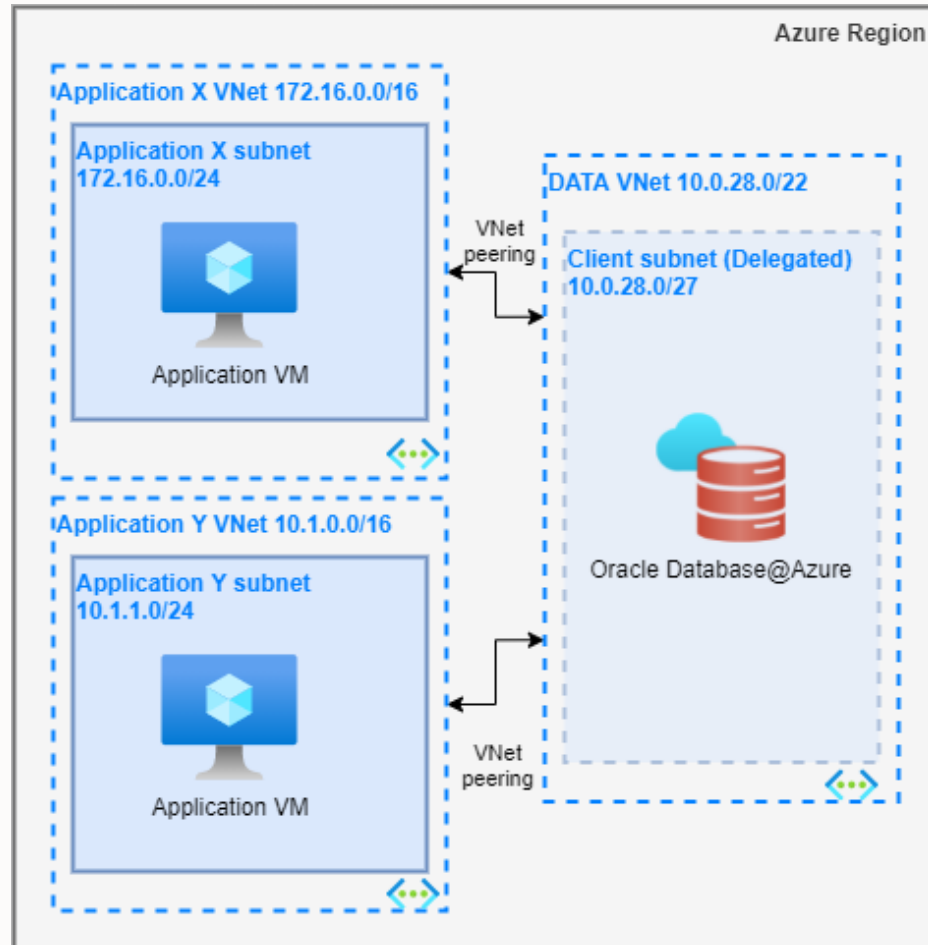
Network Topology

Local VNet Topology



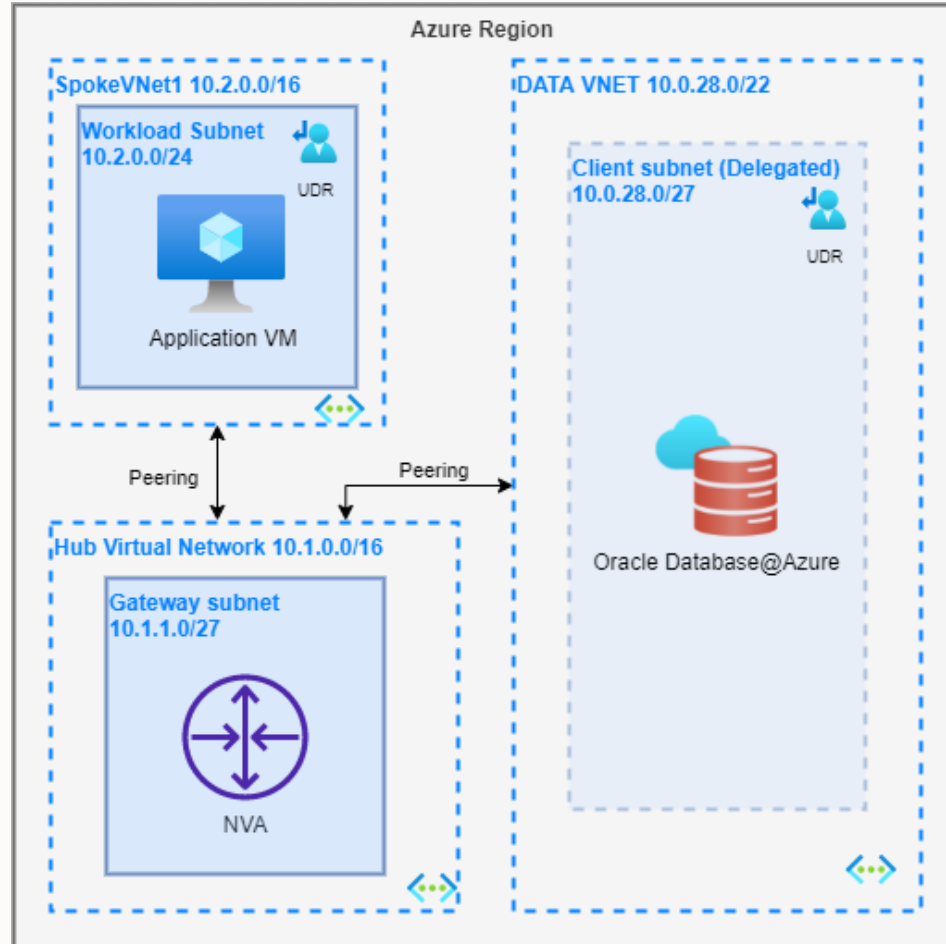
- Simple networking topology connects apps to the database in the same VNet
- App connects to database via client delegated subnet
- Lowest network latency
- No ingress/egress costs

VNet Peering Topology



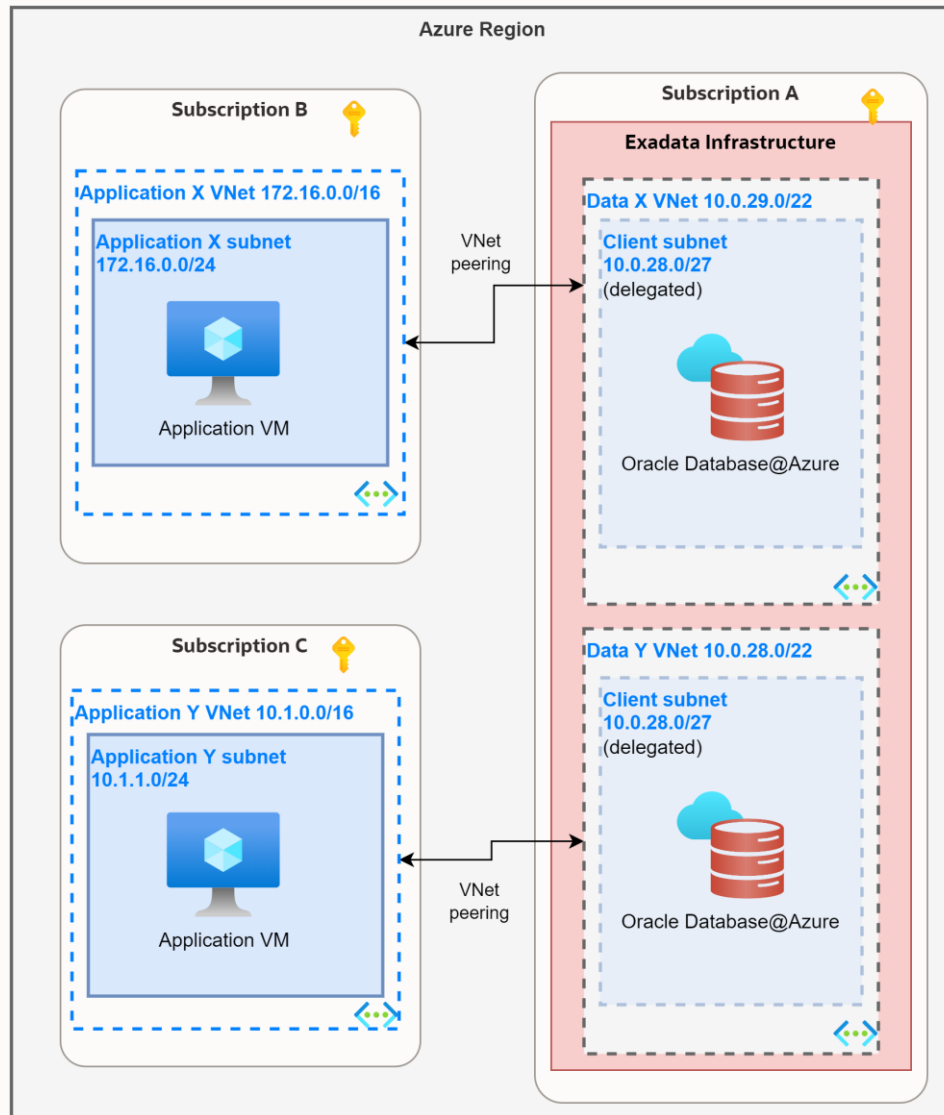
- Use local VNet Peering to connect apps in other VNet of the same region to database VNet.
- VNet Peering incurs ingress and egress costs. For more information, see Virtual Network pricing.
- Lowest network latency
- Up to 500 direct peerings possible

Hub-spoke VNet Peering Topology



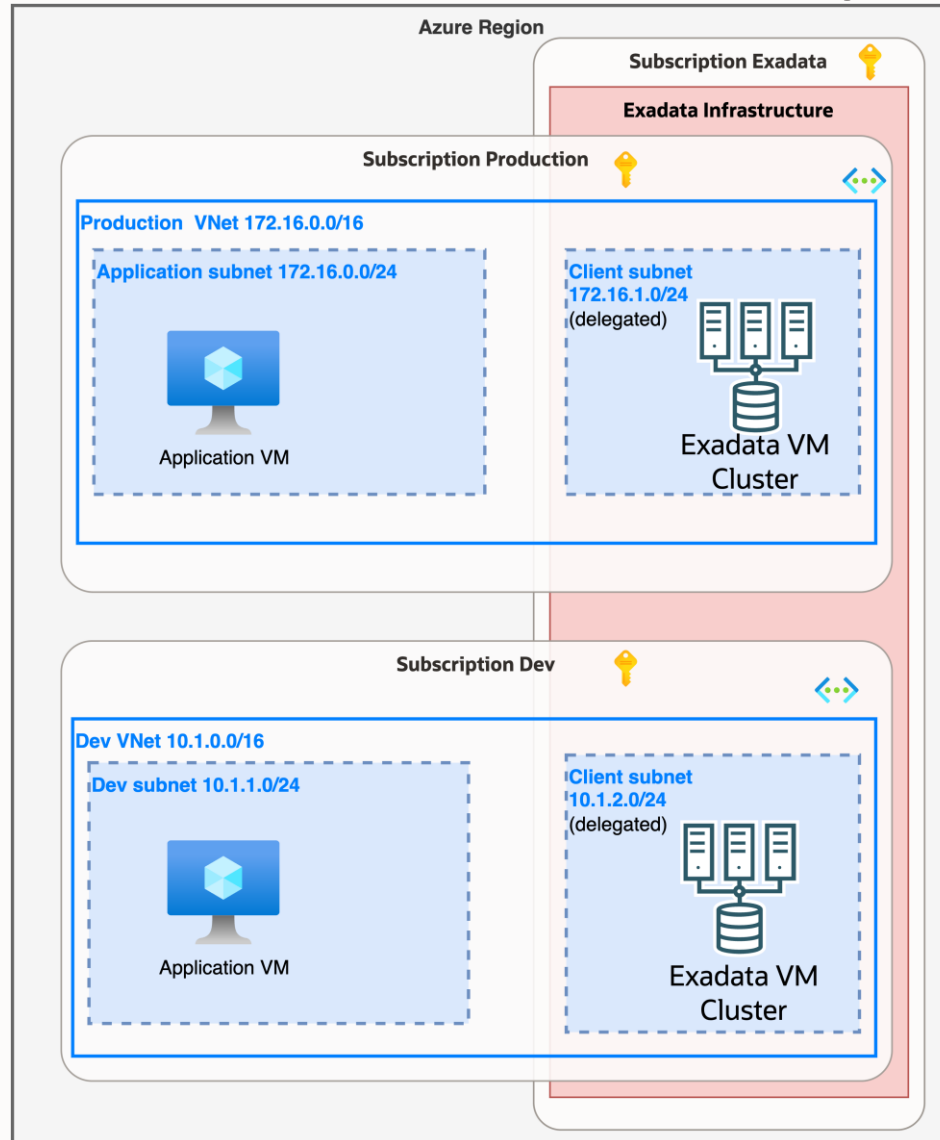
- The hub VNet is a central point of connectivity between apps and database;
- The spoke VNets peer with the hub.
- A local routing device (NVA) is prerequisite for routing traffic from the delegated subnet to hub-and-spoke
- Higher latency

Centralized DB subscription - VNet Peering Topology



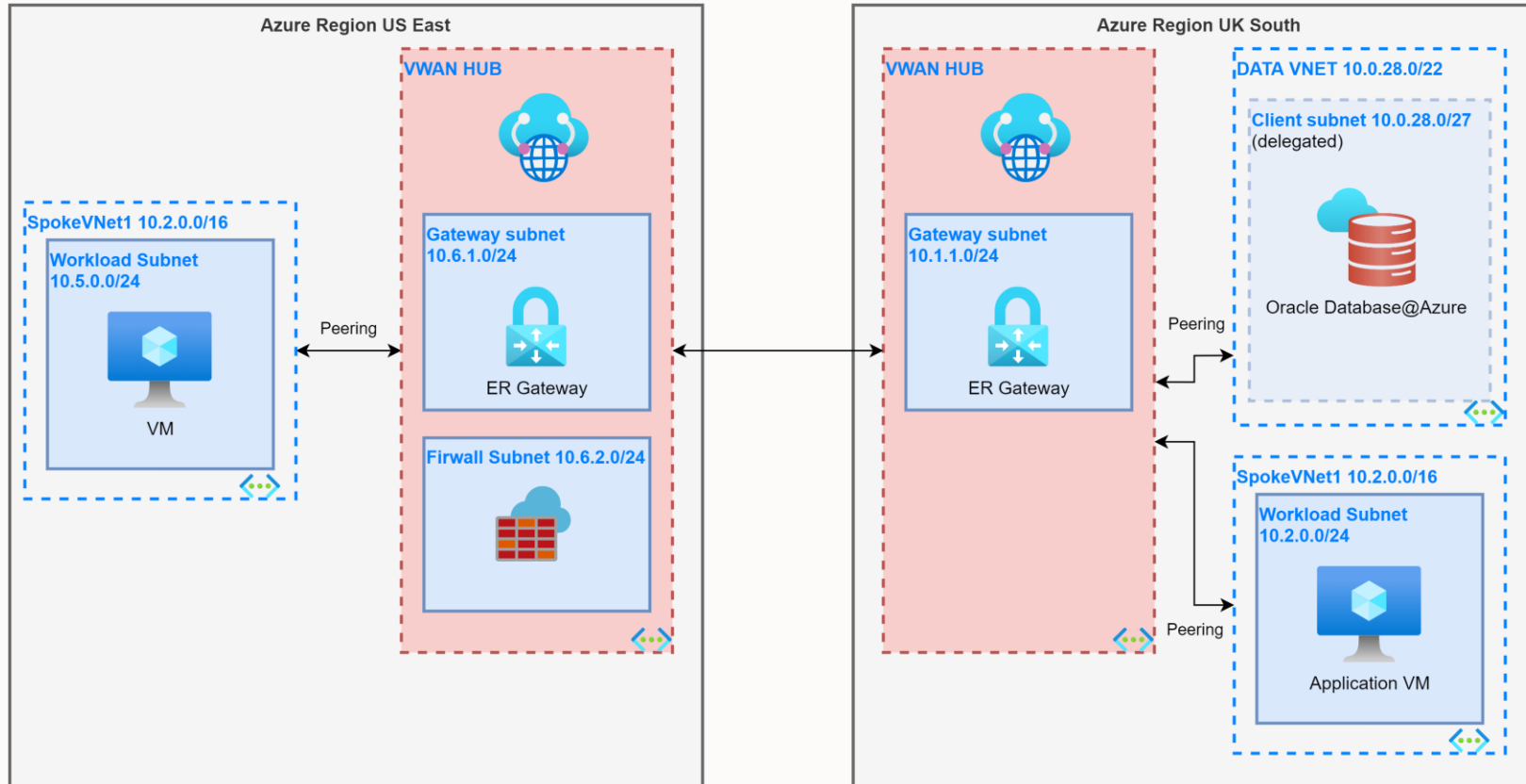
- Databases are consolidated within the same subscription for management.
- Use VNet Peering to connect apps in other VNet of the same region but different subscription to database Vnet
- VNet Peering incurs ingress and egress costs. For more information, see Virtual Network pricing.
- Lowest network latency

Share Exadata Infrastructure – Multi Subscription topology



- Exadata Infrastructure is consolidated within an Azure subscription for management.
- Each VM Cluster is deployed in a different Azure subscription and Vnet
- VM Cluster are isolated
- Lowest network latency

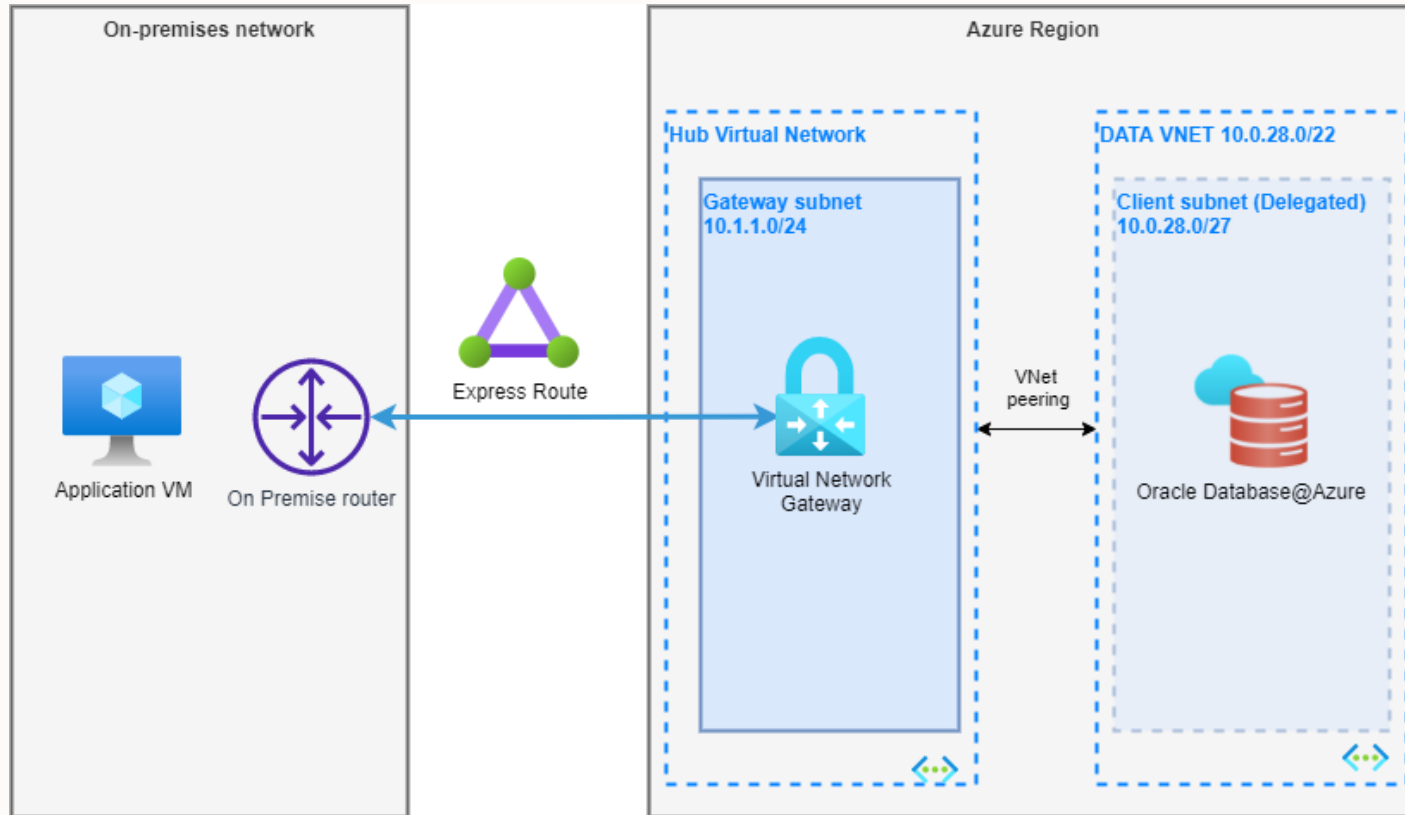
Global connectivity between regions



- VWAN Hub is configured in both regions to create global connectivity
- Firewall is configured in the Hub to inspect traffic
- Routing is managed by VWAN Hub



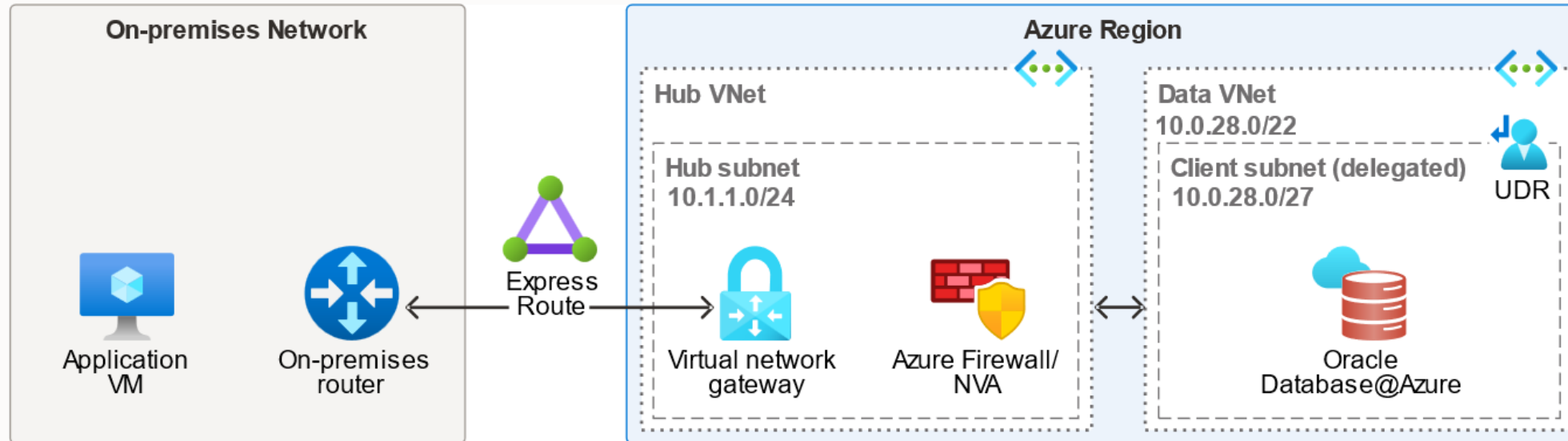
Hub-spoke On-premises without traffic inspection



- No firewall configured in the Hub
- Routes are learned through transitivity in the Gateway
- Database spoke learns routes through peering.
- VNet Peering incurs ingress and egress costs. For more information, see [Virtual Network pricing](#).

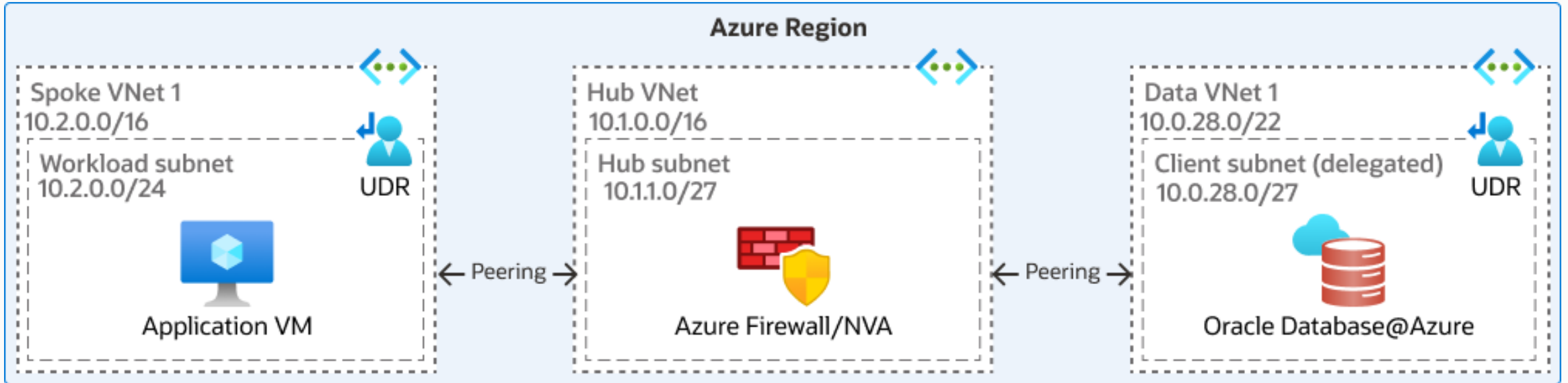


Hub-spoke On-premises with traffic inspection



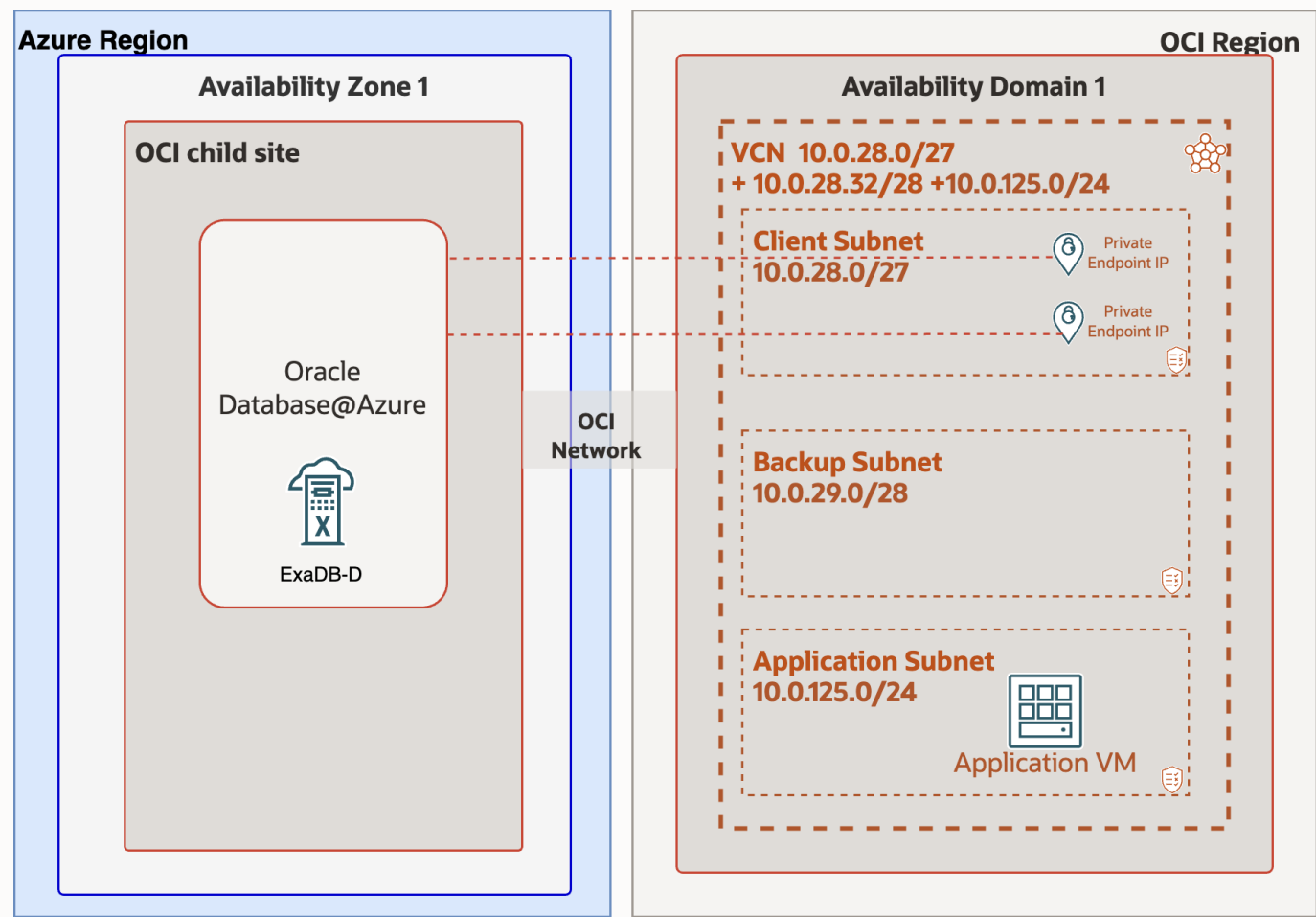
- Firewall configured in the Hub
- Routes are learned through transitivity in the Gateway
- Database spoke learns routes through peering.
- Routes are defined on each subnet to route to the Firewall
- VNet Peering incurs ingress and egress costs. For more information, see [Virtual Network pricing](#).

Hub-spoke with traffic inspection



- Firewall configured in the Hub
- Database spoke learns routes through peering.
- Routes are defined on each subnet to route to the Firewall
- VNet Peering incurs ingress and egress costs. For more information, see [Virtual Network pricing](#).

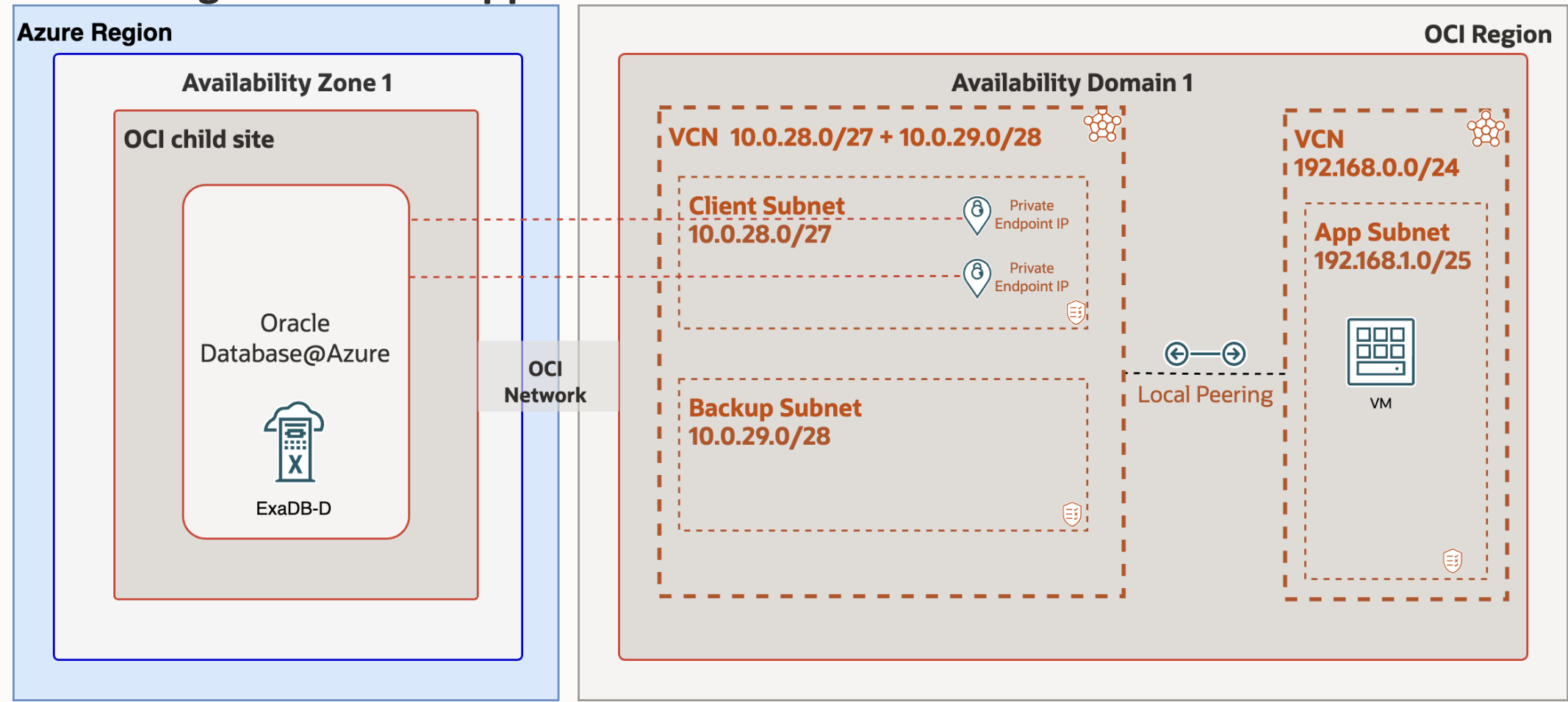
Connecting Services or application from OCI in Same VCN



OCI customers can natively connect existing resources and services to Oracle Database and Exadata resources by the private endpoint by creating an additional subnet in the shadow VCN and deploy the application in the new Subnet.



Connecting Services or Application from OCI in Different VCNs

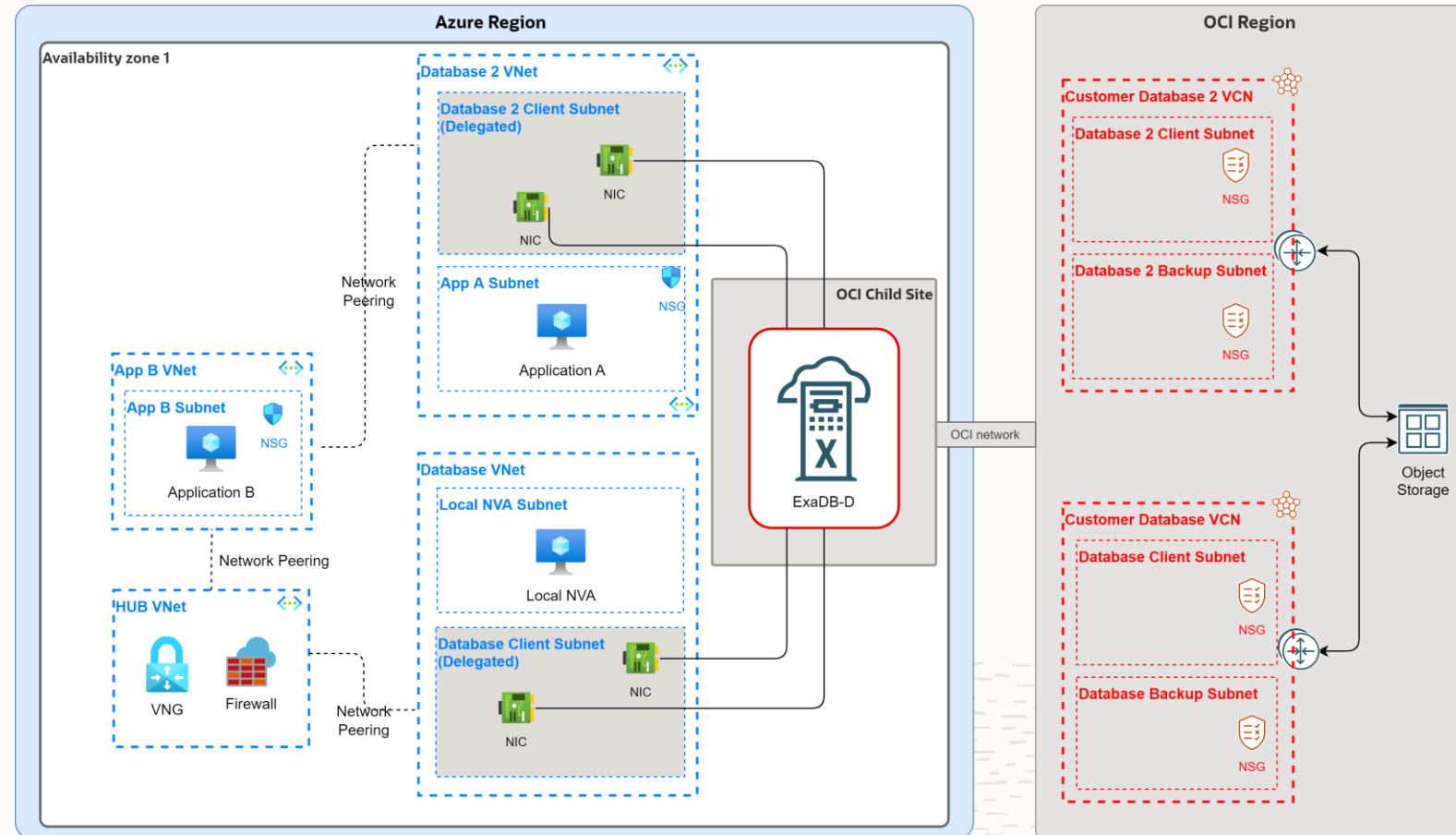


OCI customers can natively connect existing resources and services to Oracle Database and Exadata resources by the private endpoint in the VCN with local peering.



Azure Networking Examples

1. Same Exadata Cloud Infrastructure
 - 2x VM clusters
 - Can use different VNets/Subnets (this example)
2. Same Exadata Cloud Infrastructure
 - 2x VM clusters
 - Only 1 delegated subnet possible per VNet
 - Can use same VNet and same client subnet when deployed in same AZ
3. Applications can reach DB from DB VNet or other spoke VNet's through direct VNet peering or hub-and-spoke (normal Azure VNet peering charges applies)



Example : 2x VM clusters on same Exadata Cloud Infrastructure



Network Requirements

Network

Oracle Database@Azure requirement

- An Azure VNet available in the region and subscription to deploy the Exadata Cloud VM Cluster
 - The VNet address space should be in RFC 1918.
 - The VNet must at least must contain two subnets, one for database client and another for database backup.
 - Client Subnet cannot overlap 192.168.16.16/28
 - Backup Subnet cannot overlap 192.168.128.0/20
 - For Data Guard client and backup subnets CIDR cannot overlap between Primary and Standby networks
 - DRG is managed by Oracle
 - Need to enable **subnet delegation** for the client subnet to Oracle Database@Azure.
 - DNS resolution for Exadata VM Cluster machines and SCAN IPs are available in Azure DNS.
 - Minimum required subnet sizes, /27

Rack Size	Client Subnet: # Required IP Addresses	Client Subnet: Minimum Size Note	Backup Subnet: # Required IP Addresses	Backup Subnet: Minimum Size Note
Quarter Rack	(4 addresses * 2 nodes) + 3 for SCANs = 11	/27 (32 IP addresses)	3 address * 2 nodes = 6	/28 (16 IP addresses)



Azure Networking – ExaDB-D

Remember to plan for expansion

ExaDB-D – VM Cluster Networking in Azure

- Needs a VNet with 1x subnet (**Client subnet**) in Azure
 - Client subnet needs to be created and **delegated** to Oracle Database@Azure before ExaDB-D VM cluster deployment
- Need to provide **Backup subnet** CIDR block during VM cluster deployment
 - Backup subnet will be deployed in OCI, not in Azure
- Plan not to overlap network
 - Non overlapping CIDR blocks are needed for each deployment in a DR scenario

The following table shows the minimum required subnet size depending on ExaDB-D configuration you choose. Each database node require 4x IP addresses and in addition 3x addresses are reserved for Single Client Access Names (SCANs.)

Rack Size	Client Subnet # Required IP Addresses	Client Subnet : Minimum size *	Backup Subnet # (Not in Azure) Required IP Addresses	Backup Subnet : (Not in Azure) Minimum size *
Flexible infrastructure (X9M and higher)	4 addresses per database node (minimum 2 nodes) + 3 for SCANs + 13 for ODBAA service	Minimum size determined by total number of IP addresses needed for database node	3 addresses per database node (minimum 2 nodes)	Minimum size determined by total number of IP addresses needed for database nodes

*The Networking service in Azure reserves five (5) IP addresses in each subnet



Azure Networking for ExaDB-D – example

ExaDB-D Network examples : 2x or 4x Database nodes

Size of subnet	Total number of IPs in subnet
/28	16
/27	32
/26	64
/25	128
/24	256
/23	512

	Client Subnet # IP Addresses in this example	Client Subnet : Minimum size in this example	Backup Subnet # (not in Azure)
Example 2x nodes	4x2 nodes + 3 = 11 IPs ODBAA requires 13 IPs* Total = 24	/27 minimum	/28 minimum
Example 4x nodes	4x4 nodes + 3 = 19 IPs ODBAA requires 13 IPs* Total = 32	/26 minimum (recommend to to use larger for future expansions)	/27 minimum

* The Networking service in Azure reserves five (5) IP addresses in each subnet.
ODBAA service requires eight (12) IP.



Azure Networking ADB-S

Remember to plan for expansion

- Needs a VNet with 1x subnet (**DB subnet**) in Azure
- DB subnet needs to be created and **delegated** to Oracle Database@Azure before ADB-S deployment
- Can be reused (shared) for additional databases

DB Subnet # Required IP Addresses	DB Subnet Minimum size
1 address Deployed ADB-S instance. + 5 addresses reserved for for Azure networking + 13 addresses for ODBAA service	Minimum CIDR requirement for 1 deployed ADB-S is /27 (16 Addresses)

*The Networking service in Azure reserves five (5) IP addresses in each subnet.



Network delegation

Need to enable **subnet delegation** for client subnet to Oracle Database@Azure.

Oracle.Database/networkAttachments

subnet

demo_vnet_2

Name

subnet

Subnet address range *

10.0.0.0/24

+ 5 Azure reserved addresses

☐

Add IPv6 address space

NAT gateway

None

Network security group

None

Route table

None

SERVICE ENDPOINTS

Create service endpoint policies to allow traffic to specific azure resources from your virtual network over service endpoints. [Learn more](#)

Services

0 selected

SUBNET DELEGATION

Delegate subnet to a service

Oracle.Database/networkAttachments

NETWORK POLICY FOR PRIVATE ENDPOINTS

The network policy affects all private endpoints in this subnet. Select the types of network policies that control traffic going to the private endpoints in this subnet. [Learn more](#)

Private endpoint network policy

Save

Cancel

[Give feedback](#)



Oracle Database@Azure - Network considerations

1 delegated subnet by Vnet

Multiples Cluster in the same delegated subnet - shared

Multiples Cluster from **separated** Exadata infra in the **same AZ** in the same delegated subnet

Client subnet - Minimum range /27

- Check documentation to plan accordingly
 - https://docs.oracle.com/en-us/iaas/Content/multicloud/oaa_ip.htm

Backup subnet – Minimum range /28

- By default 192.168.252.0/22

Microsoft network planning

- <https://learn.microsoft.com/en-us/azure/oracle/oracle-db/oracle-database-network-plan>

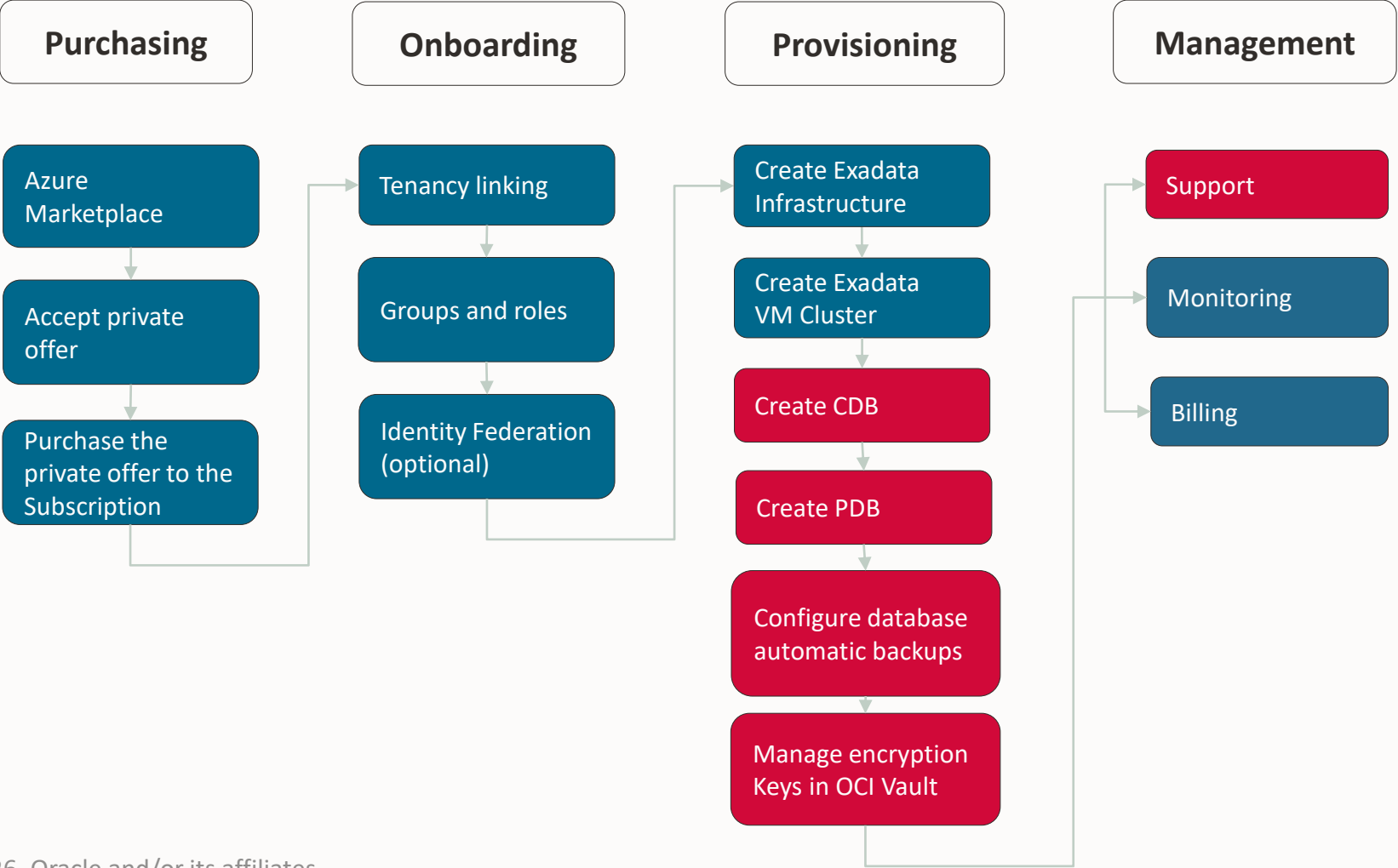
Purchase / Enablement

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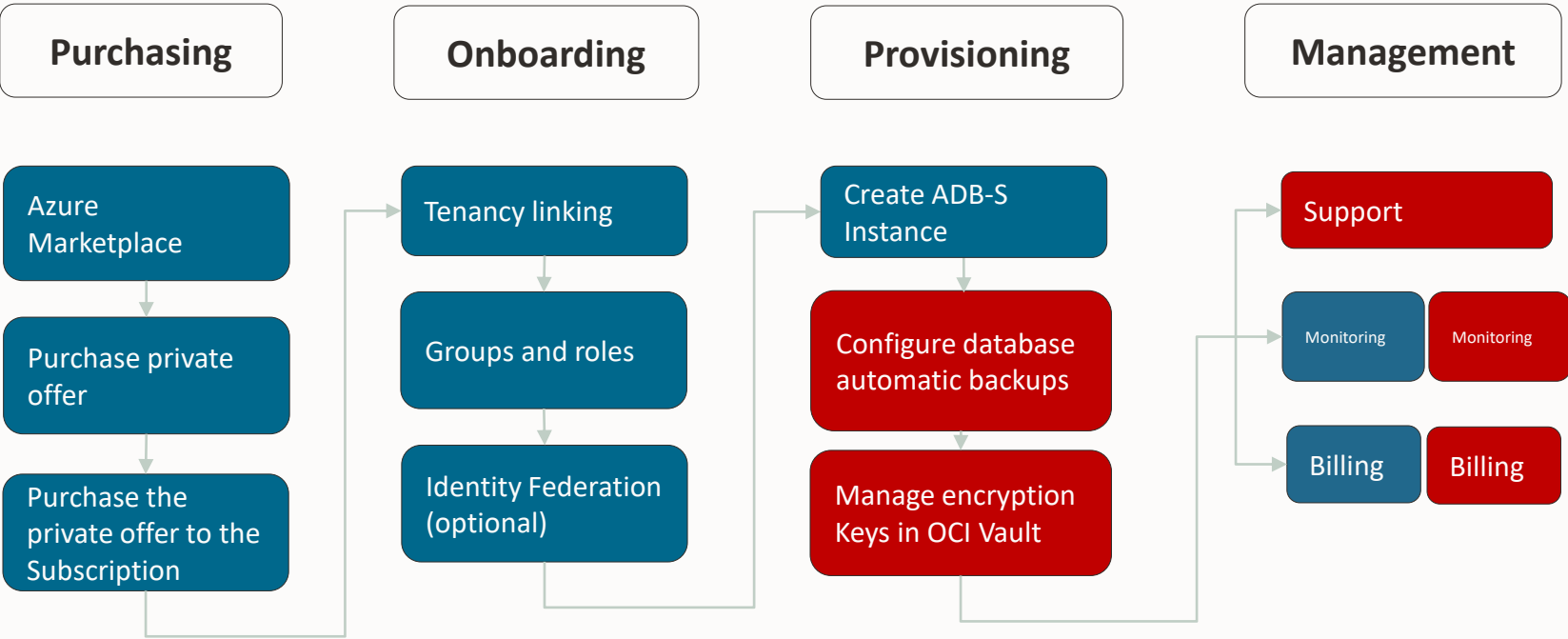
Sales process

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User Journey – Private Offer (ExaDB-D)



User Journey – Private & Pay-As-You-Go Offer (ADB-S)



Public Offer



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Resource groups (3)

Marketplace (10)

Documentation (15)

Azure Active Directory (99+)

Management groups (0)

Show less

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Marketplace

Resources

See all

recoveryservices

Subscription / resource group

App Service

recoveryservices

Subscription / resource group

App Service

recoveryservices

Subscription / resource group

App Service

servicebus

Solution

susanappservicebackup

Storage Account

randyw-vision

API Connection

preview-susanappservicebackup

Resource type

randyw-vision-prediction

Cognitive Service

Marketplace

Product name

Product name

Product name

Documentation

See all

Azure documentation - long title goes here

Azure documentation - long title goes here

Azure documentation - long title goes here

Azure documentation - long title goes here

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Searching in Contoso-pay-as-you-go (filter)

Change

Create a resource

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Categories

Databases (87)

Compute (42)

oracle database@azure

Pricing : All

Operating System : All

Publisher Type : All

Product Type : All

Publishers

☐ Azure benefit eligible only ⓘ

☐ Azure services only

Showing 1 to 20 of 137 results for 'oracle database@azure'. [Clear search](#)

 You have 1 results customized for your organization in private plans. [View private plans](#)

 Oracle Database Oracle America, Inc. Virtual Machine Oracle Database offers market-leading performance, scalability, reliability, and security. Create	 Oracle Database 19.3.0.0 Oracle America, Inc. Virtual Machine Oracle Database 19c offers market-leading performance, scalability, reliability & security in cloud Create	 Oracle Database@Azure Oracle America, Inc. SaaS Oracle database services running on Oracle Cloud Infrastructure (OCI), collocated in Microsoft Azure Starts at Free Subscribe	 Oracle Database 12.1.0.2 Enterprise Edition Oracle Virtual Machine Oracle Database 12c Enterprise Edition (12.1.0.2.0) is a next-generation database designed for the cloud. Create	 Oracle Database 12.1.0.2 Standard Edition Oracle Virtual Machine Oracle Database 12c Standard Edition is an affordable, full-featured data management solution that is ideal for midsize companies. Create
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Oracle Database@Azure

Oracle America, Inc.



Oracle Database@Azure

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Oracle America, Inc. | SaaS

Azure benefit eligible

Plan

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Media



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Oracle America, Inc.

Looking for an existing subscription to this product



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Plan	Description	Price/payment frequency	Billing term
Oracle Database@Azure Pay As You Go Subscribe	Get started using Oracle Autonomous Databases today. You do not need to manage any hardware or install any software. We handle provisioning, backups, patches, upgrades, and scaling on this fully managed service.	\$0.00/month Plus: Oracle Database@Azure Pay as You Go and Overage: \$1 per credit Oracle Database@Azure Commitment Credits: \$0 per credit	1-month subscription



Microsoft Azure

Search resources, services, and docs (G+/I)

Home > Marketplace > Oracle Database@Azure >

Create OracleSubscription ...

Basics

Consent

Review + create

Project details

Only certain customers are eligible for this product.

You can view this product in Marketplace and [Contact Sales](#) ¹⁷

Subscription *

Instance details

Region

Global

Plan *

Oracle Database@Azure Pay As You Go

Billing term *

1 Month Subscription

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Product details

Oracle Database@Azure by Oracle

[Oracle terms of use](#)

[Cloud Services Agreement](#)

This order is subject to Service Descriptions for Oracle Database@Azure Credits - Part Number B99055

[Available here](#)

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Basics

Subscription

Region

Plan

Billing term

Global

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Name

Preferred e-mail address

Preferred phone number

Basics

Subscription

Region

Plan

Billing term

Global

Oracle Database@Azure Pay As You Go

1 Month Subscription

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Home > OracleSubscriptions >

OracleSubscriptions

[+ Create](#) [Edit columns](#) [...](#)

Name ↑↓

 default	...
---	---------------------



default

OracleSubscription

[☆](#) [☆](#) [...](#)

[◊](#) [◀](#)

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-  Activity log
-  Access control (IAM)
-  Tags
- [> Settings](#)
- [> Help](#)

[Add Azure subscriptions](#) [Refresh](#)

Essentials

Subscription [\(move\)](#) : [c](#) State : Succeeded

Subscription ID : 1

Purchase details

Offer name	Oracle Database@Azure
Purchase status	 Subscribed
Plan	oracle_database_at_azure_pay_as_you_go
Purchase start date	2024-09-09T15:46:35.670Z
Purchased by	
Billing term	P1M

Oracle Cloud Account

Status	 Activated
Add subscription status	---
Added Azure subscriptions	View



Private Offer



Microsoft Azure

Marketplace

AllServices (32)Resources (99+)Resource groups (3)Marketplace (10)Documentation (15)

Azure Active Directory (99+)Management groups (0)Show less

ServicesSee all

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ResourcesSee all

recovery servicesApp Service

recovery servicesApp Service

susanappservicebackupStorage Account

preview-susanappservicebackupResource type

recovery servicesApp Service

servicebusSolution

randyw-visionAPI Connection

randyw-vision-predictionCognitive Service

Marketplace

Product name

Product name

DocumentationSee all

Azure documentation - long title goes here

Azure documentation - long title goes here

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Azure documentation - long title goes here

Searching in Contoso-pay-as-you-go (filter)Change

Create a resource

Recent resources

Name

Fabri

Test

Sam

Mob

Gizm

Lore

Prod

Lore

Gizm

Function App

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IT & Management Tools (3117)

Security (2352)

Analytics (2077)

Developer Tools (1798)

Web (1674)

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Operating System : All

Publisher Type : All

Product Type : All

Publisher name : All

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Cloud Infrastructure Services

Virtual Machine

Kafka Server on Ubuntu Server 20.04. High-performance Distributed Event Streaming Scalable Platform

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Datadog

SaaS

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Subscribe



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Barracuda Networks, Inc.

Virtual Machine

The most deployed WAF in public cloud.

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Create



CIS Ubuntu 22.04 LTS Benchmark L1

Center For Internet Security, Inc.

Virtual Machine

Hardened according to a CIS Benchmark - the consensus-based best practice for secure configuration

Starts at \$0.02/hour

Create



OpenVPN Server

yaseen's market

Virtual Machine

Deploy OpenVPN Server Easily

Starts at \$0.012/user/3 years

Create

Popular Azure services in your organization











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Check eligibility

Showing 1 of 1 Private Offers.

Private Offer name	Status	Prepared by	Products included	Expiry date	Start date	End date	Billing account
Oracle for Woodgrove Bank	Accepted	Oracle	1 product	08/31/2023	6/15/2023	12/31/2023	5607965
Product name Oracle Database@Azure	Status Ready	Plan name PlanName	Publisher Oracle	Product type SaaS	Azure benefit eligible Yes		



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Showing 1 of 1 Private Offers.

Private Offer name ↑↓

Oracle for Woodgrove Bank

View

Product name ↑↓

Plan name ↑↓

Oracle Database@Azure

PlanName

Purchase

Oracle Database@Azure

Oracle | SaaS

★ 5 (390 ratings)

Private Offer

Azure benefit eligible

Plan PlanName

Subscription MyAzureSubscription

Overview

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Usage Information + Support

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Private Offer name

Oracle for Woodgrove Bank

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Product name

Plan name

Oracle Database@Azure

PlanName

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Oracle Database@Azure

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5 (390 ratings)

Private Offer

Azure benefit eligible

Plan

Subscription

PlanName

MyAzureSubscription

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Plans + pricing

Usage Information + Support

Reviews

Plan	Description	Price + Payment options	Billing term
PlanName Private Offer Subscribe	Private plan	\$820,000.00/3 years Plus: Overage Cloud Credits: \$0 cloud credits Cloud credits: \$0 cloud credits	3-year

Purchase

Microsoft Azure

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Connie Wilson

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Showing 1 of 1 Private Offers.

Private Offer name ↑↓

Oracle for Woodgrove Bank

View

Product name ↑↓

Plan name ↑↓

Oracle Database@Azure

PlanName

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★ 5 (390 ratings)

Private Offer Azure benefit eligible

Plan ⓘ

Subscription ⓘ

PlanName

MyAzureSubscription

Overview Plans + pricing Usage Information + Support Reviews

Usage Information

Publisher ID

Product ID

oci_exadata

Plan ID ⓘ

PlanName

Categories

Security

Legal

Privacy policy

License Agreement

Purchase

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Product documentation

Support

Contact sales

Microsoft Azure

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Basics

Consent

Tags

Review + create

Project details

After the creation, the billing of this resource starts immediately.

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Subscription *

MyAzureSubscription

Instance details

Region

Global

This resource is a global resource that works across Azure regions.

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Create Oracle purchase  

Basics

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Review + create

Basics

Subscription	MyAzureSubscription
Resource group	myBuyer_group
Plan	PlanName
Billing term	3-year
Recurring billing	Yes
Subtotal	\$300,000.00

Tags

None	None
------	------

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Home >

 **Microsoft.Oracle_1687973914143** | Overview  

Deployment

 Delete  Cancel  Redeploy  Download  Refresh

 Overview

 Activity log

 Access control (IAM)

 Tags

 **Your deployment is complete**

 Deployment name: Microsoft.Oracle_1687973914143

Subscription: [M](#)

Start time: 6/28/2023, 10:38:39 AM

Correlation ID:

> Deployment details

∨ Next steps

[Go to resource](#)

Give feedback

 Tell us about your experience with deployment



Oracle Database@Azure Deployment

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Pre-requirements for Deployment of Oracle Database@Azure

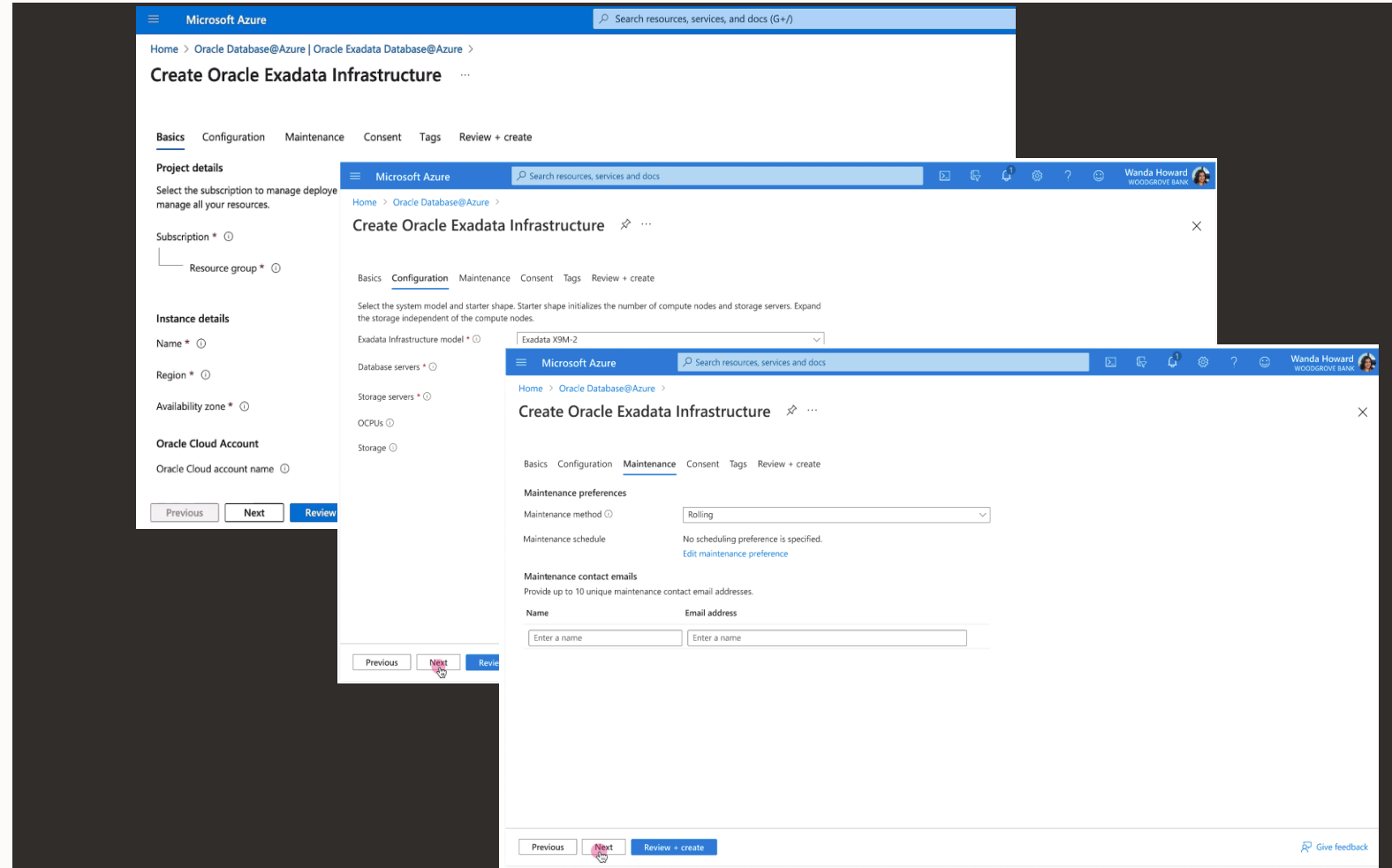
- [Azure permissions to accept the private offer](#)
- An existing Azure subscription
- An Azure VNet with a subnet delegated to the Oracle Database@Azure service (Oracle.Database/networkAttachments)
- Azure groups and roles
 - odbaa-exa-infra-administrators to provision Exadata Infrastructure
 - odbaa-vm-cluster-administrators to provision VM cluster
- Oracle Cloud account
 - OCI ExaDB-D Service limits
 - OCI IAM Policies for Database Service Resource
 - OCI IAM Policies for Object Storage
 - OCI IAM Policies for OCI Vault
- Identity Federation



Exadata Database Service Provisioning

Steps to provision Exadata Infrastructure

1. Enter basic information like name, AZ, Subscription and resource group you want to deploy in
2. Choose the Exadata Infrastructure with the number of Database Servers and Storage
3. Select the managed patching for the infrastructure as well as the schedule
4. Once these details are entered, you can consent and provision



The screenshot displays the 'Create Oracle Exadata Infrastructure' wizard in the Microsoft Azure portal. The wizard is divided into several tabs: Basics, Configuration, Maintenance, Consent, Tags, and Review + create. The 'Basics' tab is active, showing fields for Subscription, Resource group, Name, Region, Availability zone, and Oracle Cloud account name. The 'Configuration' tab is also visible, showing fields for Exadata Infrastructure model, Database servers, Storage servers, OCPUs, and Storage. The 'Maintenance' tab is active, showing fields for Maintenance method (Rolling), Maintenance schedule (No scheduling preference is specified), and Maintenance contact emails (Name and Email address). The 'Review + create' tab is also visible, showing a summary of the configuration.



Steps to provision Exadata VM Cluster

1. Enter basic information like name, AZ, subscription and resource group you want to deploy in.
2. Select licensing type (BYOL or License Included)
3. Allocate VM cluster resource (memory, nodes, storage, etc.).
4. Select the VNet it will be deployed and the two **Delegated Subnets**

The screenshot displays the 'Create Oracle Exadata VM Cluster' wizard in the Microsoft Azure portal. The wizard is divided into several tabs: Basics, Configuration, Networking, Diagnostics collection, Consent, Tags, and Review + create. The 'Basics' tab is currently selected, showing fields for Name, Region, Exadata infrastructure, License type, Grid infrastructure Version, Security (SSH public key source), and Key pair name. The 'Configuration' tab is also visible, showing 'Database servers selection' and 'VM cluster allocation' sections. The 'Networking' tab is selected, showing 'Virtual network', 'Client subnet', 'Backup subnet', and 'DNS' sections. The 'Review + create' tab is also visible, showing a summary of the configuration and a 'Review + create' button.



Create Database (CDB and PDB) in OCI

- Must be part of the Oracle Database@Azure CDB or PDB admin groups, depending on the operation
- In the Azure VM cluster resource page, click the OCI Database URL **Go To OCI** using the identity federation
- On the OCI console, create CDB and/or PDB

ORACLE Cloud Search resources, services, documentation, and Marketplace Netherlands Northwest (Amsterdam) Help

Create database

Basic information for the database

Provide the database name
DB1109

Provide a unique name for the database Optional ⓘ

Select a Database version
19c

Provide a PDB name Optional ⓘ

Specify a database home

Database Home source
☒ Select an existing Database Home ☐ Create a new Database Home

Database Home display name
Select a Database Home for the new database

Create administrator credentials

Username Read-only
sys

Password ⓘ

Confirm password

Create database Cancel

Resources

Databases (3)

Database homes (3)

Virtual IP address (0)

Virtual Machines (2)

Metrics

Work requests (0)

Databases

Create database

Name	State	Database
	Available	
	Available	
	Available	



DNS

Microsoft Azure

Search resources, services, and docs (G+/)

[Home](#) > [Oracle Database@Azure](#) | [Oracle Exadata Database@Azure](#) >

Create Oracle Exadata VM Cluster

Basics

Configuration

Networking

Diagnostics Collection

Consent

Tags

Review + create

Virtual network *

Client subnet *

Backup subnet CIDR

vnet-emea-specialists-1

subnet-emea-specialists-delegated

DNS

Custom DNS

Host name prefix *

☐

Previous

Next

Review + create

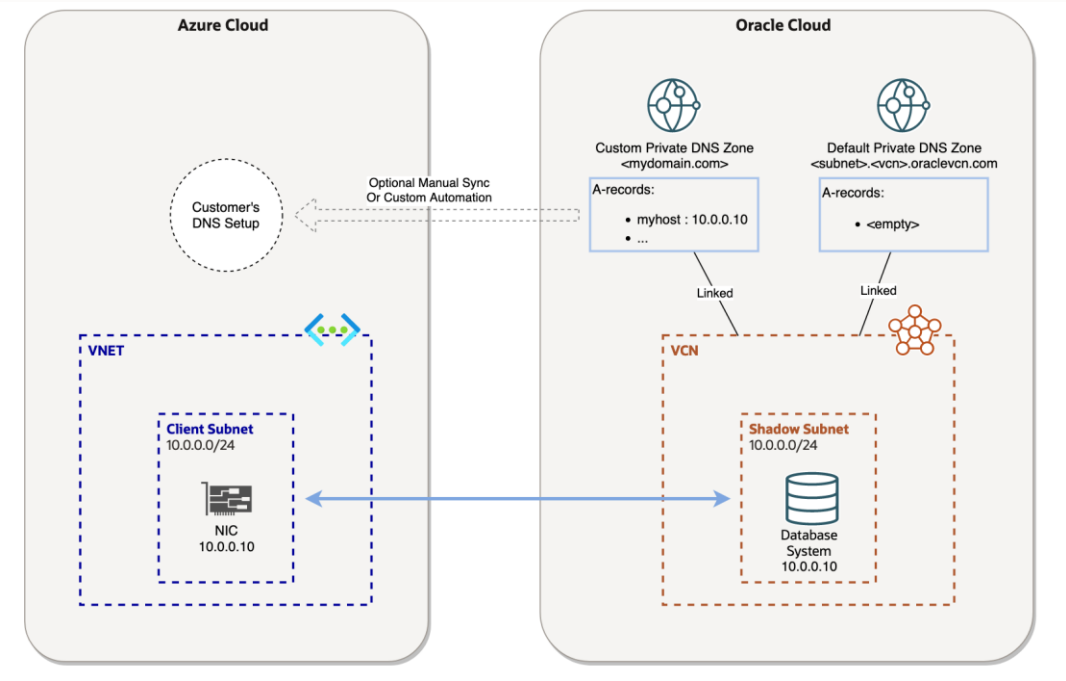
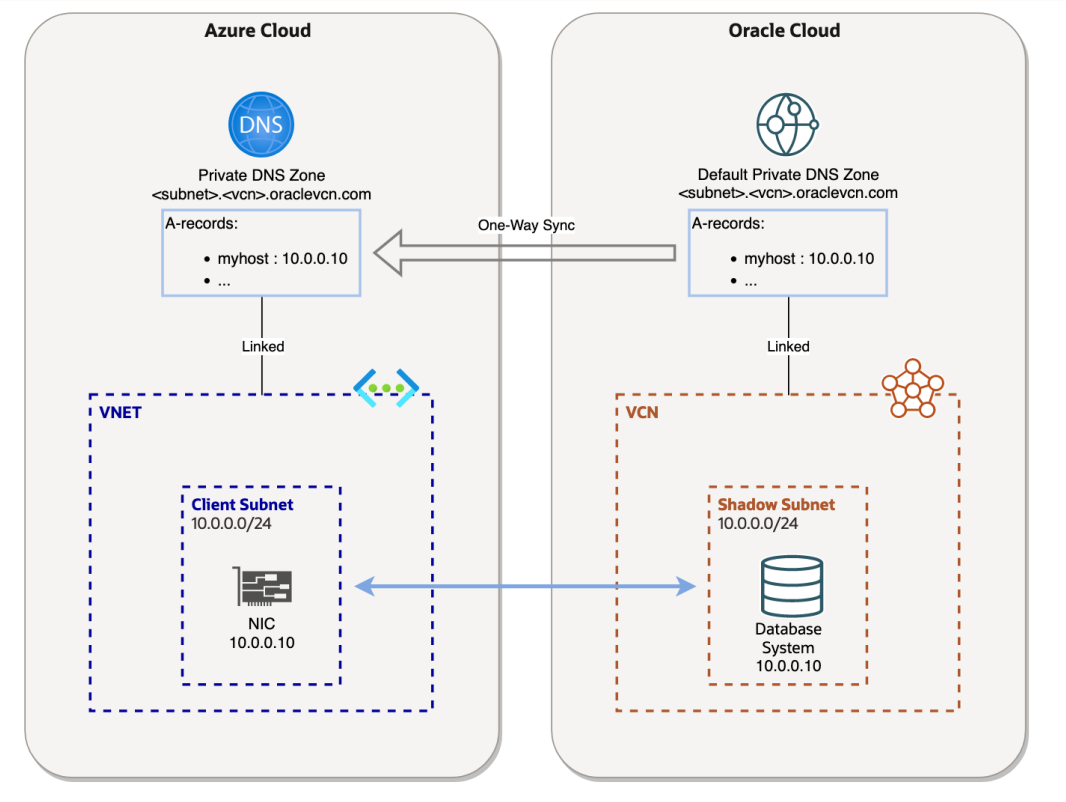


DNS resolution in OCI and Azure – Use-cases

Use Cases	DNS in Azure	VM Cluster - Networking	FQDN	Azure Subscription	Steps
Fully managed	DNS servers Default (Azure-provided)	Uncheck private DNS service	*.oraclevcn.com	Same Exadata subscription	Fully managed in Azure and OCI
Custom FQDN	DNS servers Default (Azure-provided)	Check private DNS service	*.acme.com	Custom subscription	- Managed in OCI DNS Private Zone - Create Private DNS zone and records in Azure
Custom DNS	DNS servers Custom (on-prem)	Check private DNS service	*.acme.com	N/A	- Managed in OCI DNS Private Zone - Create Zone and records in custom DNS



DNS setup



Autonomous Database Provisioning

Deployment: Basic

 Microsoft Azure

Home > Oracle Database@Azure | Oracle Autonomous Database@Azure >

Create Oracle Autonomous Database ...

 Changes on this step may reset later selections you have made. Review all options prior to deployment.

Basics

Configuration

Networking

Maintenance

Consent

Tags

Review + create

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Resource group *

jsitest1_group

Create new

Instance details

Name *

jsiadb1

Region *

East US

Oracle Cloud Account

Oracle Cloud account name

Previous

Next

Review + create

- Resource group
- Name
- Region

Deployment: Configuration

Microsoft Azure

Home > Oracle Database@Azure | Oracle Autonomous Database@Azure >

Create Oracle Autonomous Database

Basics **Configuration** Networking Maintenance Consent Tags Review + create

Workload type * Transaction Processing

Database configuration

Database version * 19c

ECPU count * 2

Compute auto scaling ☒

Storage * 1024

Storage unit size * ☒ GB ☐ TB

Storage auto scaling ☒

Backup retention

Automatic backups are managed by Oracle. Backup storage is billed separately and in addition to database storage. When a backup ages beyond the backup retention setting, it is deleted. [Learn more](#)

Backup retention period in days * 60

Microsoft Azure

Home > Oracle Database@Azure | Oracle Autonomous Database@Azure >

Create Oracle Autonomous Database

Storage auto scaling ☒

Backup retention

Automatic backups are managed by Oracle. Backup storage is billed separately and in addition to database storage. When a backup ages beyond the backup retention setting, it is deleted. [Learn more](#)

Backup retention period in days * 60

Administrator credentials

Set the password for your Autonomous Database ADMIN user here.

Username

Password *

Confirm password *

License

License type * License included

Advanced options ☒

Management

Character set AL32UTF8

National character set AL16UTF16

Previous Next **Review + create**

- Workload type: ATP or ADW
- Database version: 19c
- ECPU count
- Auto scaling
- Storage size
- Storage auto scaling
- Backup retention
- Username
- Password
- Licence type
- Advanced options
 - Character set
 - National Character set



Deployment: Networking

The screenshot shows the 'Create Oracle Autonomous Database' wizard in the Microsoft Azure portal, specifically the 'Networking' tab. The breadcrumb navigation at the top indicates the path: Home > Oracle Database@Azure | Oracle Autonomous Database@Azure >. The wizard has several tabs: Basics, Configuration, Networking (selected), Maintenance, Consent, Tags, and Review + create. Under the 'Networking' tab, the 'Network access' section contains the following fields:

- Access type:** A dropdown menu with 'Managed private virtual network IP only' selected.
- Require mutual TLS (mTLS) authentication:** An unchecked checkbox.
- Virtual network *:** A dropdown menu with 'jsivnet' selected.
- Subnet *:** A dropdown menu with 'adb1' selected.

Below these fields, a yellow warning box states: 'The subscription and resource group must be selected in the "Basics" tab in order to list virtual networks.' At the bottom of the wizard, there are three buttons: 'Previous', 'Next', and 'Review + create'.

- Access type: Managed private virtual network IP only
- Require mutual TLS (mTLS) authentication
- Virtual network
- Subnet

Deployment: Maintenance

[Home](#) > [Oracle Database@Azure](#) | [Oracle Autonomous Database@Azure](#) >

Create Oracle Autonomous Database ...

[Basics](#) [Configuration](#) [Networking](#) **[Maintenance](#)** [Consent](#) [Tags](#) [Review + create](#)

Maintenance preferences

Maintenance patch level

Regular

Maintenance contact emails

Contacts in this list receive operational notices and announcements as well as unplanned maintenance notifications.

Email address

Enter an email address

- Maintenance patch level
- Maintenance contact emails

[Previous](#)

[Next](#)

[Review + create](#)



Deployment: Tags

Microsoft Azure

Search resources, ser

Home > Oracle Database@Azure | Oracle Autonomous Database@Azure >

Create Oracle Autonomous Database ...

Basics

Configuration

Networking

Maintenance

Consent

Tags

Review + create

Tags are name/value pairs that enable you to categorize resources and view consolidated billing by applying the same tag to multiple resources and resource groups. [Learn more about tags](#)

Note that if you create tags and then change resource settings on other tabs, your tags will be automatically updated.

Name ⓘ	Value ⓘ	Resource
	:	Autonomous Database

Previous

Next

Review + create

- Tag name and value

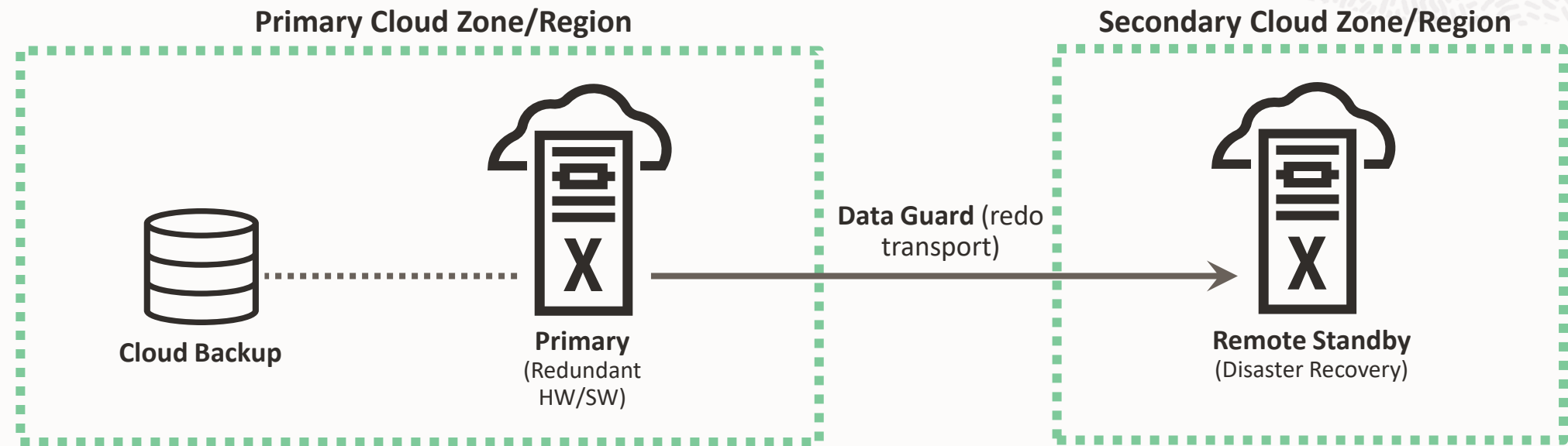


High Availability and Disaster Recovery

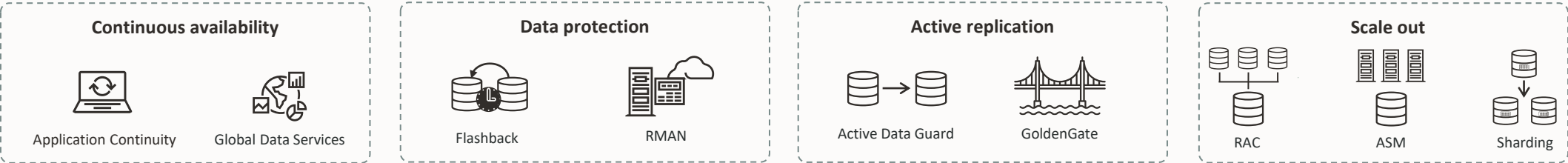
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Oracle Maximum Availability Architecture (MAA)

High-Availability Blueprint in the Cloud



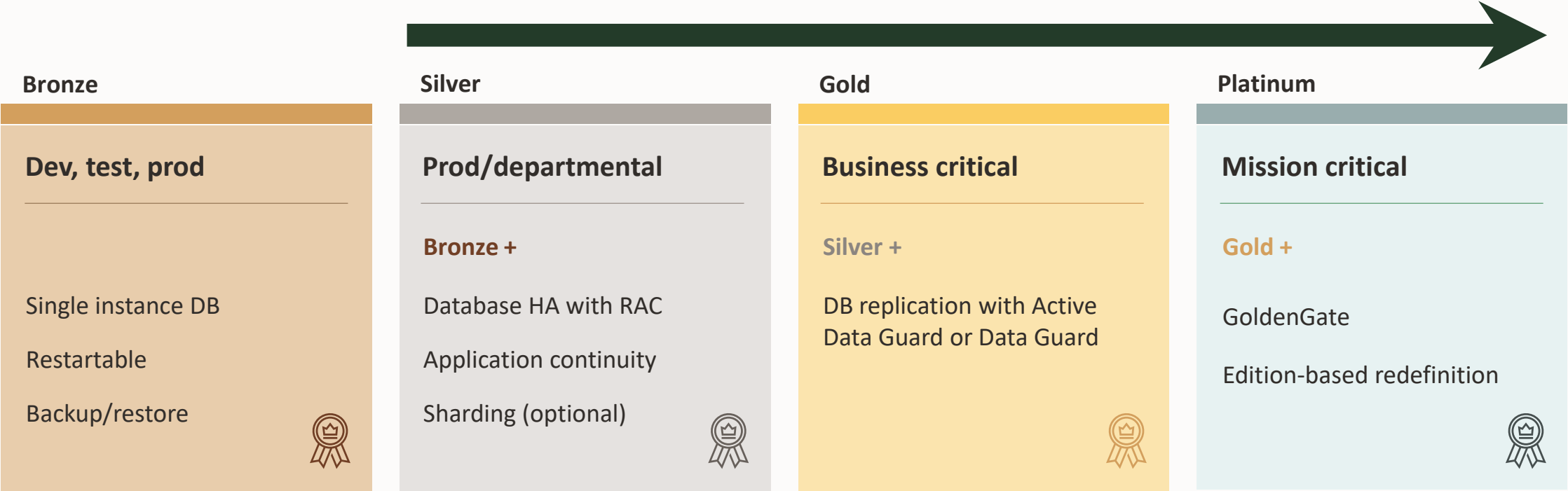
Key Cloud MAA Technologies



MAA reference architectures

Availability service levels

Oracle Database@Azure



ExaDB@Azure / DR different AZ / App DR same AZ

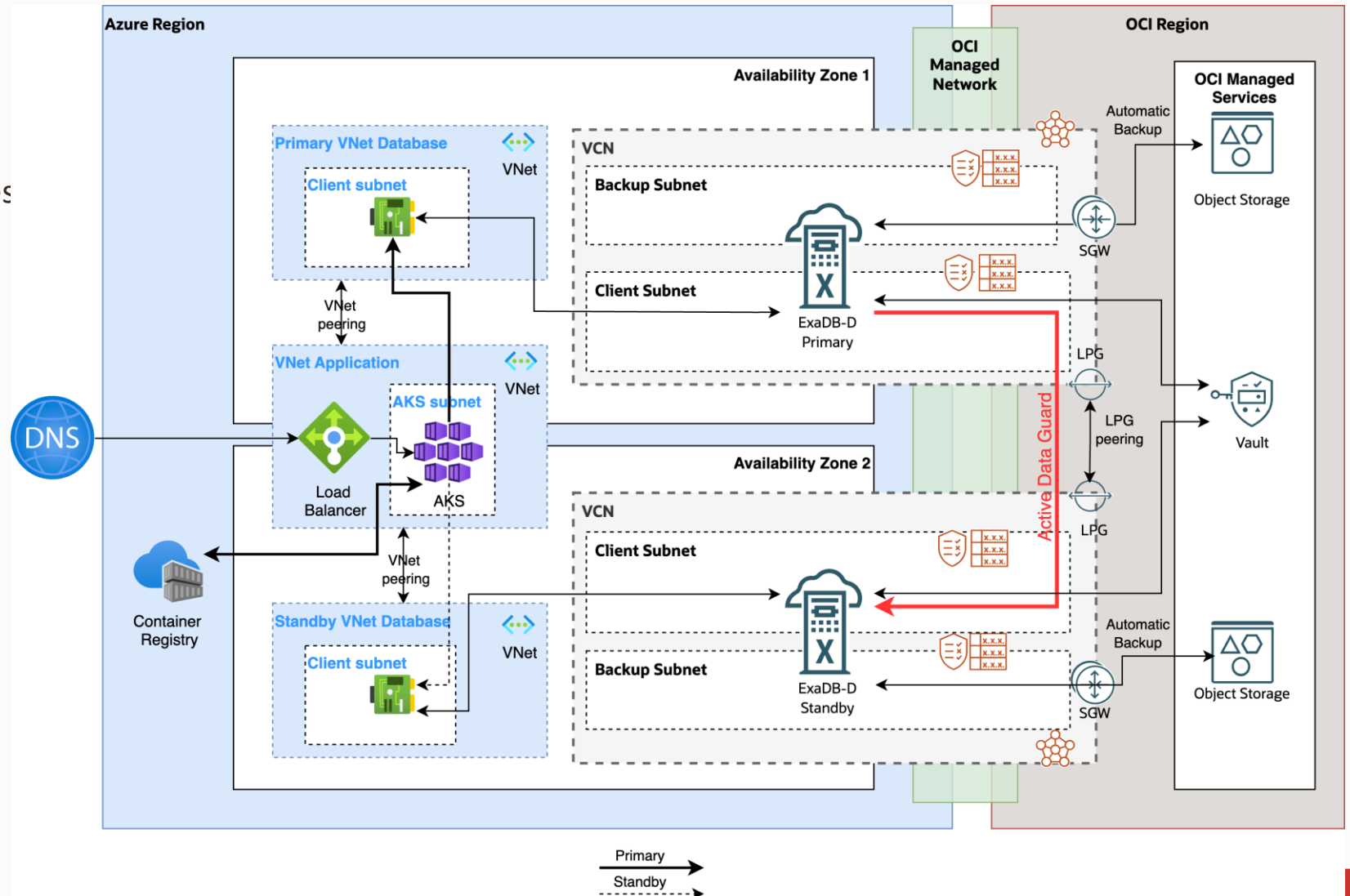
Gold MAA – OCI network – Low level

Application layer on Azure

- Application is deployed in Regional Private Azure Kubernetes Services (AKS) multi-AZs
- Vnet Application peered with Primary and Standby Vnet Databases

Database layer on OCI

- Primary ExaDB in AZ1
- Standby ExaDB in AZ2
- Redo log shipment via the OCI Network (no cross-AZs traffic)
- Routing uses local peering gateway between VCNs



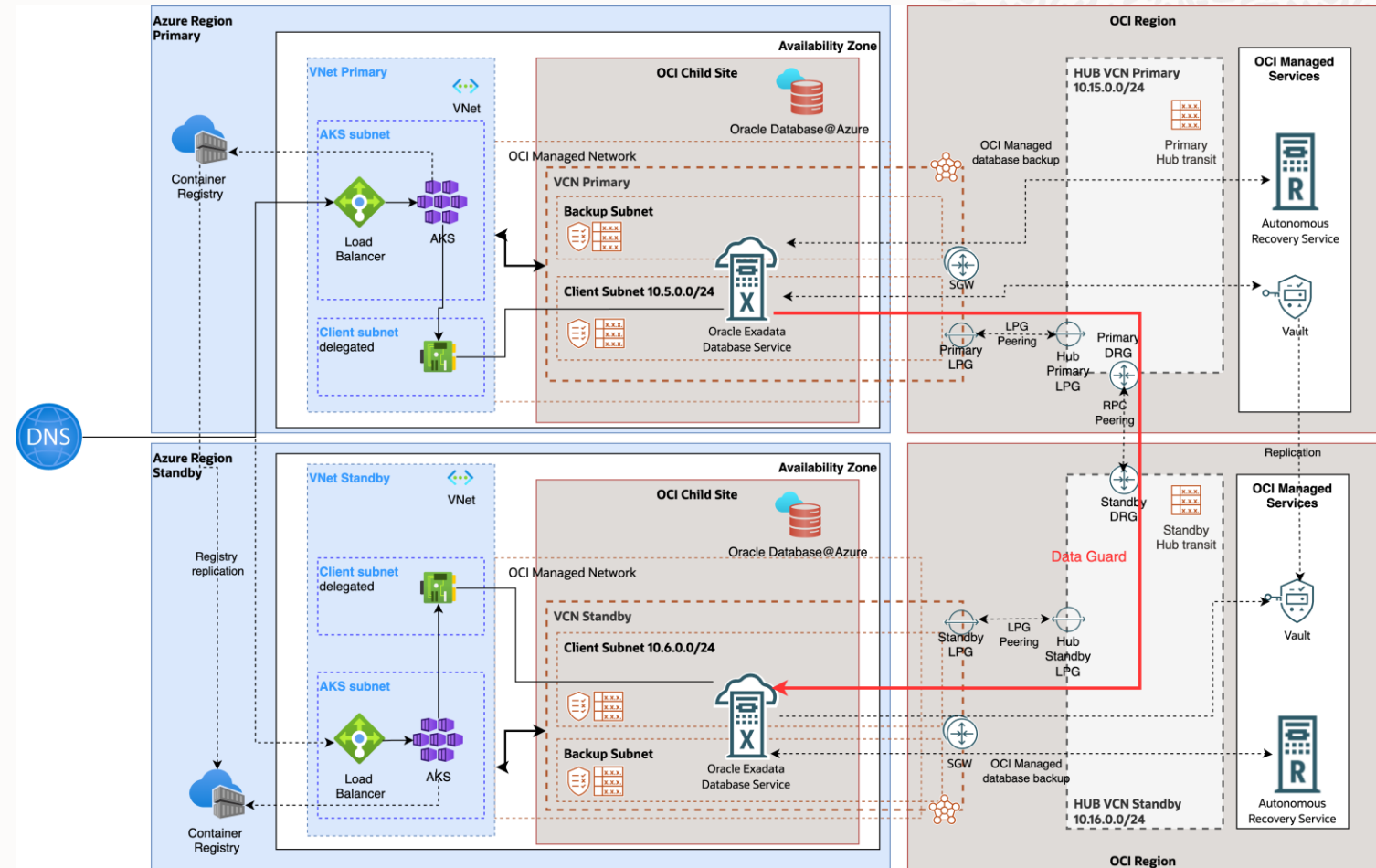
Oracle Database@Azure | Cross-region Replication Overview

Application layer on Azure

- Application is deployed in Regional Azure Kubernetes Services (AKS)
- Container images are replicated between regions
- Deploy Dataguard Observer on Azure VM

Database layer on OCI

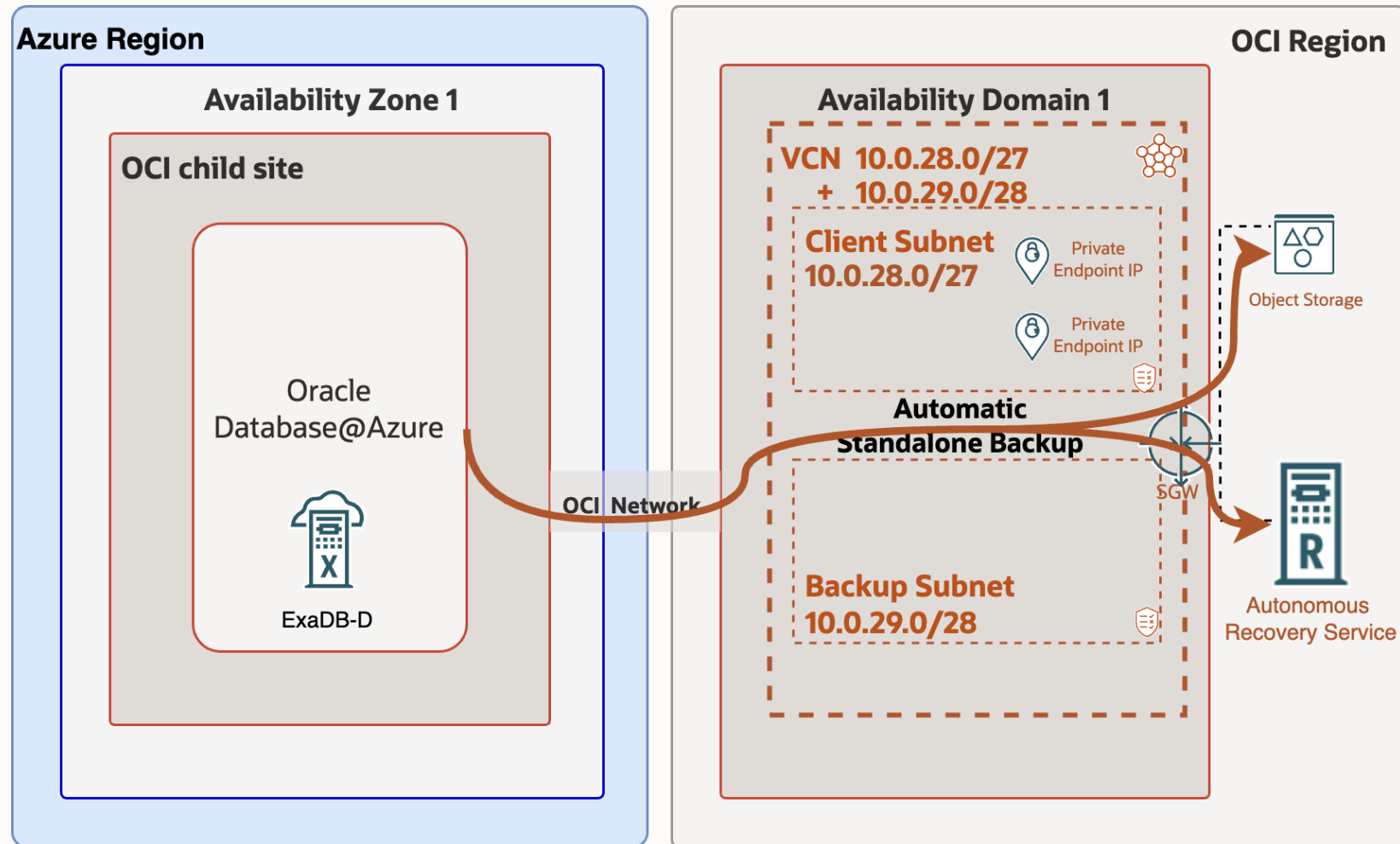
- Primary DB is hosted in Region 1
- Standby DB is hosted in in Region 2
- Redo log shipment is via the OCI backbone (incurs egress charges)
- Routing uses local peering to hub VCNs and DRGs
- OCI Vault cross region replication (incurs egress charges)
- Observers are place in Azure application subnet.



Backup

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Automatic and Standalone Backup to OCI Object Storage



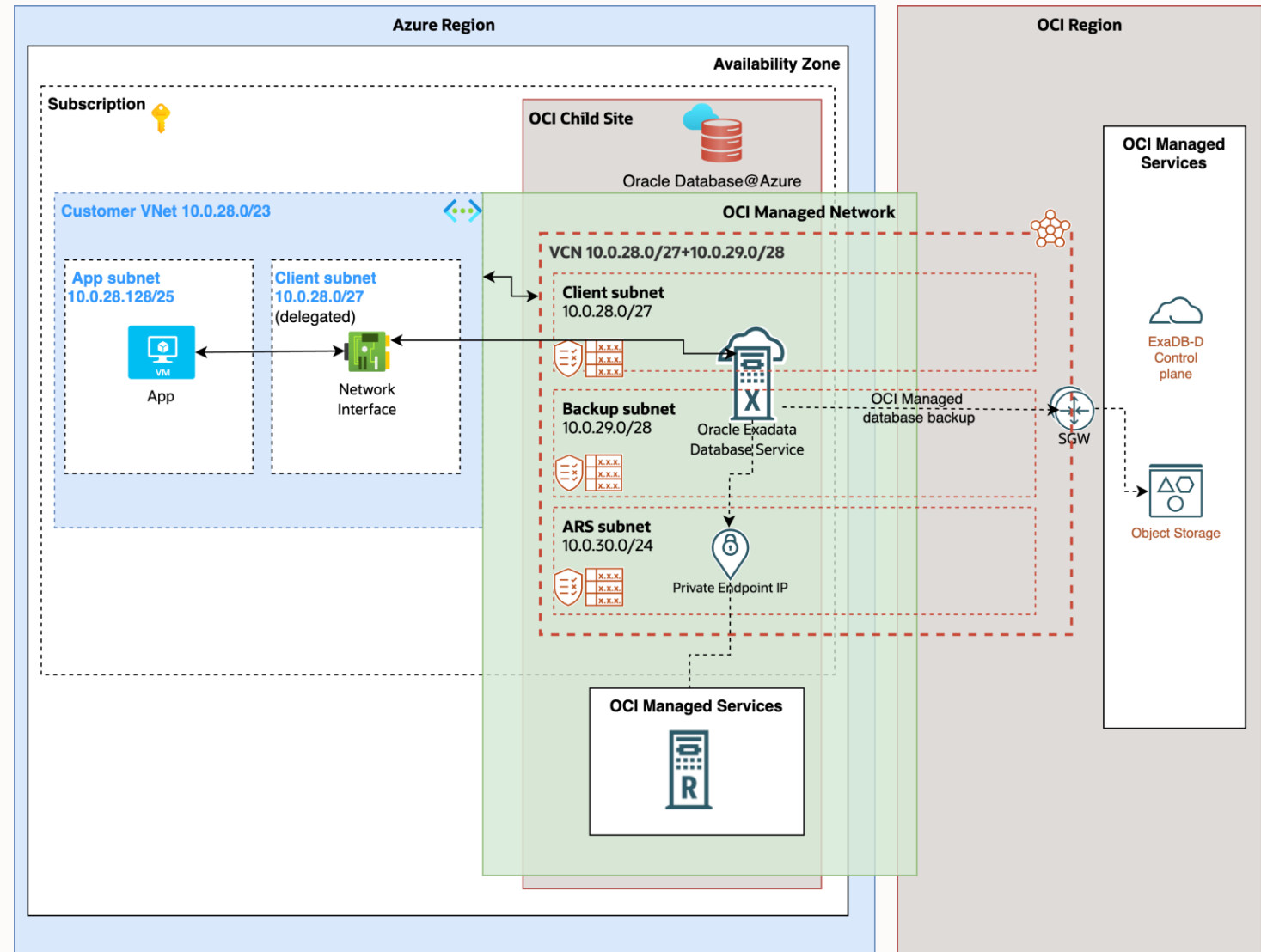
Automatic backup with a Maximum retention period of 60 days to object storage.
Standalone/Full backup for long term backup retention through OCI API/CLI/Console.



Autonomous Recovery Service

Oracle Database Autonomous Recovery Service (ARS) is a fully managed, standalone, and centralized cloud backup solution for Oracle Cloud Infrastructure (OCI) databases

- Draw down from Azure commitment
- Dedicated OCI Subnet required, /24 CIDR
- Protection policy
 - Bronze, Silver, Gold, Platinum



Migration

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Migration Methods

	Migration	Online	Downtime *
Recommended: ZDM Oracle Zero Downtime Migration	Physical/Logical		Minimal
Other methods			
Oracle Data Guard	Physical		Minimal
Oracle GoldenGate	Logical		Minimal
RMAN	Physical		Required
Oracle Data Pump	Logical		Required
Oracle Database Multitenant Movement	Physical		Required

* Downtime – The duration depends on the volume and use case.



Migrating Databases to Oracle Cloud Infrastructure

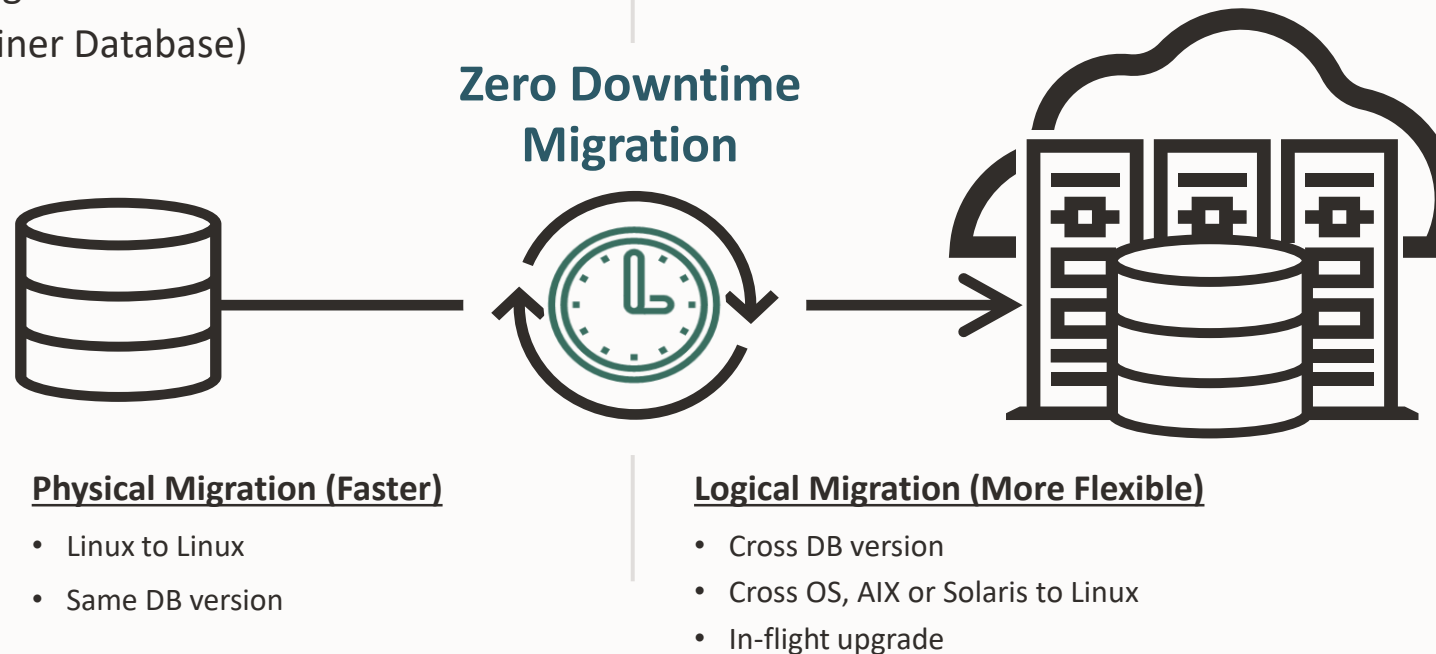
Oracle Zero Downtime Migration – Simple and Free!

Oracle Database Sources

- On-premises deployments
- Cloud deployments
- Oracle Database Enterprise & Standard Editions
- Linux, AIX, Solaris
- RAC, RAC One Node, Single-Instance
- Non-CDB or CDB (Container Database)

Oracle Cloud Targets

- Oracle Exadata Database Service
- Autonomous Database Serverless

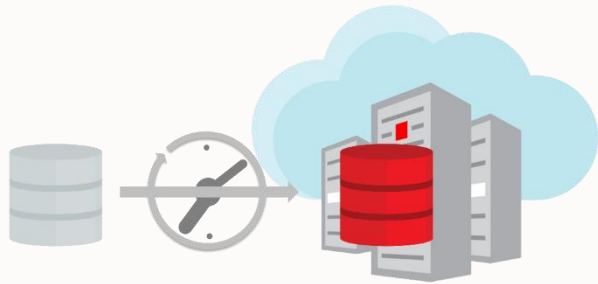


Oracle Zero Downtime Migration (ZDM)

Overview

Simple

- Single-button approach



Fleet Scale

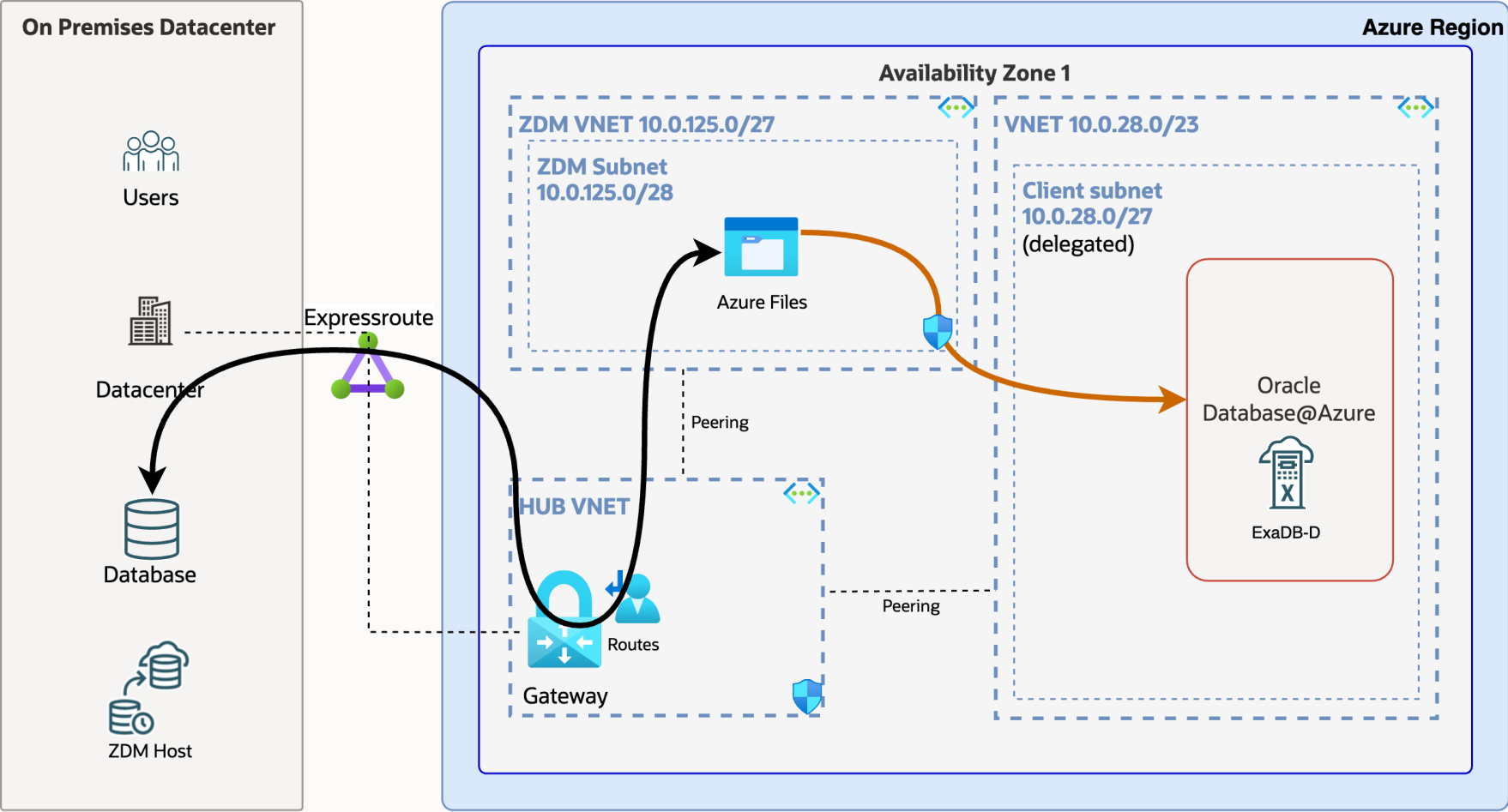
- Centralized
- Scheduled Operations
- Audit Trail
- Migrations in Parallel
- Jobs Framework

Comprehensive

- MAA Compliant
- Resumable
- Dry-run option
- Customizable User Actions
- Automated Encryption

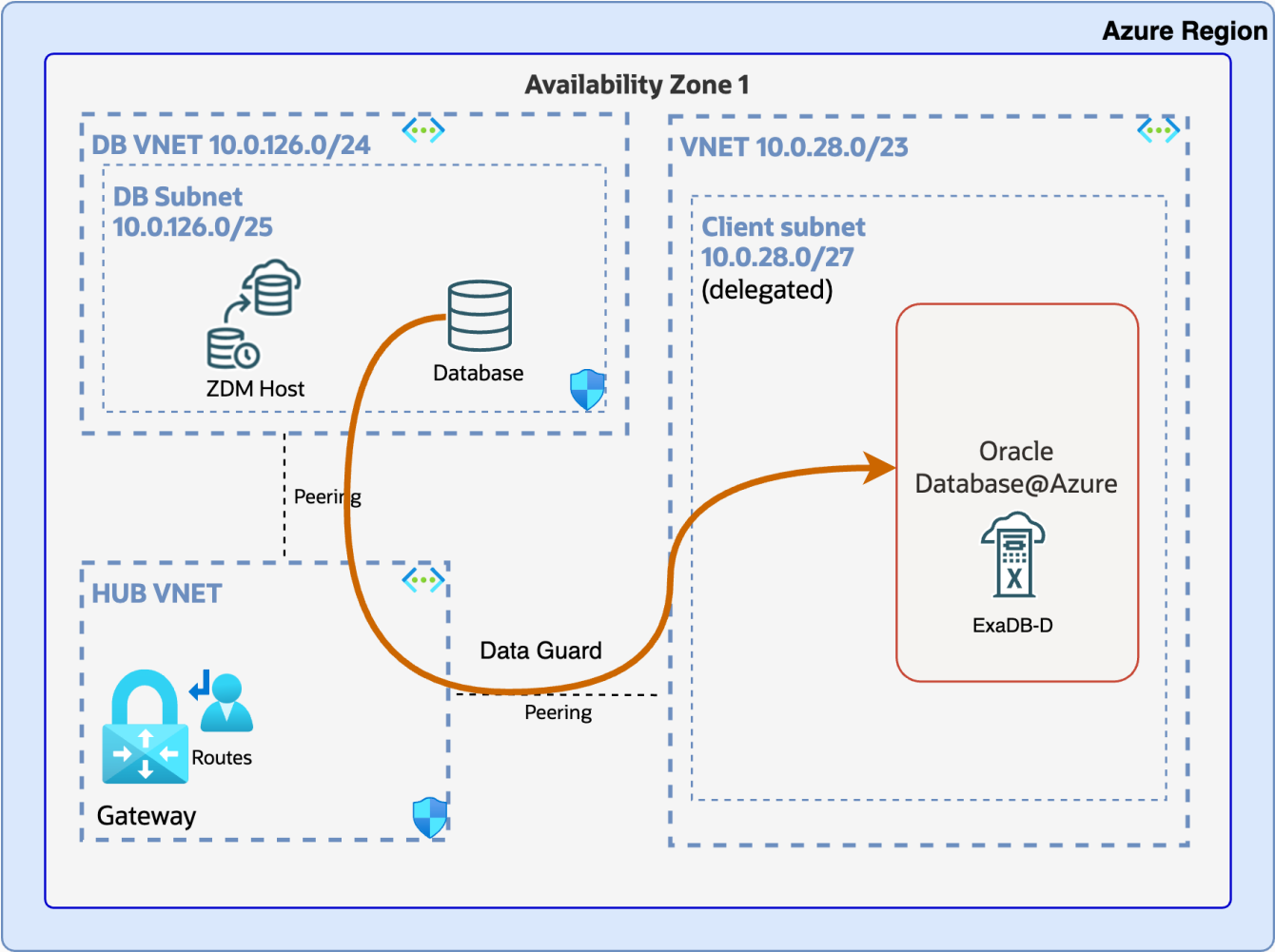
Migration with ZDM on Azure from On Premises

Physical Offline Migration

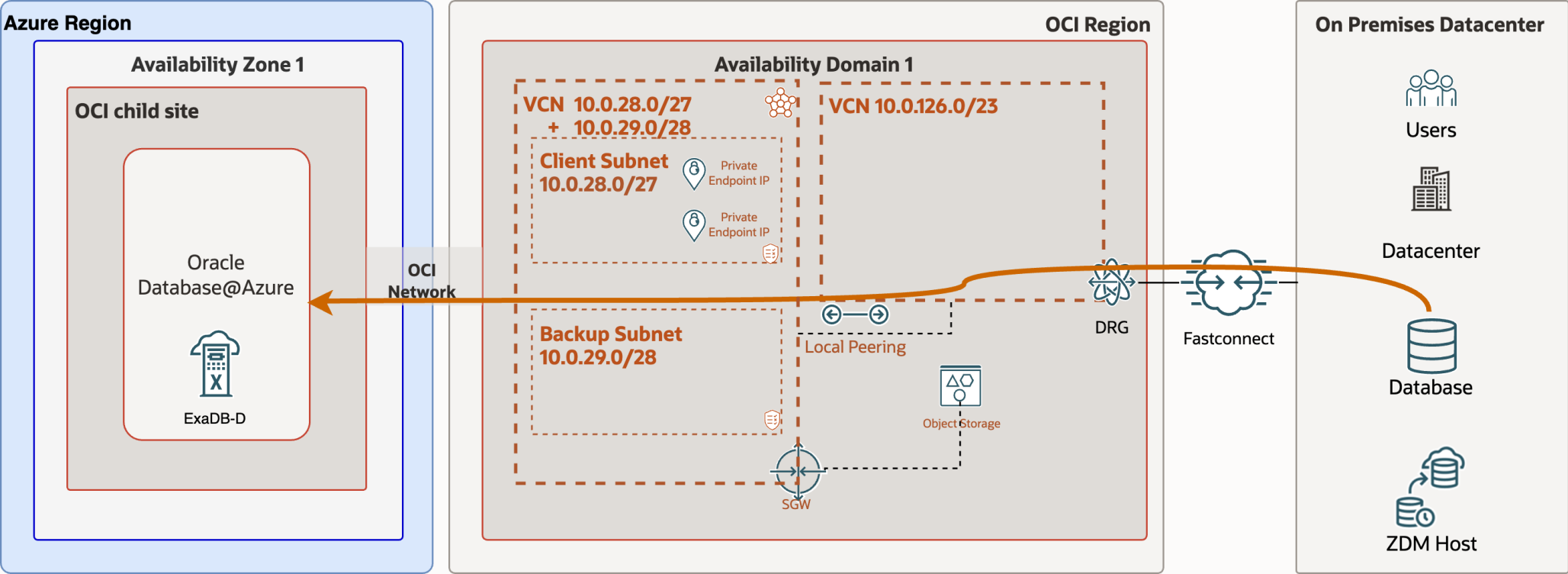


Migration with ZDM on Azure

Physical Online Migration – direct method (restore from service)



Migration with ZDM from On Premises through OCI



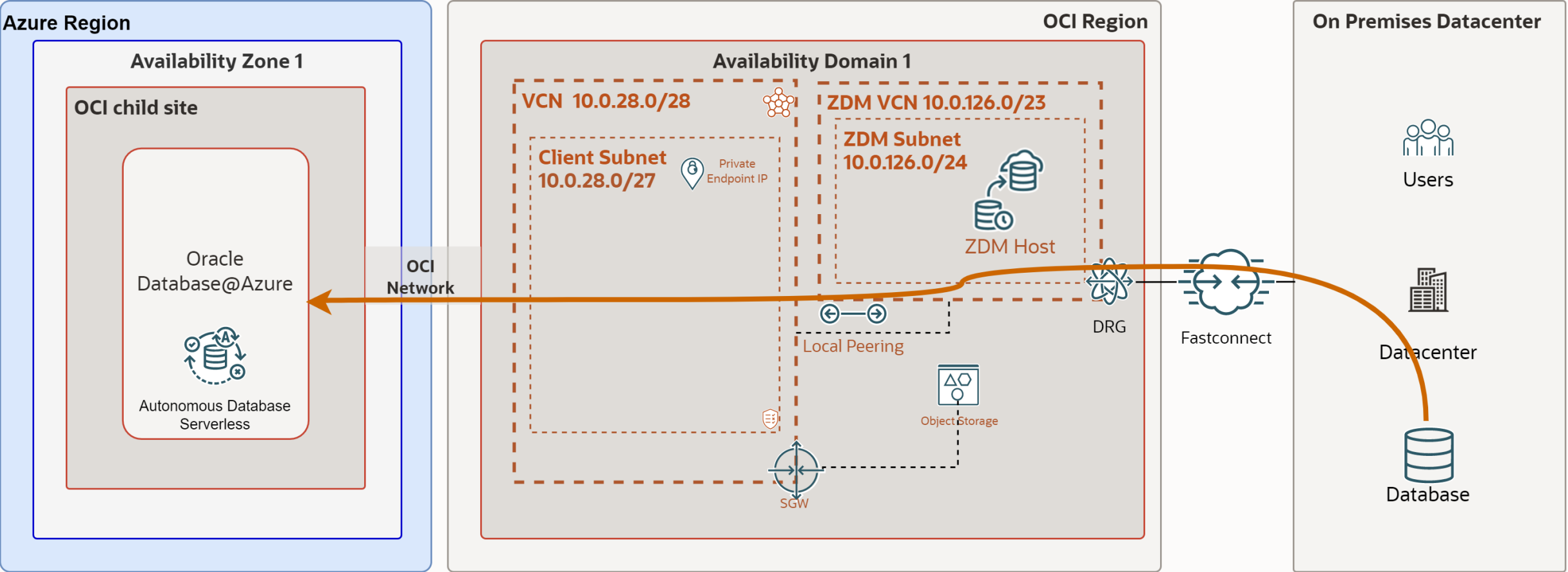
Migration Methods – ADB-S

	Migration	Online	Downtime *
Recommended: ZDM Oracle Zero Downtime Migration	Logical		Minimal
Other methods			
Oracle GoldenGate	Logical		Minimal
Oracle Data Pump	Logical		Required

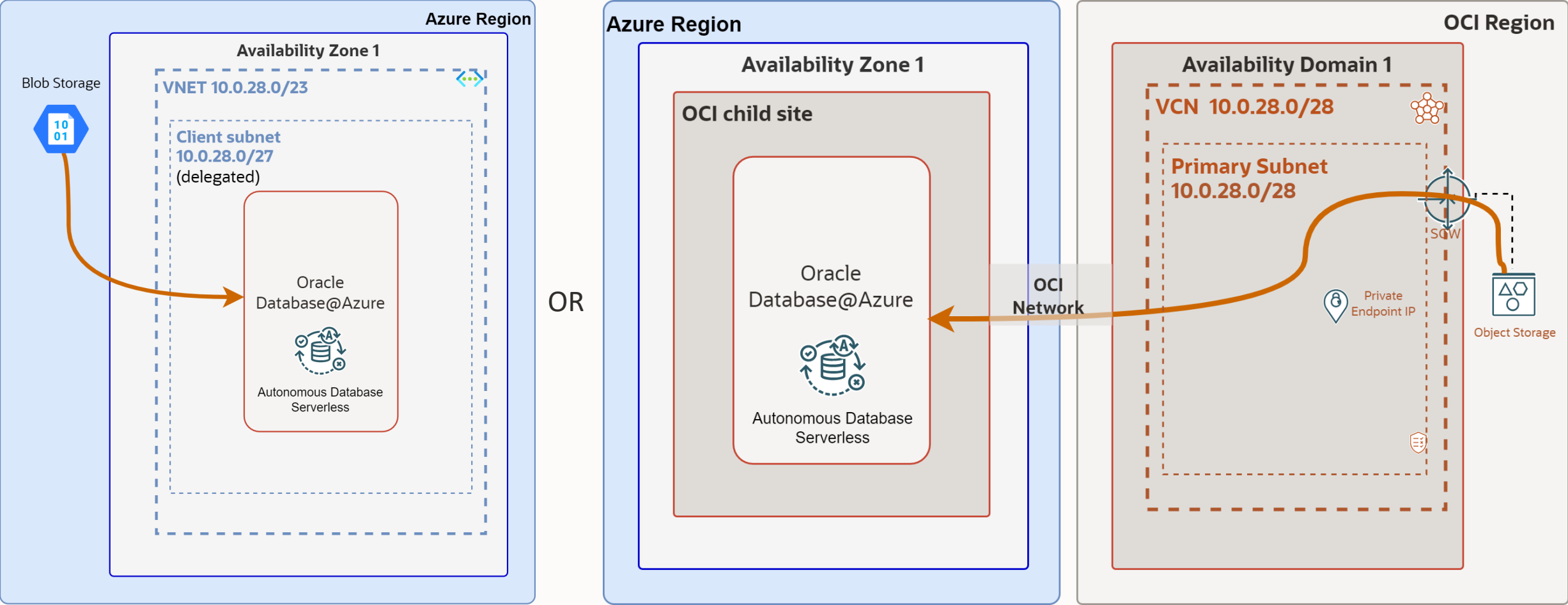
* Downtime – The duration depends on the volume and use case.



Migration with ZDM from On Premises through OCI – ADB-S



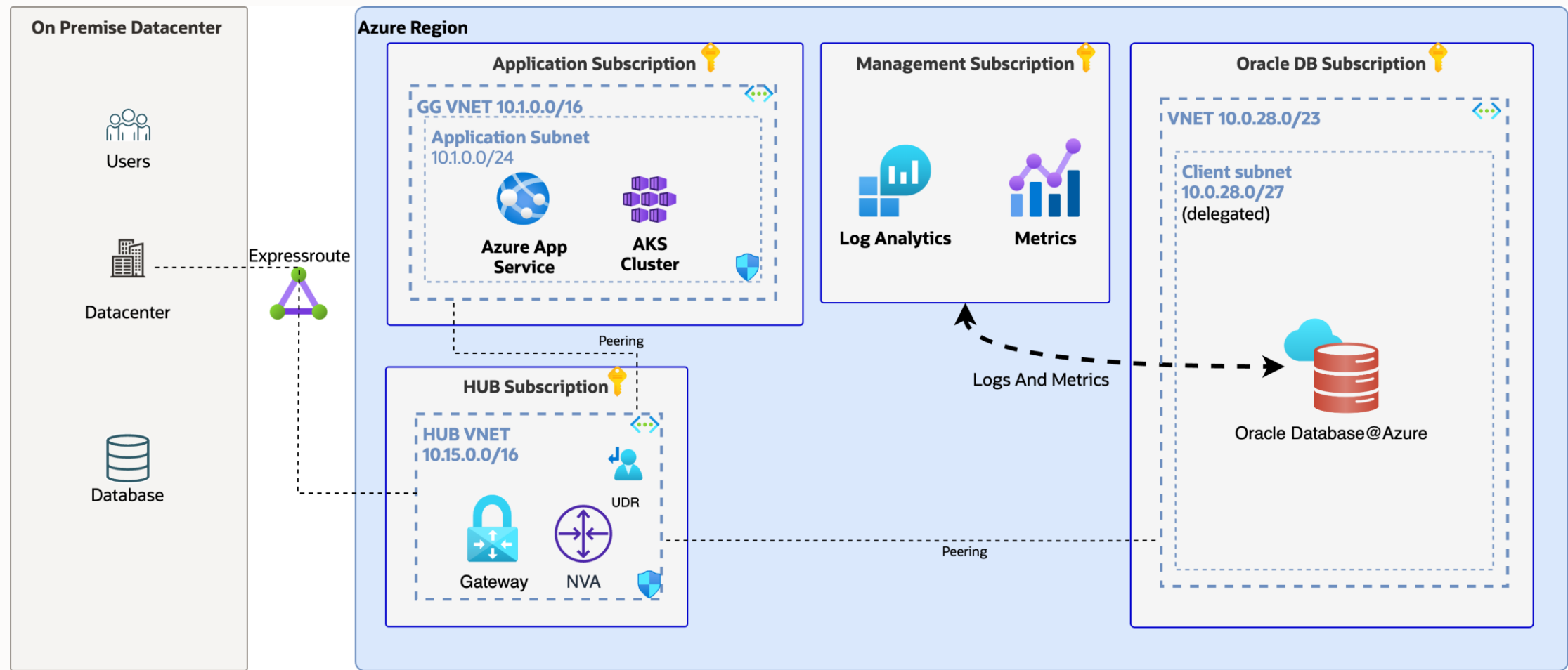
Migration with Data Pump – ADB-S



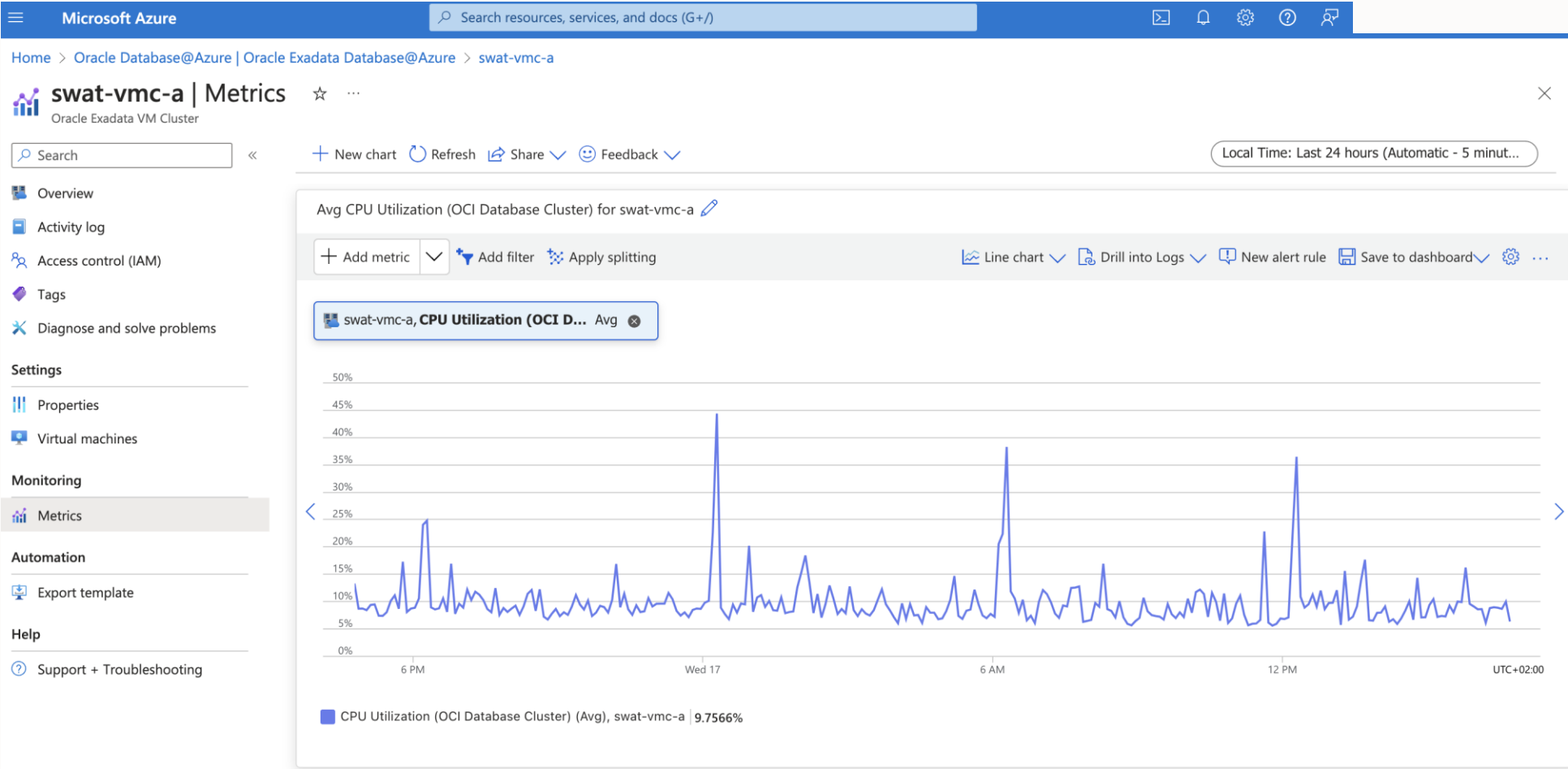
Monitoring

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Azure Subscription Segmentation: Monitoring subscription



Azure Monitor Metrics



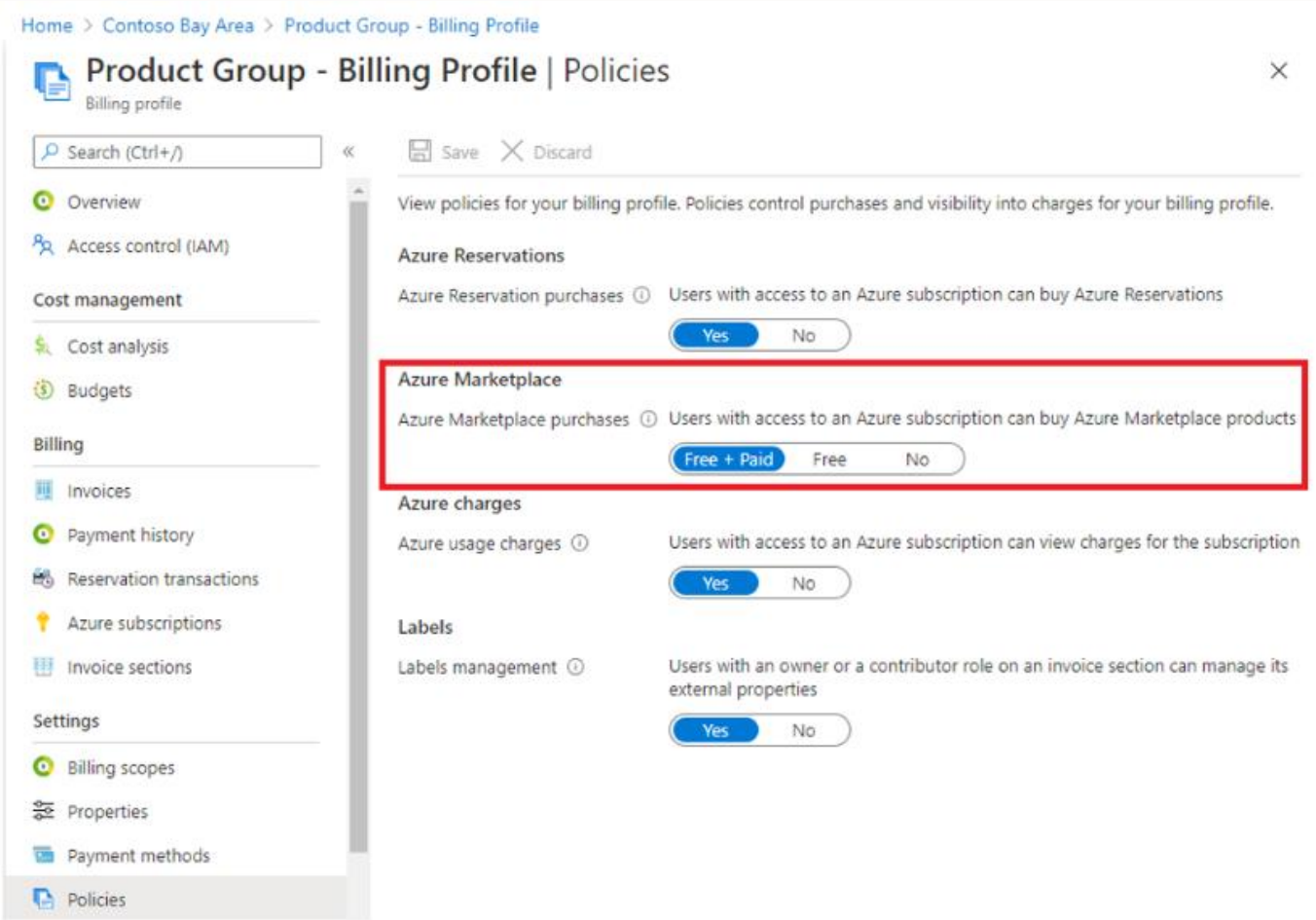
Identity & Permissions

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Identity - Azure roles

Azure permissions allowing to accept "Free + Paid" offers in Azure Marketplace

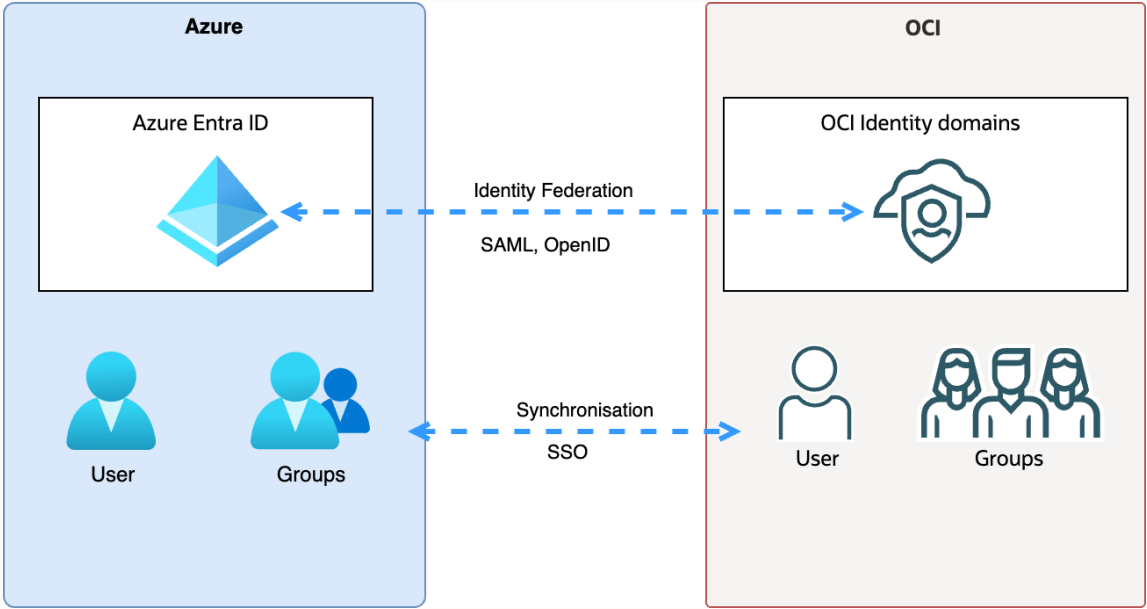
Agreement type	Permissions to accept offer	Permissions to purchase or subscribe
Microsoft Customer Agreement (MCA)	Billing account owner or contributor	Subscription owner or subscription contributor
Enterprise Agreement (EA)	Enterprise administrator	Subscription owner or subscription contributor



Identity Roles and Groups

Group name	Role Assigned	Description
odbaa-exa-infra-administrators	odbaa-exa-infra-administrator	User of this group will be able to administrate ExaDB-D resources in Azure
odbaa-vm-cluster-administrators	odbaa-vm-cluster-administrator	User of this group will be able to administrate VM Cluster resources in Azure
odbaa-db-family-administrators	N/A	This group is replicate in OCI as part of the federation process, OCI policies will be attached to it.
odbaa-db-family-readers	N/A	This group is replicate in OCI as part of the federation process, OCI policies will be attached to it.
odbaa-exa-cdb-administrators	N/A	This group is replicate in OCI as part of the federation process, OCI policies will be attached to it.
odbaa-exa-pdb-administrators	N/A	This group is replicate in OCI as part of the federation process, OCI policies will be attached to it.

Group name	Policy	Description
odbaa-db-family-administrators	manage database-family	Group to manage DB family actions.
odbaa-db-family-readers	read database-family	Group to read DB family actions.
odbaa-exa-cdb-administrators	manage db-homes manage databases manage db-backups	Group to manage CDB actions.
odbaa-exa-pdb-administrators	manage pluggable-databases	Group to manage PDB actions.



Identity Federation configuration

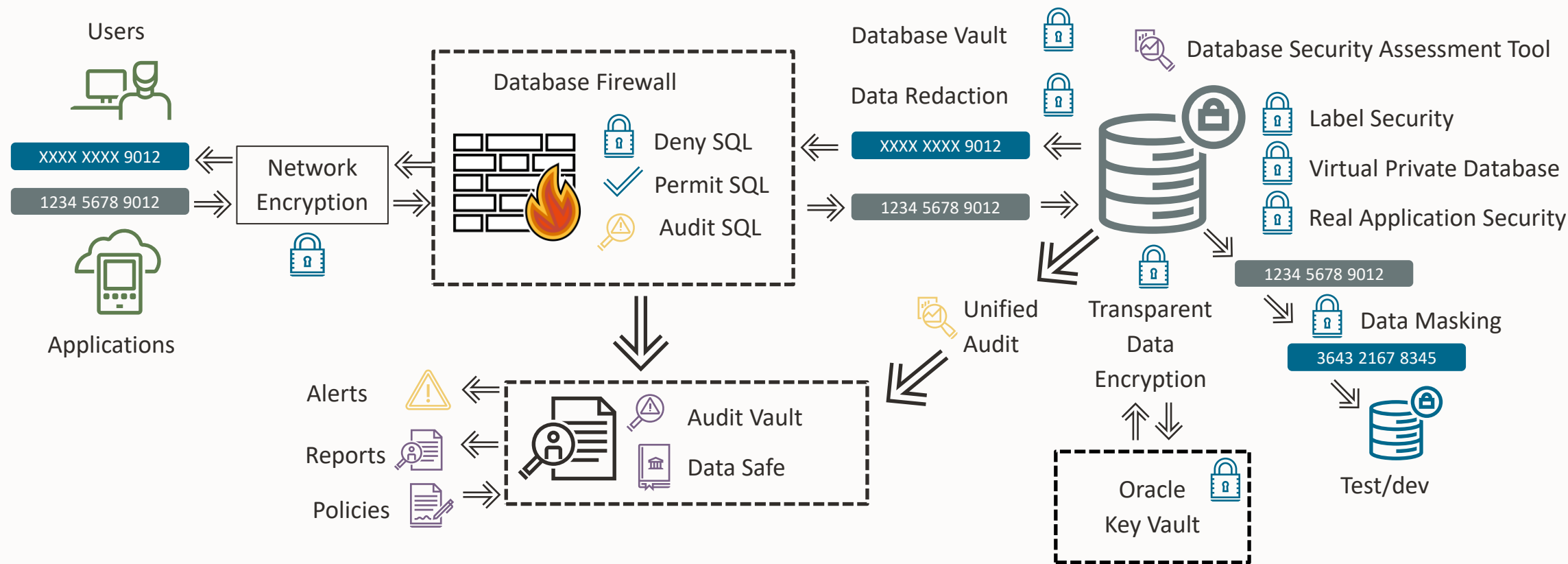


Security

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Oracle Maximum Security Architecture

Defense in Depth to protect your data



Securing your Databases

Control User and Network Access: Use passwords, private subnets, and network security groups to control user and network access.

Restrict Permissions for Deleting Database Resources: To prevent inadvertent or malicious deletion of databases, grant the delete permissions (DATABASE_DELETE and DB_SYSTEM_DELETE) to a minimum set of users and groups.

Data Encryption: All databases created in OCI are encrypted using transparent data encryption (TDE). Ensure that any migrated databases are also encrypted. Rotate the TDE master key every 90 days or less.

Patching: Apply Oracle Database security patches (Oracle Critical Patch Updates) to mitigate known security issues, and keep the patch levels up-to-date.

Use DB Security Tools: Established Data Masking and Auditing tools to secure your data are available for Oracle Database@Azure.

Enable Data Safe: Data Safe is a unified control center for Oracle cloud and on-premises databases. Use Data Safe to assess database and data security configuration, detect associated risk for user accounts, identify existing sensitive data, implement controls to protect data, and audit user activity.



Securing database traffic – in transit

Configuring Oracle Database Native Network Encryption and Data Integrity

Configure native Oracle Net Services data encryption and data integrity for both servers and clients.

Oracle Database enables you to encrypt data that is sent over a network

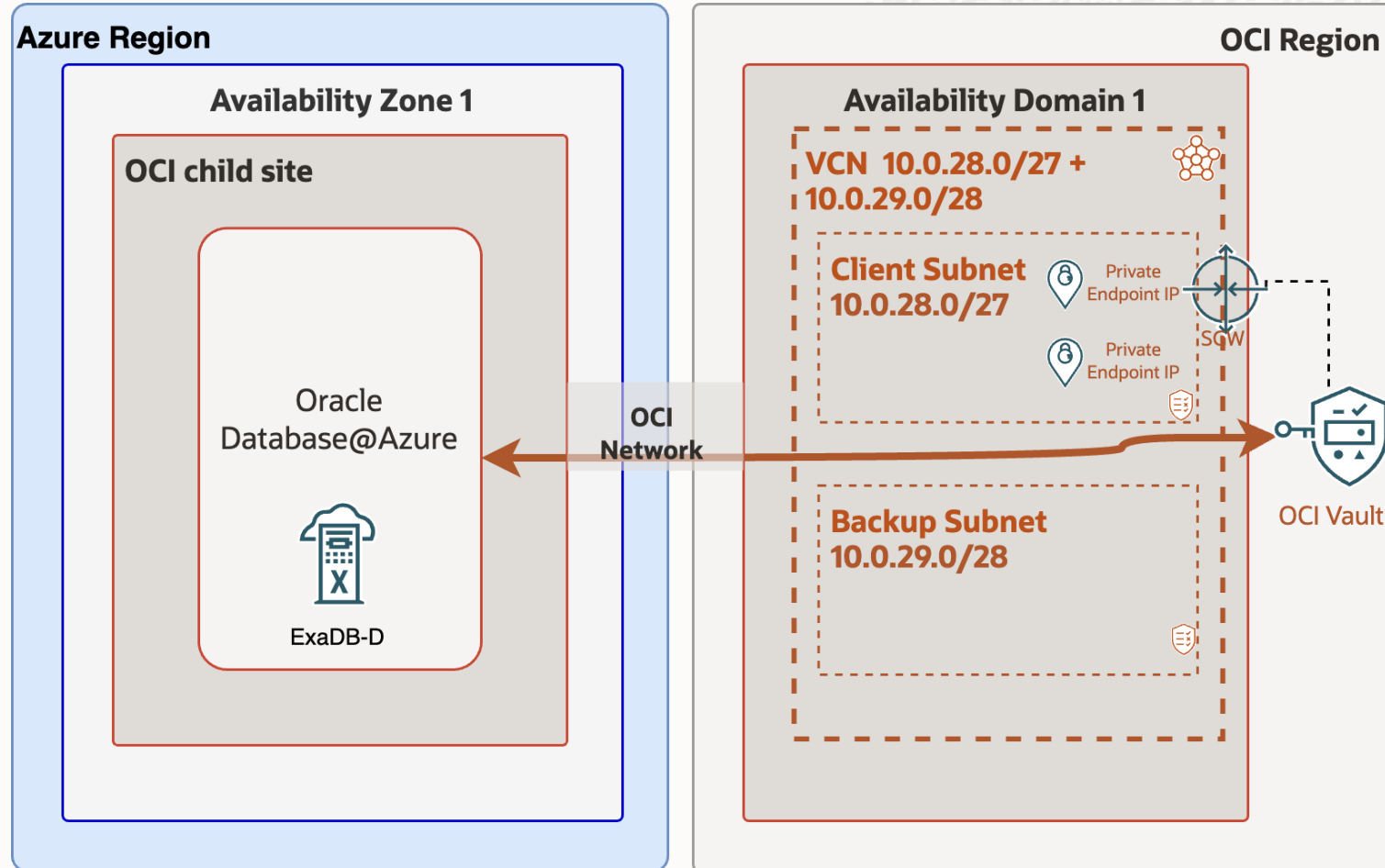
- Oracle Database provides **native data network encryption and integrity** to ensure that data is secure as it travels across the network
- Oracle Database supports the **Federal Information Processing Standard (FIPS)** encryption algorithm, **Advanced Encryption Standard (AES)**
- **Triple-DES encryption (3DES)** encrypts message data with three passes of the DES algorithm
- Oracle offers two ways to encrypt data over the network, native network encryption and **Transport Layer Security (TLS)**
- The Physical layer between the Oracle Database@Azure and OCI are **encrypted**.

<https://docs.oracle.com/en/database/oracle/oracle-database/19/dbseg/configuring-network-data-encryption-and-integrity.html>

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Encryption using OCI Vault – ExaDB-D

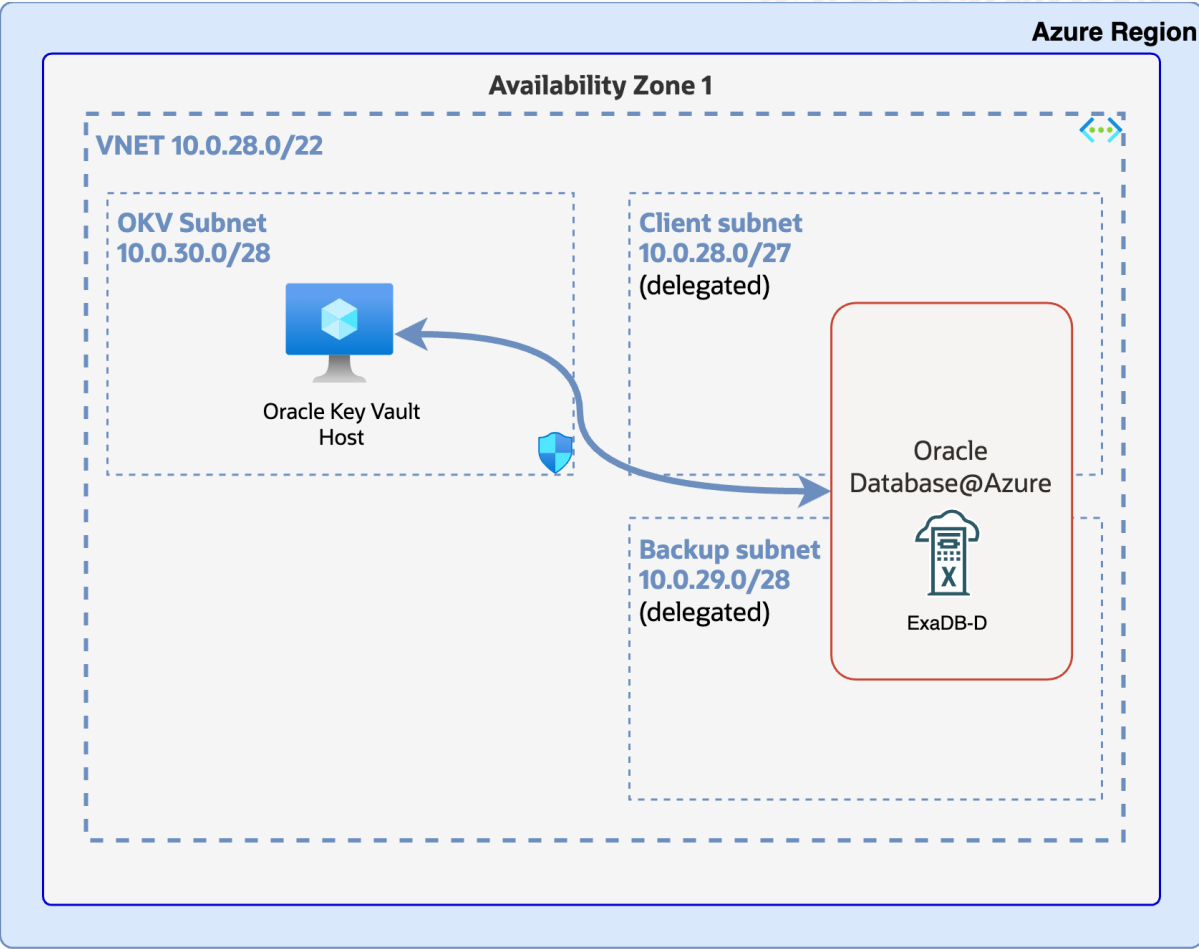


OCI Vault is the default Key Management system for Oracle Database@Azure. For using OCI Vault, the customer needs UCM.

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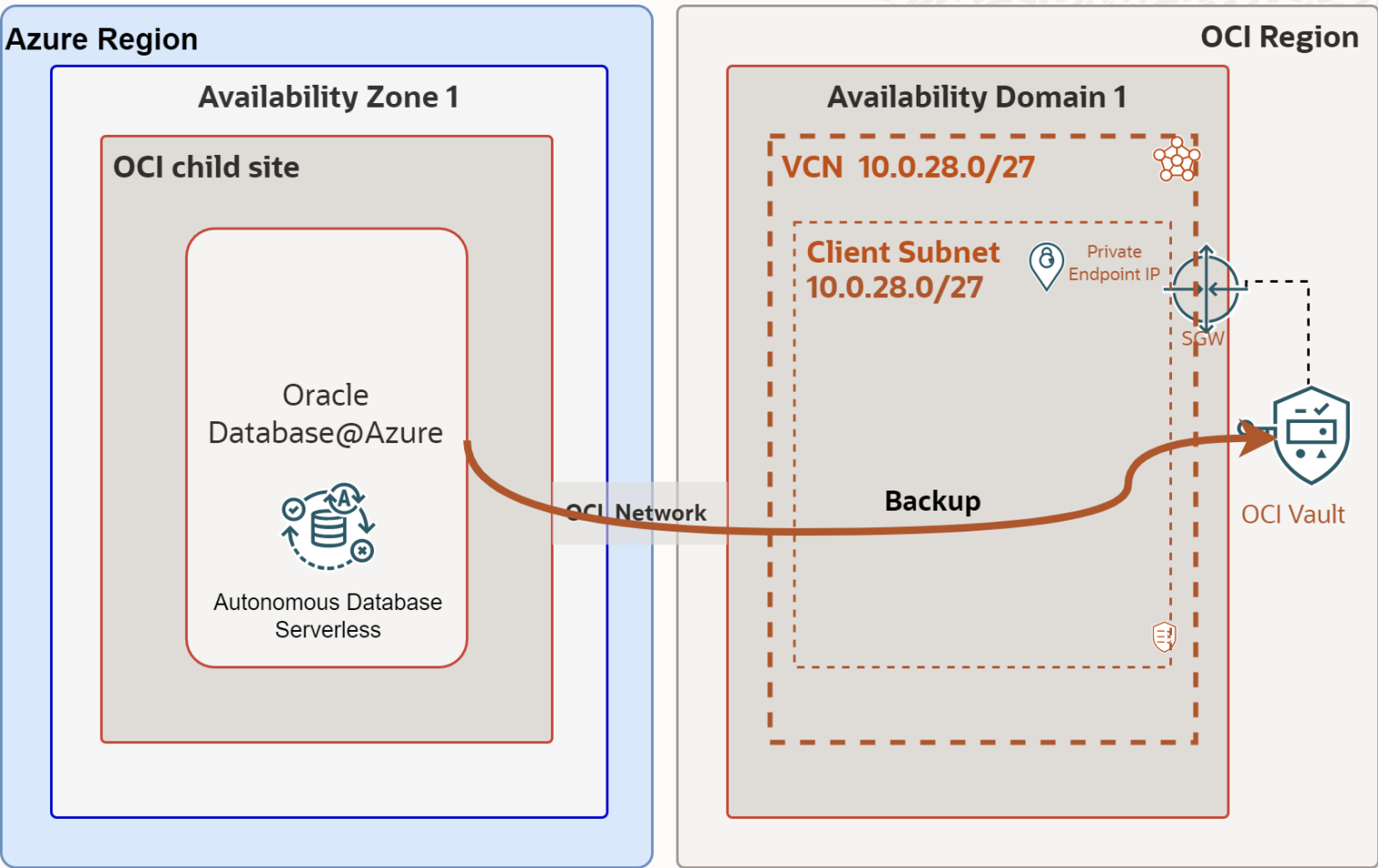
Encryption using Oracle Key Vault on VM Host – ExaDB-D



Standard images for OKV exist or Install an OKV instance on a Azure VM host.



Encryption using OCI Vault - ADBS



By default, Oracle Managed Keys and OCI Vault are available for ADB-S.
To use OCI Vault, the customer needs UCM.



Compliances



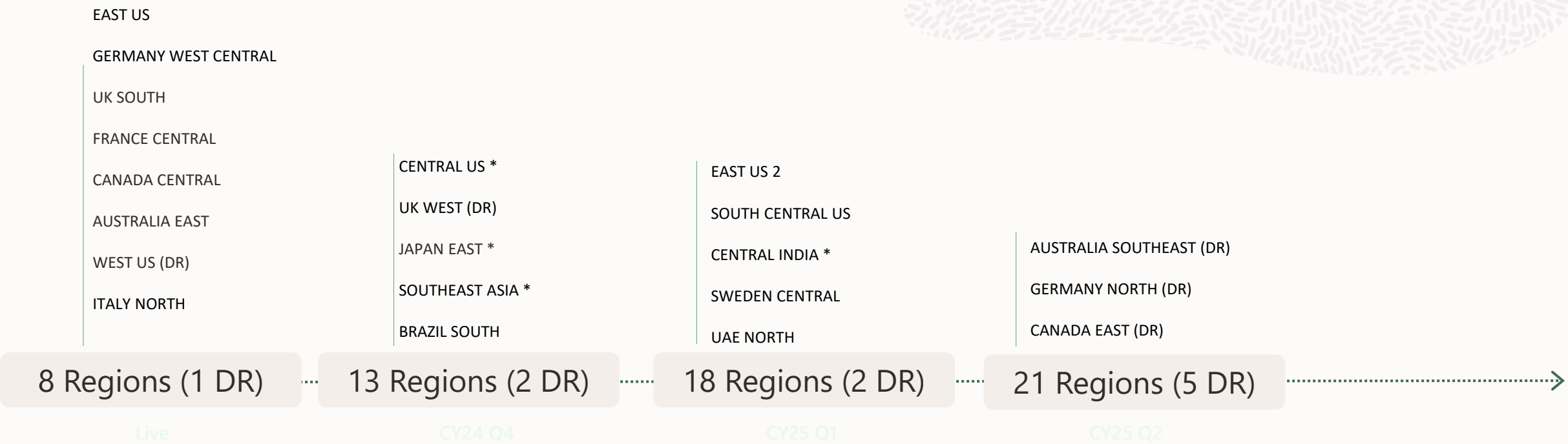
Certifications

- PCI-DSS, SOC 1,2,3, HIPAA, C5, ISO, CSA STAR, HDS
- Microsoft: <https://servicetrust.microsoft.com/>
- OCI: <https://www.oracle.com/corporate/cloud-compliance/>

Roadmap

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Oracle Database@Azure Region Roadmap: Onboarding Ready Dates



* First AZ available to onboard customer

More Planned Regions

- North Europe
- West Europe
- West US 2
- West US 3
- Spain Central

More Planned DR Regions

- Brazil Southeast
- France South
- Japan West
- North Central US
- South India



Automation

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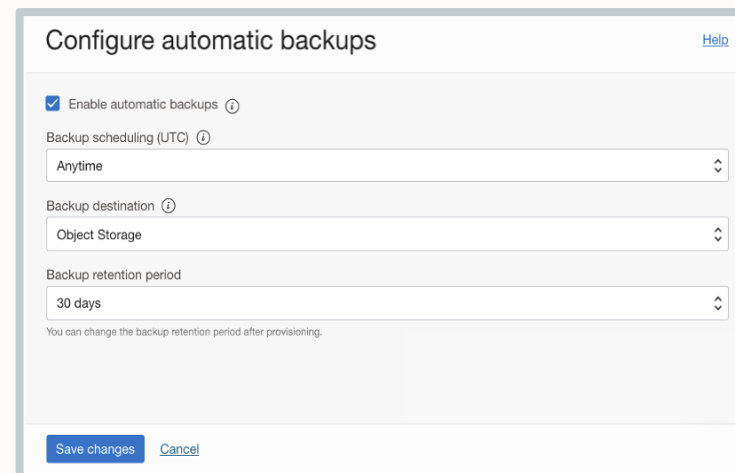
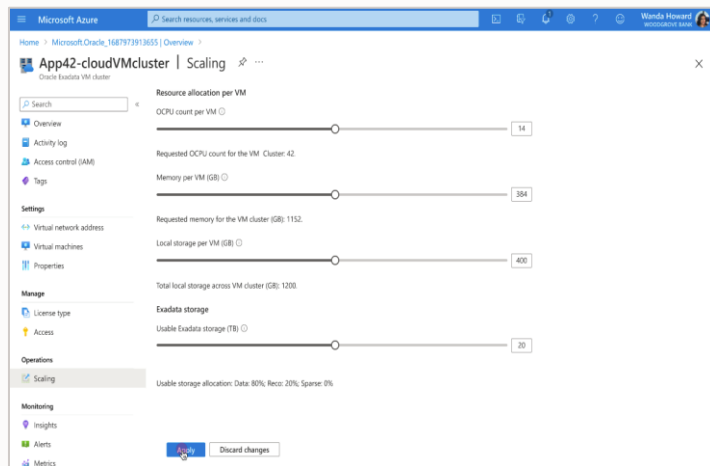
Cloud Automation for Lifecycle Management

Microsoft Azure Portal UI REST APIs, SDK, CLI, Terraform

- Create Oracle Exadata Infrastructure and Exadata VM Clusters
- Schedule Infrastructure Maintenance
- Scale Servers and OCPUs

Oracle Cloud UI REST APIs, SDK, CLI, Terraform

- Create Database Homes and Databases
- Update Operating System, Grid Infrastructure, and Databases
- Backup and Recover Databases
- Enable Data Guard



Terraform Automation

Combine AZURE and OCI provider to create the Exadata infrastructure, the VM cluster and the database in the same Terraform

```
terraform {
  required_providers {
    azurerm = {
      source = "hashicorp/azurerm"
      version = "= 3.69.0"
    }
    oci = {
      source = "oracle/oci"
      version = "5.10.0"
    }
  }
  required_version = ">= 0.14.9"
}
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

```
# azurerm_resource_group_template_deployment.exa_cluster will be created
+ resource "azurerm_resource_group_template_deployment" "exa_cluster" {
  + deployment_mode = "Incremental"
  + id              = (known after apply)
  + name            = "Terraform-Deploy-DemoExa-01-VMCluster-01"
  + output_content  = (known after apply)
  + parameters_content = jsonencode(
    {
      + clusterName = {
        + value = "DemoExa-01-VMCluster-01"
      }
    }
  )
}

# azurerm_resource_group_template_deployment.exa_infra will be created
+ resource "azurerm_resource_group_template_deployment" "exa_infra" {
  + deployment_mode = "Incremental"
  + id              = (known after apply)
  + name            = "Terraform-Deploy-DemoExa-01-Infra-02"
  + output_content  = (known after apply)
  + parameters_content = jsonencode(
    {
      + infraName = {
        + value = "DemoExa-01-Infra-02"
      }
    }
  )
}
```



Terraform Automation

The OCI provider create the database in Oracle Database@Azure

```
    }  
  }  
}  
  
# oci_database_db_home.exa_db_home will be created  
resource "oci_database_db_home" "exa_db_home" {  
  + compartment_id      = (known after apply)  
  + database_software_image_id = (known after apply)  
  + db_home_location    = (known after apply)  
  + db_system_id        = (known after apply)  
  + db_version          = "19.20.0.0"  
  + defined_tags        = (known after apply)  
  + display_name        = "TFDBHOME"  
  + freeform_tags       = (known after apply)  
  + id                  = (known after apply)  
  + is_desupported_version = (known after apply)  
  + kms_key_id          = (known after apply)  
  + kms_key_version_id  = (known after apply)  
  + last_patch_history_entry_id = (known after apply)  
  + lifecycle_details   = (known after apply)  
  + source              = "VM_CLUSTER_NEW"  
  + state               = (known after apply)  
  + time_created        = (known after apply)  
  + vm_cluster_id       = (known after apply)  
  
  + database {  
    + admin_password      = (sensitive value)  
    + backup_id           = (known after apply)  
    + backup_tde_password = (sensitive value)  
    + character_set       = (known after apply)  
    + connection_strings  = (known after apply)  
    + database_id         = (known after apply)  
    + database_software_image_id = (known after apply)  
    + db_name             = "TFCDB"  
    + db_unique_name      = (known after apply)  
    + db_workload         = "OLTP"  
    + defined_tags        = (known after apply)  
    + freeform_tags       = (known after apply)  
    + id                  = (known after apply)  
    + kms_key_id          = (known after apply)  
    + kms_key_version_id  = (known after apply)  
    + lifecycle_details   = (known after apply)  
    + ncharacter_set      = (known after apply)  
    + one_off_patches     = (known after apply)  
    + pdb_name            = "TFPDB"  
    + sid_prefix          = (known after apply)  
    + state               = (known after apply)  
    + tde_wallet_password = (sensitive value)  
    + time_created        = (known after apply)  
    + time_stamp_for_point_in_time_recovery = (known after apply)  
    + vault_id            = (known after apply)  
  }  
}
```

Plan: 3 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?

Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value:

Support

Service Levels



- SLA Oracle Database Exadata 99.95% <https://www.oracle.com/cloud/sla/>
- SLA Oracle Autonomous Database Serverless 99.995% <https://www.oracle.com/cloud/sla/>
- SLO Oracle Database@Azure 99.9% <https://docs.oracle.com/en-us/iaas/Content/General/Reference/servicelevelobjectives.htm>
- Support process Response – support level parity
- [SLA Breach - Service pillars](#)

Azure support request

Microsoft Azure

Search resources, services, and docs (G+)

[Home](#) > [Oracle Database@Azure | Overview](#) > [OracleSubscription](#) > [default](#) | [Support + Troubleshooting](#) > [Help + support](#) >

New support request

1. Problem description

2. Recommended solution

3. Additional details

4. Review + create

Tell us your issue, and we'll help you resolve it.

Provide information about your billing, subscription, quota management, or technical issue (including requests for technical advice).

Issue type *

Technical

Subscription *

Can't find your subscription? [Show more](#)

Service

☐ My services

☒ All services

Service type *

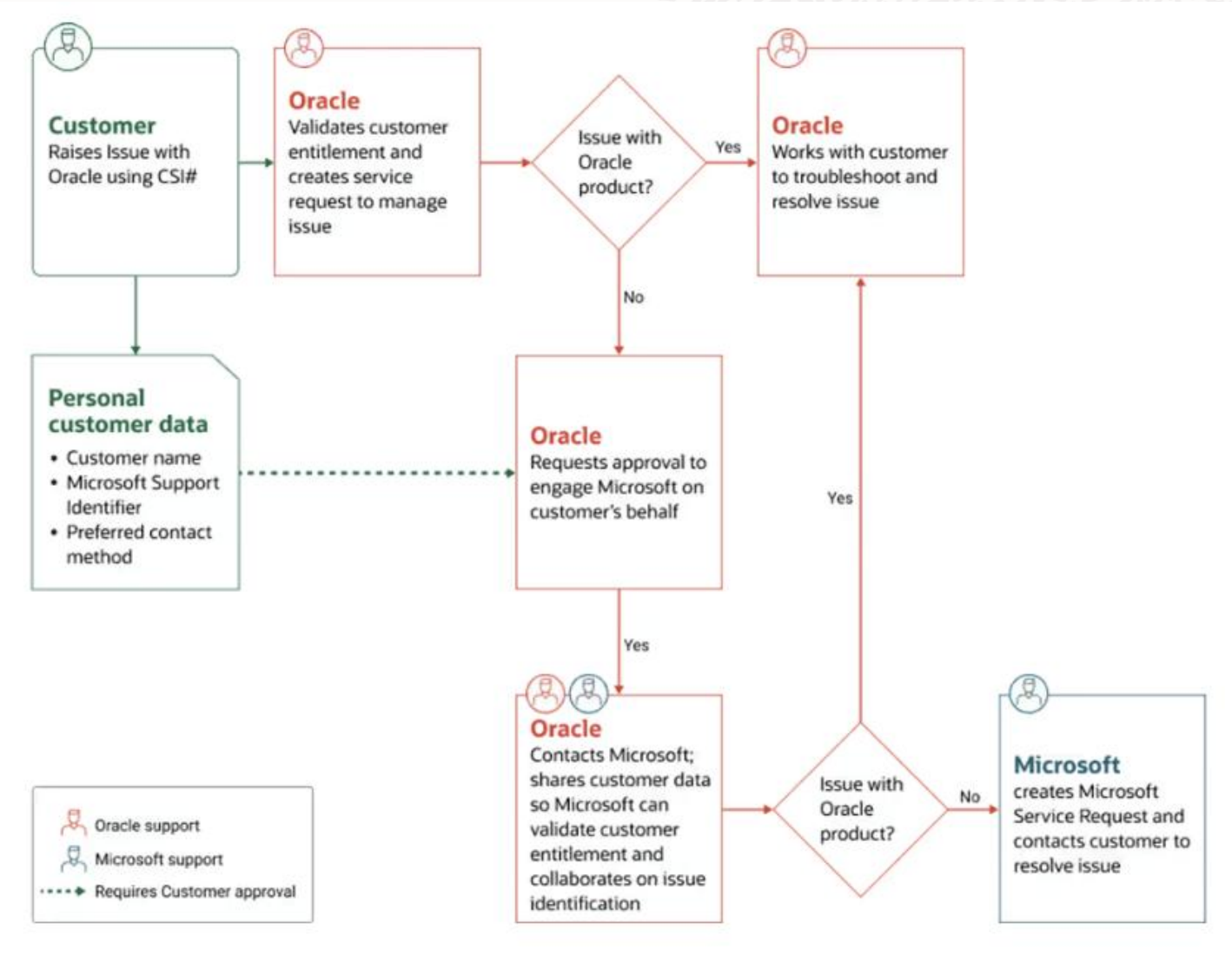
Oracle Database@Azure

Support for Oracle Database@Azure is provided by the Oracle Support team. Please visit the [OCI Support portal](#). If you're a first time user, you will need to register with the Oracle account in the OCI Support portal and have it approved by your Cloud Administrator. For any issues with accessing support for your account, please contact Oracle Support via email or phone, [details here](#)

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Collaborative Model Support



Best practices - Troubleshooting

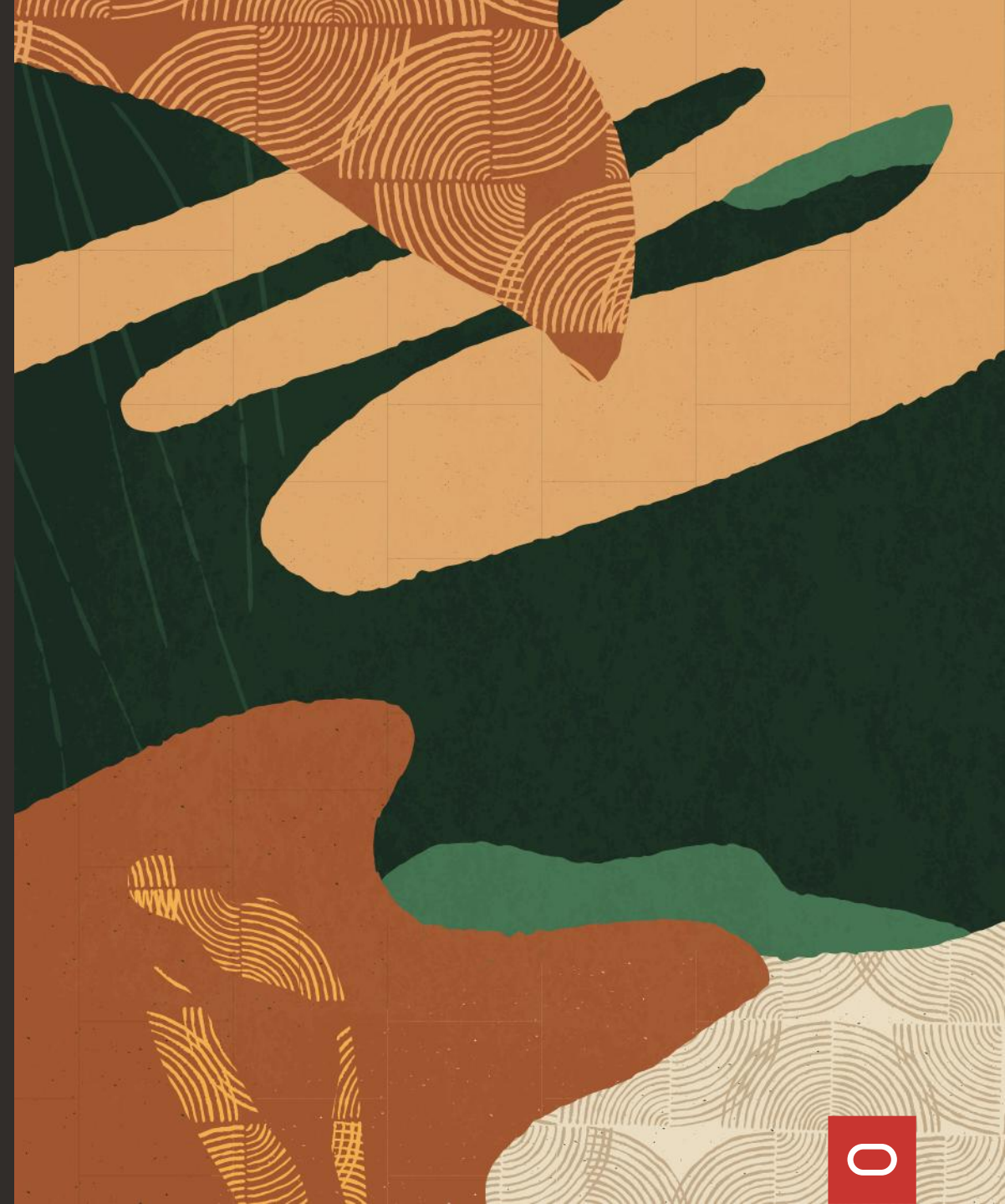


- Failing to deploy Cluster VM
 - Select Azure Subscription – Oracle Database@Azure
 - Select Azure resource group inside Azure Subscription
 - Select VNET in resource group
 - Delegated client subnet inside selected VNET
- If you cannot create resources – make sure there is no Azure lock
- Remove existing resources in client subnet and settings (delegated subnet)
- Can you access the database from the VNET ?
- Can you resolve the FQDN ?
- Make sure OCI Tenancy is IAM Domains enabled
- Cannot go to Go to OCI link
 - Check SSO or OCI credentials

Thank you & Next Steps

OCI Strategic Partnership and Growth

Ready to assist you in planning your migration



Oracle Cloud Infrastructure for Microsoft Azure professionals

Concept	Microsoft Azure	Oracle Cloud Infrastructure
Cluster of data centers and services	Region	Region
Abstracted data center	Availability Zone	Availability Domain
Account	Account	Tenancy
Organizing resources	Subscriptions Resource Groups	Compartments
Virtual Network	Virtual Network (VNET)	Virtual Cloud Network (VCN)
Dedicated Private Connectivity	ExpressRoute	FastConnect
Identity and Access Management	Active Directory	OCI IAM Domains

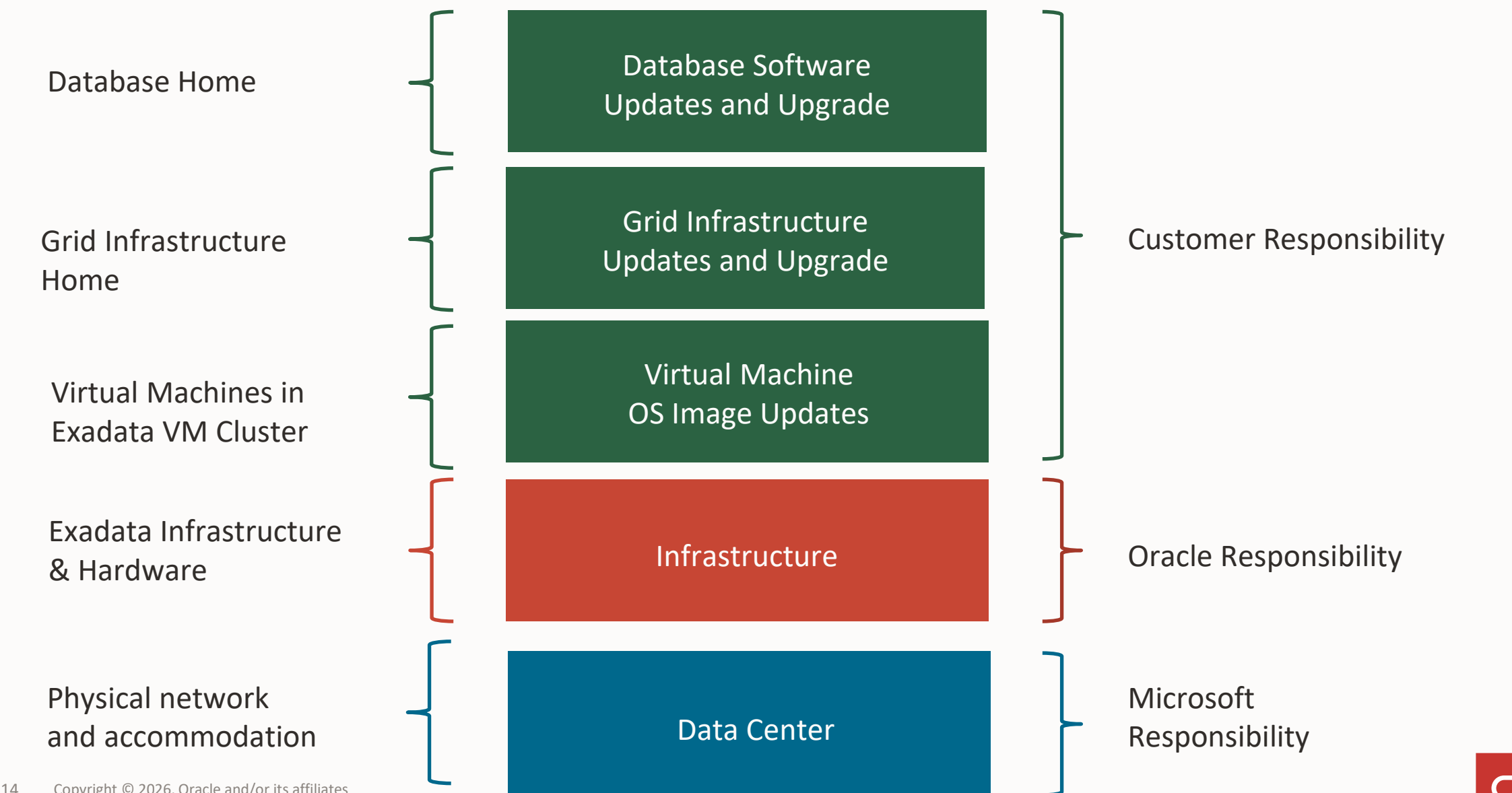


Multicloud Services

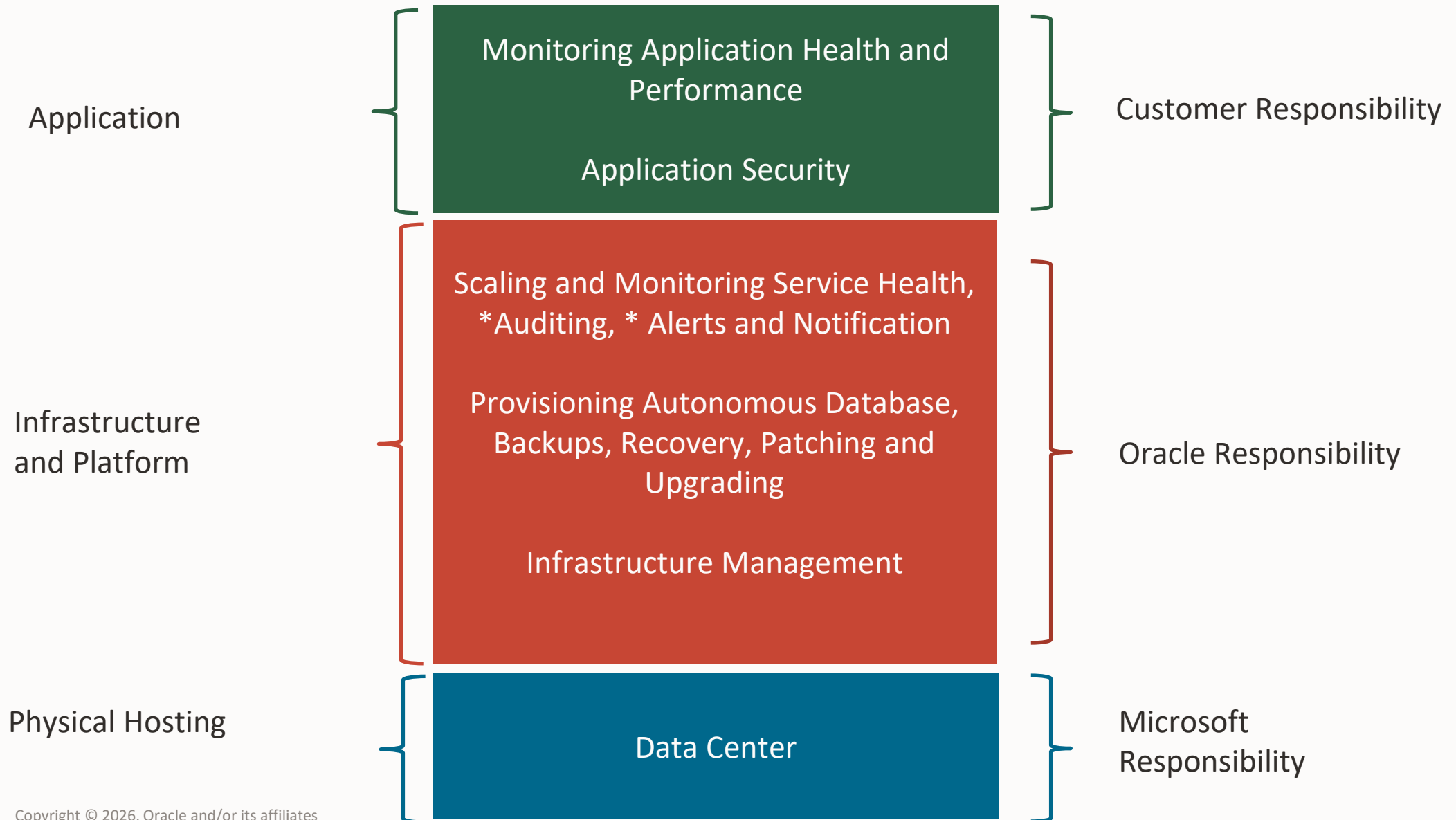
Services	OCI database services	Oracle Database@Azure	Oracle Interconnect for Microsoft Azure	Oracle Database Service for Azure
Connectivity	Native in OCI	Native in Azure	Dedicated Tunnel, Customer managed	Dedicated tunnel, Oracle managed
Network cost	Zero (local in OCI)	Zero (local to Azure)	ExpressRoute port and FastConnect port charges	Azure Vnet peering egress and ingress cost
OCI services available	ExaDB-D, ADB, Base DB, MySQL Heatwave	ExaDB-D	All OCI services	ExaDB-D, ADB, Base DB, MySQL Heatwave
Common use cases	App DB full stack in OCI	App DB full stack in Azure	App integration, App DB split stack	App DB split stack
SLA	ExaDB-D 99.95%; ADB 99.995%; BaseDB 99.9%; MySQL 99.95%	ExaDB-D 99.95%; ADB 99.995%;	Customer managed	SLO for Availability
Region availability	66 OCI regions	18 Regions	12 regions	12 regions
Latency		Azure native	< 2ms	~ 2ms
Support	OCI support	OCI and Azure collaborative	OCI and Azure collaborative	OCI and Azure collaborative
Automation	Yes	Yes	Yes	No
Decision points	High performance + low latency + no or low MACC	Large unmet MACC + ExaDB-D + low latency	Azure and OCI multicloud integrations / Need ADB and BaseDB immediately	Rapid deployment for quick POC



Oracle Database@Azure Responsibility Matrix



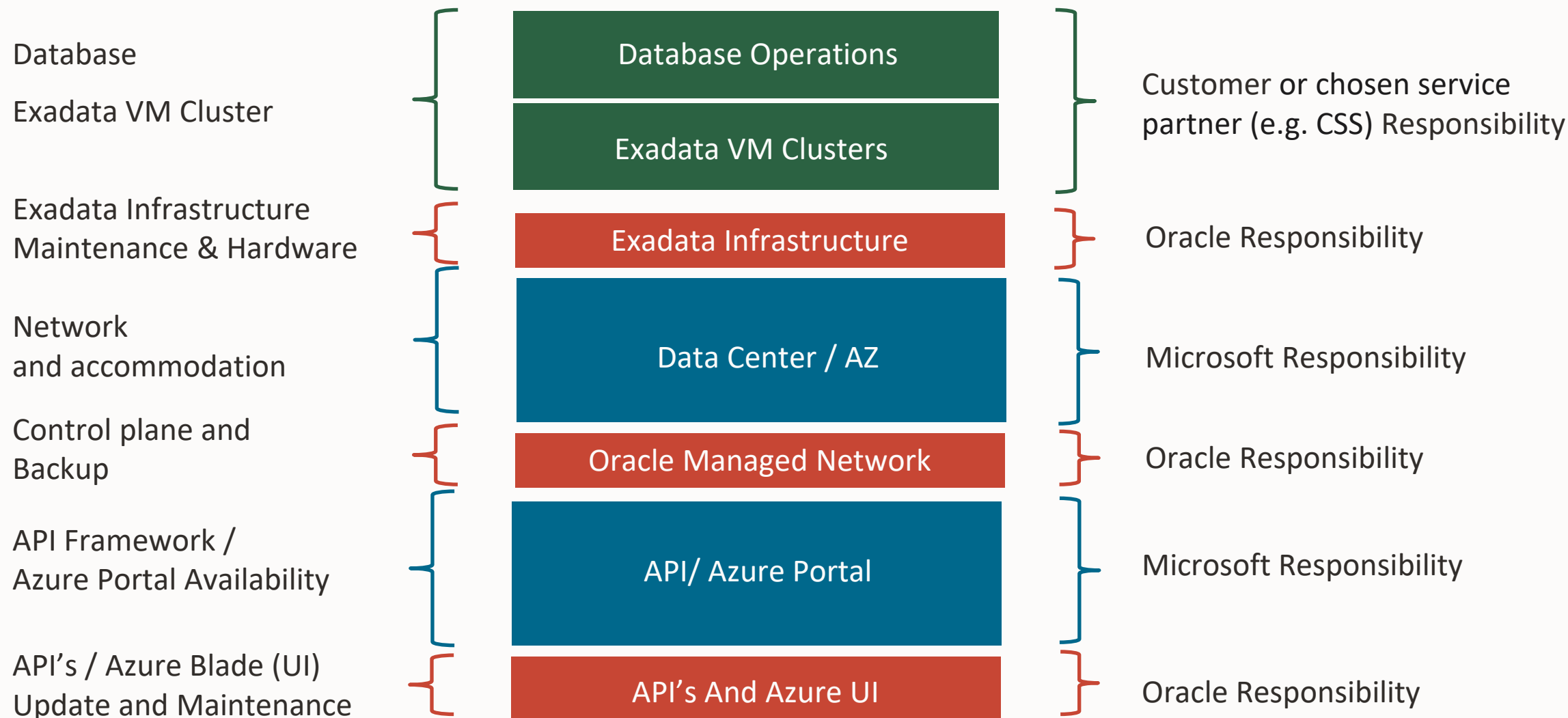
Oracle Database@Azure ADB-S Responsibility Matrix





Oracle Database@Azure

Oracle Database@Azure Responsibility Matrix



Oracle Database@Azure Operational Resources

	Azure	OCI
Resource & Lifecycle Management	Exadata instance management (Infra. & VM cluster)	VM Cluster & DB management Scaling, Maintenance DB CRUD, backup, keys, Database connection string information
Observability	VM and Databases Metrics	VM Databases logs and metrics
Billing, Costs & Usage	Invoice Cost management	Detailed usage reporting
Support	Collaboration Service Request first line to OCI	



Autonomous Database - Serverless Operational Resources

	Azure	OCI
Resource & Lifecycle Management	ADB-S creation	ADB-S Management Daily operations
Observability	Database logs and metrics*	Databases logs and metrics
Billing, Costs & Usage	Invoice Cost management	Detailed usage reporting
Support	Collaboration Service Request first line to OCI	





Oracle Database@Azure

Oracle Database@Azure Operational Resource Dependencies

- Oracle Database@Azure requires Client connectivity through Azure network
- Oracle Database@Azure is connected through OCI managed network for Exadata Database Service Control plane and automatic/standalone backup
- Oracle Database@Azure connectivity setup between OCI VCN and Azure edge router (MeetMe).
- Oracle Database@Azure Exadata Database resources are created and managed through API's and Azure Blade (UI) who are created and maintained by Oracle.
- APIs for creating and managing Exadata Database Service resources are published on the Azure API Framework.
- Blade (Azure UI) for Oracle Database@Azure is made available on Azure portal





Runtime vs Operational Outage Impact and SLA's affected

	Microsoft	OCI
Operations – Manageability is affected by following items:		
Exadata Resource Management- Azure Blade(UI) and API Framework	Affected when Portal/API framework is unavailable on Azure	N/A
Database / Exadata Operations - OCI Console and API's	N/A	Affected when OCI control plane is down – OCI Manageability SLA's
Runtime – Availability is affected by the following items:		
Client Connectivity on Azure Vnet to Exadata Database Service	Affected – Azure SLA's	N/A
Backend connectivity to OCI region	N/A	Backup affected – OCI SLA's
Connectivity between Azure edge router and OCI VCN	N/A	Affected OCI – SLA's
Azure AZ availability	Affected – Azure SLA's	N/A
OCI AD availability	N/A	Backup affected – OCI SLA's

<https://www.oracle.com/uk/cloud/sla/>

<https://www.microsoft.com/licensing/docs/view/Service-Level-Agreements-SLA-for-Online-Services?lang=1>



What is available in Azure and OCI console

	ADB-S @Azure	
	Azure portal	OCI console
Type	ATP and ADW	
Network Access	Private access	
Autonomous Data Guard	Autonomous Data Guard Backup-based disaster recovery	
Wallet	Download	Download Rotate
Connection strings	Yes	
Resource Scaling	Yes	
Elastic Pool	No*	Yes
Start/Stop	No	Yes
NSG	No*	Yes
Performance Hub	No	Yes
Encryption Keys	No	Yes
Backup/Restore	No	Yes
Database Actions	No*	Yes
Admin Password	Yes (During Provisioning)	Yes (After provisioning)
Backup Retention	Yes (During Provisioning)	Yes (After provisioning)
Licensing	Yes (During Provisioning)	Yes (After provisioning)
Enable mTLS	Yes (During Provisioning)	Yes (After provisioning)

* Feature on the roadmap